WORLDWIDE INFLUENZA CENTRE

WHO CC for Reference & Research on Influenza

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Report prepared for the WHO annual consultation on the composition of influenza vaccine for the Northern Hemisphere 2017-2018

27th February – 01st March 2017

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Acknowledgements

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The epidemiological section of this report was prepared using data reported to the European Surveillance System (TESSy) and presented in Flu New Europe (http://www.flunewseurope.org) the joint ECDC-WHO Europe weekly influenza update, with the assistance of members of the Influenza and Other Respiratory Pathogens Group at the WHO European Office:

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CONTENTS

WHO Vaccine Recommendation Meeting for the Northern Hemisphere 2017-2018 influenza season.

WHO CC, London - Final Report for WHO meeting: 27 February- 01 March 2017.

Influenza Activity in the WHO European Region, October 2016 – February 2017

Weekly reporting on influenza activity started in week 40/2016. The course of the 2016-2017 season, to week 06/2016, has been characterized by an early increase in levels of transmission compared to previous years **(Figure E1-A)** and no clear west to east pattern in the spread of influenza across the Region **(Figure E2).** Influenza A viruses have predominated accounting for 96% of all sentinel detections with the great majority (99%) of subtyped influenza A viruses from sentinel sites being A(H3N2). Confirmed cases of influenza virus type A infection reported from hospitals have predominantly been in adults aged over 65 years. Substantial excess all-cause mortality has been observed in people aged 15–64 years and markedly in people aged 65 years or older in the majority of the 19 countries reporting such data. Two-thirds of the A(H3N2) viruses genetically characterized belong to a recently emerged genetic subclade (3C.2a1). Vaccine effectiveness estimates are similar to those in recent years. No reduced susceptibility to oseltamivir or zanamivir has been observed for any of the 918 viruses tested so far this season.

The proportions of respiratory specimens from patients presenting with influenza-like illness (ILI) or acute respiratory infection (ARI) symptoms that tested positive for influenza virus increased between weeks 46/2016 and 51/2016 with the overall positivity rate in specimens from sentinel sources reaching approximately 50% until week 4/2017 before two consecutive weeks with of decline, to 42% in week 6/2017. The apparent peak in positivity this season was earlier than those in recent years and was higher than peaks observed in the 2015-2016 and 2013-2014 seasons but similar to that observed in the 2014-2015 season, also a predominantly influenza A(H3N2) season **(Figure E1-A)**. Increasing trends in ILI and ARI were reported by at least one country in each sub-region by week 49/2017 and each sub-region has experienced at least one week of high or very high intensity of influenza activity to date. However, there have been no sustained periods of high or very high intensity in western or central Asian parts of the region and countries in these areas have reported predominantly baseline levels of activity in recent weeks unlike elsewhere. So far this season Southern Europe is the only sub-region to report sustained periods of very high intensity influenza activity **(Figure E2)**.

For the whole WHO European Region, over 100,000 influenza detections have been reported to the European Surveillance System (TESSy) - with type A detections (96%) predominating over type B (4%) **(Table E1)**. This number of influenza detections is substantially more than that observed at a similar point last season (over 30,000 influenza detections; week 04/2016) and is likely, partially at least, due to the earlier start to the season. Within type A, the A(H3N2) subtype (99%) has predominated over A(H1N1)pdm09 viruses (1%) **(Table E1)** and this pattern is evident for both sentinel and non-sentinel influenza virus detections **(Figures E1-B & C)**. Across the Region, of those type B viruses for which lineage-determination was performed, 57% were of the Yamagata lineage **(Table E1).** While Yamagata lineage type B viruses were predominant (60% and over**)** in southern, western, eastern and northern European countries**,** in west and central Asian countries Victoria lineage type B viruses were dominant, though the number of viruses with a determined lineage from western Asia was very low (n=9). While all countries in the Region have reported influenza type A virus detections to outnumber those of influenza B; the proportions of influenza B detections have been considerably greater in some countries in central Asia compared to those in other sub-regions, yielding a type A:B ratio of 62%:38%. It is of note that A(H3N2) detections have greatly outnumbered A(H1N1)pdm09 detections across the Region; there were no A(H1N1)pdm09 detections in central Asia and no more than 2% of subtyped influenza A viruses were A(H1N1)pdm09 in other regions. **(Table E1 and Figures E2 & E3)**.

Since week 40/2016, nine countries (the Czech Republic, Finland, France, Ireland, Romania, Slovakia, Spain, Sweden, and the United Kingdom) have reported a total of 5999 laboratory-confirmed hospitalized influenza cases (2929 cases from ICUs and 3070 from other hospital wards) with cases from ICU peaking over the New Year period **(Figure E4-A).** Among cases admitted to ICUs influenza type A detections (98%) have predominated over type B (2%). Within type A, viruses of the A(H3N2) subtype (90%) have predominated over A(H1N1)pdm09 viruses (10%). Similarly, the observed proportion of influenza B detections (<1%) was low in all other hospital wards and A(H3N2) was the dominant subtype (>99%) among influenza A detections. For hospitalized patients where age was reported, those aged 65 years and older accounted for the greatest number of cases **(Figure E4-B).** A(H3N2) viruses were dominant in all age groups.

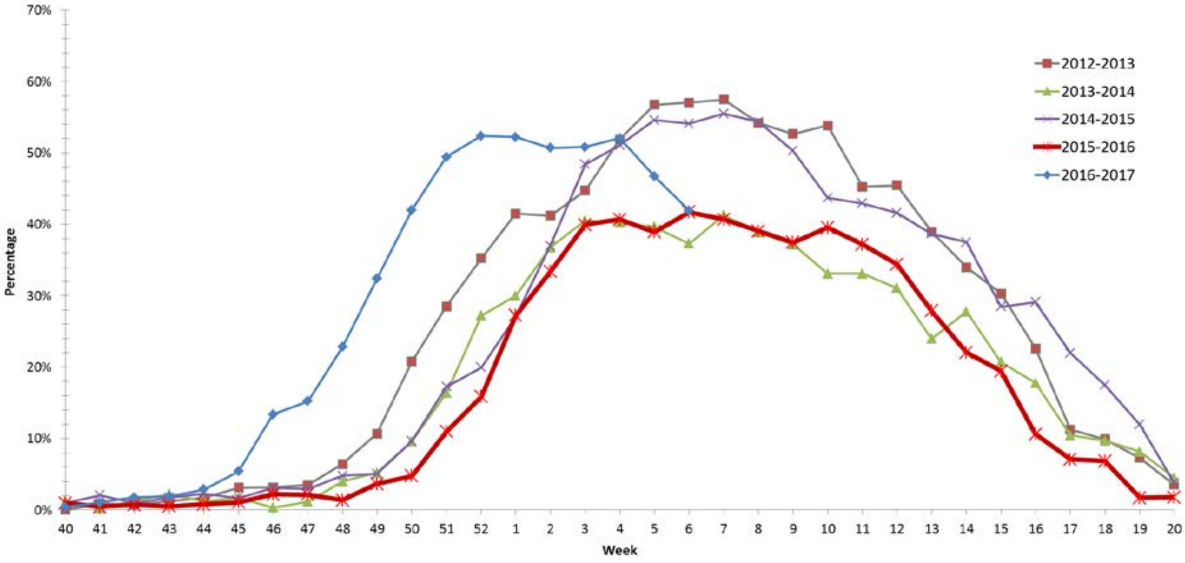
Since week 40/2016, 15 European countries with sentinel surveillance systems for severe acute respiratory infection (SARI) have reported a total of 23 903 cases. These cases occurred mainly among children but in some countries, including Albania and Serbia, increases in those aged 65 years and older have been observed **(Figure E4-C)**. Of 6 373 cases tested for influenza virus, 2 353 (37%) were positive: type A viruses (85%) predominated over type B (15%) and, of the influenza A viruses subtyped, the vast majority (>99%) were A(H3N2). Virology by age data are not available.

Neuraminidase inhibitor susceptibility has been assessed for 918 viruses (867 A(H3N2), 12 A(H1N1)pdm09 and 39 type B) with collection dates since week 40/2016. None has shown evidence of reduced inhibition to oseltamivir or zanamivir.

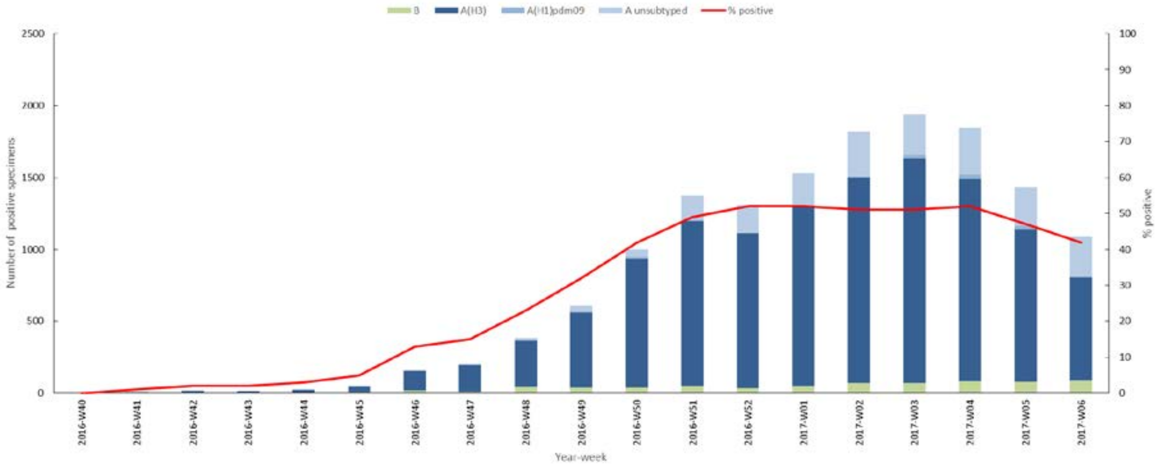
Many of the 17 countries (Belgium, Denmark, Estonia, Finland, France, Greece, Hungary, Ireland, Italy, Malta, the Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, and the United Kingdom (England, Scotland and Wales only)) participating in the European project for monitoring excess mortality for public health action (EuroMOMO – http://www.euromomo.eu) have this season observed a marked increase in all-cause excess mortality among the elderly aged 65 years and older. A substantial increase has also been observed in the 15–64 years age group. In the most recent influenza A(H3N2) dominated season (2014-2015) an excess all-cause mortality was also observed in those aged 65 years and older but with a smaller signal among the 15-64 years age group.

Figure E1. WHO European Region Influenza Activity: 2016-2017 (weeks 40/2016-06/2017)

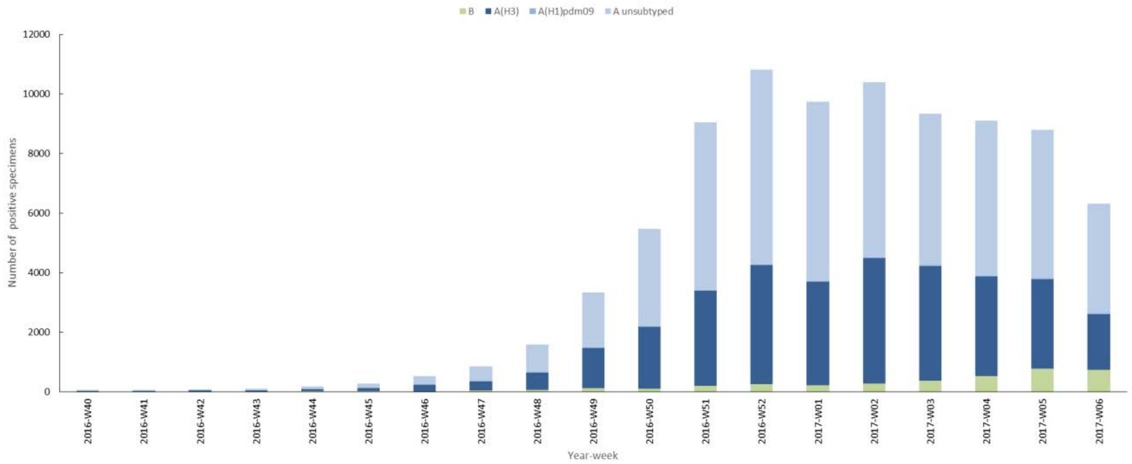
A. Percentage of sentinel ILI/ARI specimens testing positive for influenza by season



B. Sentinel Influenza Detections



C. Non-sentinel Influenza Detections



Source: European Centre for Disease Prevention and Control/WHO Regional Office for Europe. Flu News Europe, Joint ECDC–WHO/Europe weekly influenza update

Figure E2. Development of the 2016-2017 Influenza Season in the WHO European Region

Intensity of Influenza Activity and Dominant Virus (sub)type

Country / Week	40	41	42	43	44	45	46	47	48	49	50	51	52	1	2	3	4	5	6
Iceland	-	-	-	-	-	-	-	-	-	-	-	AH3	AH3	AH3	-	AH3	AH3	AH3	AH3
Ireland	-	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Portugal	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	-	-	-
Spain	-	-	AH3	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
United Kingdom (Northern Ireland)	AH3	AH3	AH3	A	A/B	A	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
United Kingdom (Wales)	-	-	AH3	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	-
United Kingdom (Scotland)	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	A
United Kingdom (England)	-	-	-	-	-	-	A	-	A	A	A	A	A	A	A	A	A	A	AH3
France	-	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Belgium	-	-	-	-	-	-	-	AH3	-	A	A	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Netherlands	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Luxembourg	-	-	-	-	-	-	-	AH3	-	AH3	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Switzerland	A	-	A	A	A	A	A	A	A	A	AH3	AH3	A	A	AH3	AH3	AH3	AH3	AH3
Germany	-	-	-	-	-	-	AH3	AH3	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Denmark	-	-	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Norway	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	-
Italy	-	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Austria	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3
Malta	-	-	-	-	-	-	-	-	-	-	-	AH3	AH3	AH3	-	-	-	-	-
Slovenia	-	-	-	-	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	B/AH3	B/AH3
Sweden	-	-	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Croatia	-	-	-	-	-	-	-	-	AH1	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Czech Republic	-	-	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Bosnia and Herzegovina	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Slovakia	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Albania	-	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	A	-	-	-
Hungary	-	-	-	-	-	-	-	-	-	A	-	AH3	AH3	AH3	-	AH3	AH3	AH3	AH3
Poland	-	-	-	-	-	-	-	-	-	A	A	A	A	A	A	A	A	A	A
Montenegro	-	-	-	-	-	-	-	-	AH3	AH3	A	A	A	-	-	-	-	-	-
Serbia	-	-	-	-	-	-	-	-	-	-	-	-	AH3	AH3	AH3	A	AH3	AH3	AH1H3
The former Yugoslav Republic of Macedonia	-	-	-	-	-	-	-	-	-	AH3	-	-	AH3	AH3	-	AH3	AH3	-	-
Greece	-	-	-	-	-	AH3	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Lithuania	-	-	-	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Bulgaria	-	-	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	-
Latvia	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	A	B/AH3	AH3	AH3	A	AH3
Romania	-	-	-	-	-	AH3	AH3	AH3	AH3	A	A	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Estonia	-	-	-	-	-	-	-	-	-	-	-	AH3	A	-	AH3	AH3	-	-	-
Finland	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Belarus	-	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	B/AH3	B/AH3	-	-
Republic of Moldova	-	-	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	-	-	AH3	-	-
Ukraine	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	-	-	-
Cyprus	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Israel	-	-	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Turkey	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	B/AH3
Georgia	-	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	-	-	B/AH3	B
Armenia	-	-	-	-	-	-	AH3	AH3	-	-	-	-	AH3	-	AH3	AH3	AH3	-	-
Azerbaijan	-	-	-	-	-	-	-	-	-	A	A	AH3	-	-	-	AH3	AH3	AH3	-
Uzbekistan	-	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	-	-	-	-
Kazakhstan	-	-	-	-	-	-	-	-	B	B/AH3	AH3	B/AH3	AH3	AH3	AH3	AH3	AH3	-	-
Russian Federation	-	-	-	-	-	-	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Tajikistan	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Kyrgyzstan	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

WEST

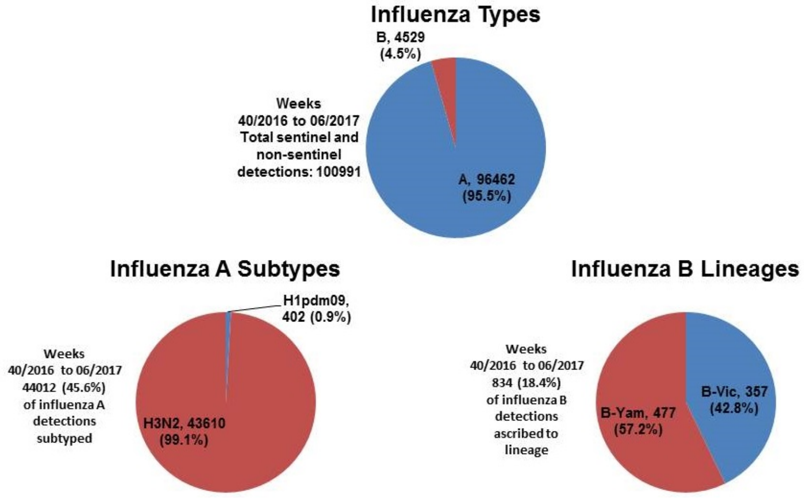
EAST

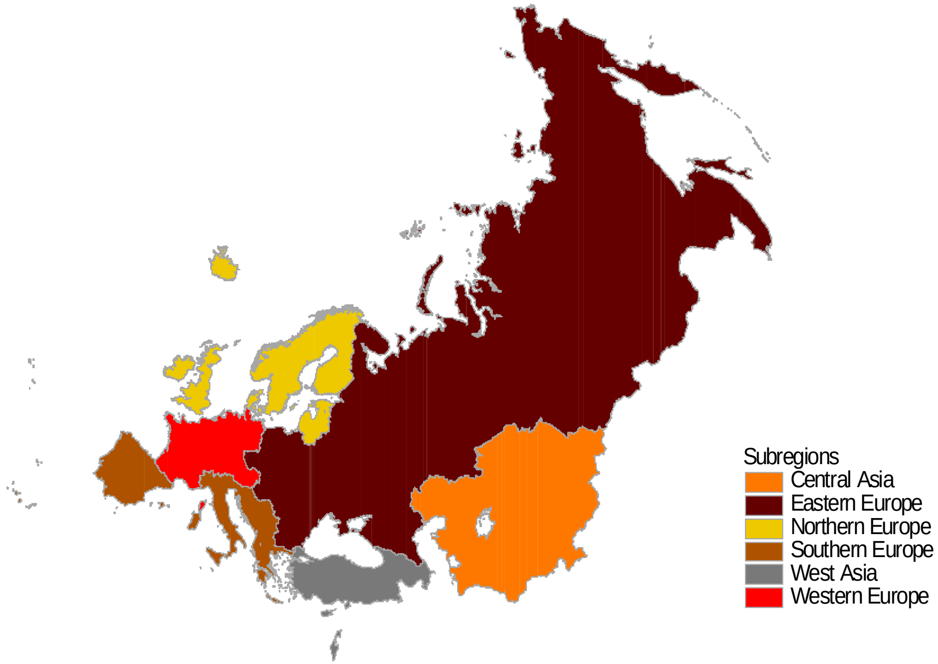
* Baseline influenza activity is the level at which clinical influenza activity remains throughout the summer and most of the winter.

Very High = exceptionally high level of activity
High = levels of activity higher than usual
Medium = usual levels of activity
Low = no activity or activity at baseline* level

Table E1. Distribution of Influenza Detections in the WHO European Region[1]: 2016-2017 Season (weeks 40/2016-06/2017)

Proportions of Influenza Types/Subtypes/Lineages based on Sentinel and Non-sentinel Detections





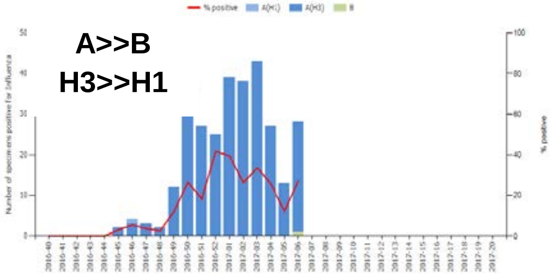
Influenza Detections*	Southern Europe		Western Europe		Eastern Europe		Northern Europe		West Asia		Central Asia		WHO European Region	
Influenza A	**16571**	**94%**	**22747**	**99%**	**15021**	**93%**	**38880**	**96%**	**2438**	**93%**	**805**	**62%**	**96462**	**96%**
Influenza A subtyped	11002		8333		12684		9531		1657		805		44012	
A (H1N1)pdm09	*72*	*1%*	*63*	*1%*	*212*	*2%*	*51*	*1%*	*4*	*0%*	*0*	*0%*	*402*	*1%*
A (H3N2)	*10930*	*99%*	*8270*	*99%*	*12472*	*98%*	*9480*	*99%*	*1653*	*100%*	*805*	*100%*	*43610*	*99%*
Influenza B	**1093**	**6%**	**208**	**1%**	**1115**	**7%**	**1444**	**4%**	**178**	**7%**	**491**	**38%**	**4529**	**4%**
B lineage determined	226		41		39		354		9		165		834	
B/Yamagata lineage	*185*	*82%*	*32*	*78%*	*23*	*59%*	*222*	*63%*	*0*	*0%*	*15*	*9%*	*477*	*57%*
B/Victoria lineage	*41*	*18%*	*9*	*22%*	*16*	*41%*	*132*	*37%*	*9*	*100%*	*150*	*91%*	*357*	*43%*
Total	***17664***		***22955***		***16136***		***40324***		***2616***		***1296***		***100991***	

*combined ILI, ARI sentinel and non-sentinel respiratory specimens positive for influenza

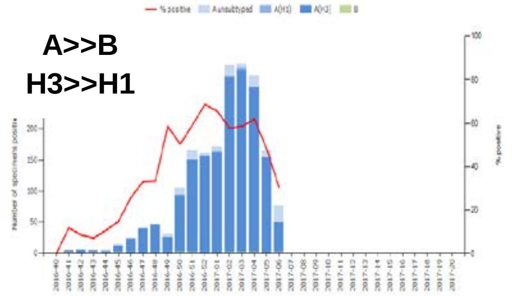
[1] WHO European Region Member States grouped into sub-regions according to United Nations geographic composition. Available from: http://unstats.un.org/UNSD/METHODS/M49/M49REGIN.HTM

Figure E3. Patterns of Influenza Type and Subtype Circulation in the WHO European Region

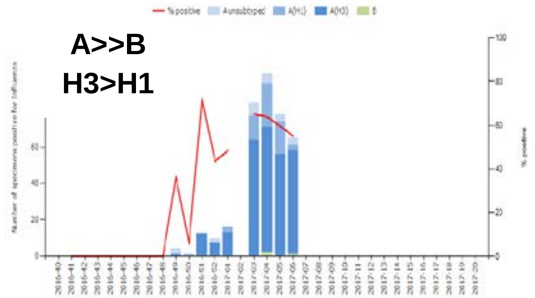
United Kingdom-England (Sentinel)



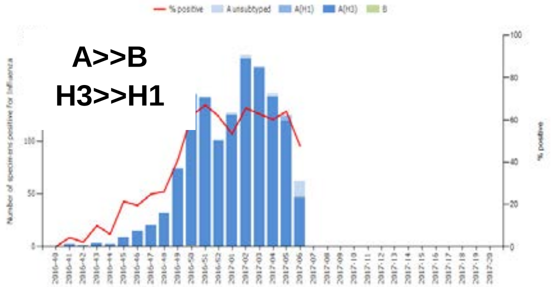
Spain (Sentinel)



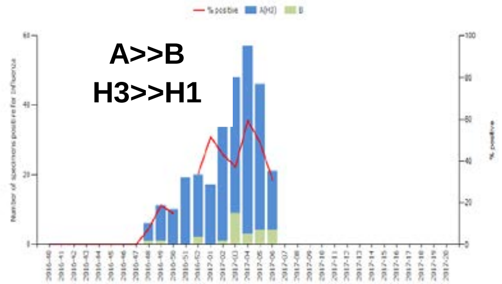
Hungary (Sentinel)



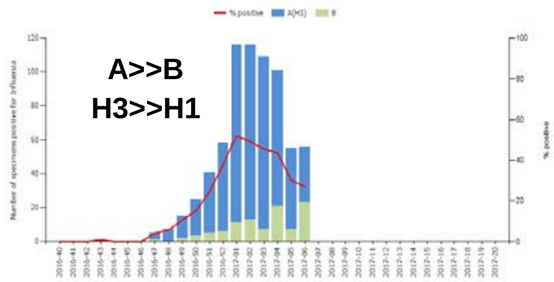
France (Sentinel)



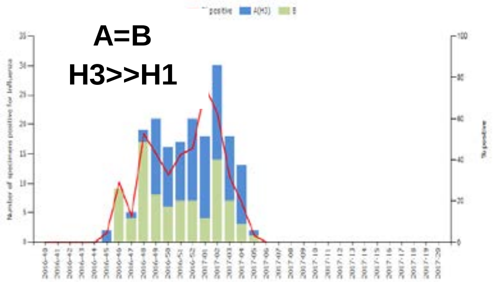
Russian Federation (Sentinel)



Turkey (Sentinel)



Kazakhstan (Sentinel)

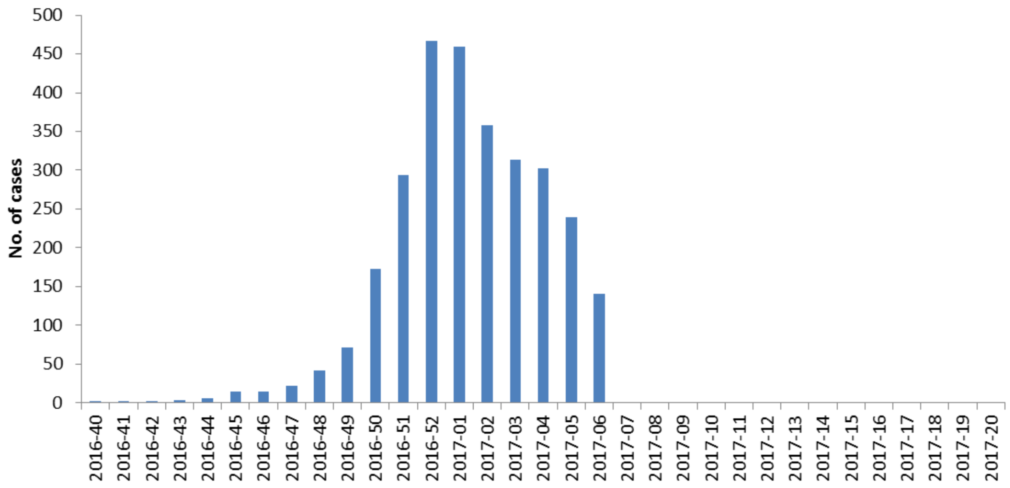


= represents similar ratio; > represents a ratio of at least 1.5:1 but less than 5:1; >> represents a ratio of ≥5:1

Source: European Centre for Disease Prevention and Control/WHO Regional Office for Europe. Flu News Europe, Joint ECDC–WHO/Europe weekly influenza update

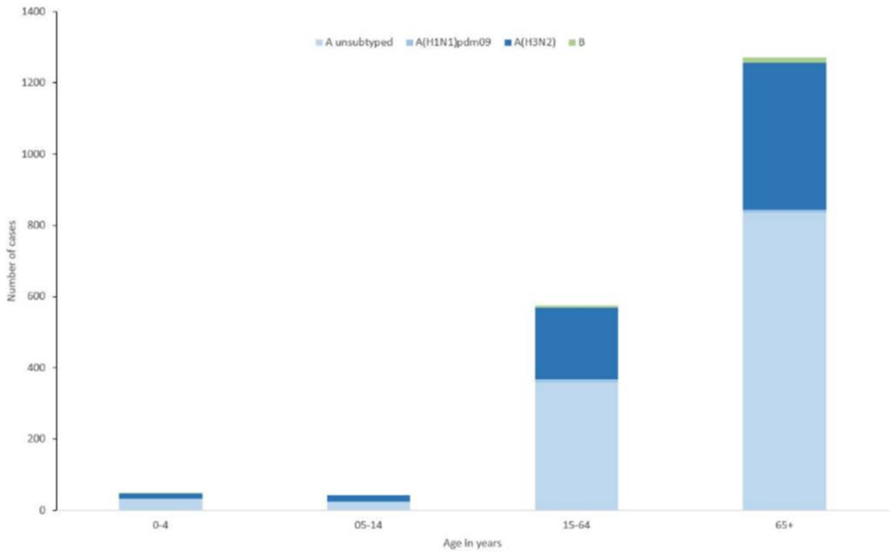
Figure E4. Hospitalized influenza cases

A. Laboratory-confirmed cases admitted to ICU by week*



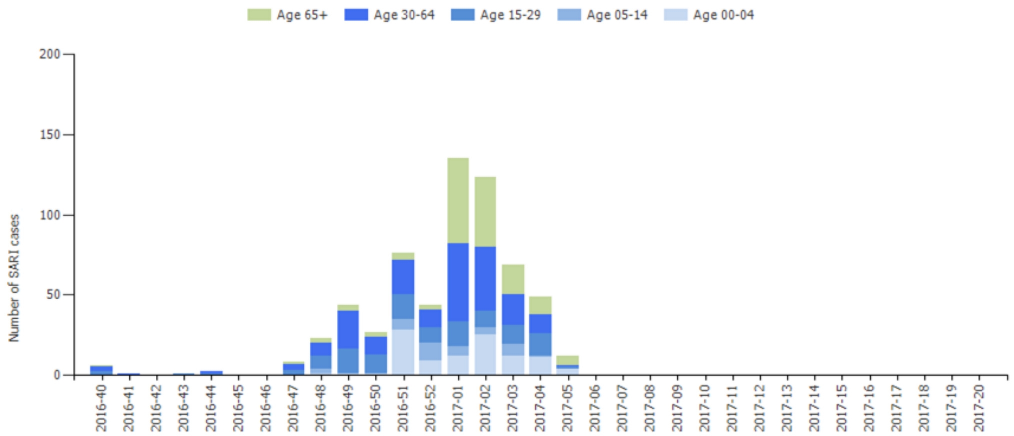
* The Czech Republic, Finland, France, Ireland, Romania, Slovakia, Spain, Sweden, and the United Kingdom

B. Distribution of virus (sub)type by age group in laboratory-confirmed hospitalized cases (ICU)*



* The Czech Republic, Finland, France, Ireland, Romania, Slovakia, Spain, Sweden, and the United Kingdom

C. SARI cases by age group (years) at sentinel hospitals*



* Data reported for Albania only

Source: European Centre for Disease Prevention and Control/WHO Regional Office for Europe. Flu News Europe, Joint ECDC–WHO/Europe weekly influenza update

Influenza specimens received at the Francis Crick Institute.

Samples, viruses or clinical samples, with collection dates after 2016-08-31 have been received from 56 countries in Europe, Africa, the Middle East, the Indian Ocean, Central Asia and the Far East. Eighty six *per cent* of the samples received were type A viruses, with influenza A(H3N2) viruses outnumbering A(H1N1) viruses at a ratio of nearly 20:1 (**Table 1**). Of the type B viruses, viruses of the B/Victoria lineage predominated over those of the B/Yamagata lineage at a ratio of 1.5:1. These proportions are somewhat different to those reported within the WHO European region (99:1 for H3N2:H1N1 and 0.7:1 for B/Vic:B/Yam: **Figure E1-B & C, Table E1**) and probably result from NICs sharing all A(H1N1)pdm09 specimens received with the Francis Crick Institute and the high number of influenza type B specimens received from non-European NICs, notably from Africa.

Table 2 shows the recovery/isolation rates from specimens received with collection dates after 2016-08-31. Both viruses and clinical samples were recovered efficiently. For H3N2 viruses, efficient recovery was obtained as assessed by detection of virus neuraminidase activity using the MUNANA sialidase assay, but the vast majority (approximately 90%) of these could not be assessed by HI assay due to HA titres being low (<4 HA units in the presence of 20nM oseltamivir).

Table 3 summarises the reactivity of A(H1N1), A(H3N2) and viruses of the B/Victoria and B/Yamagata lineages with post-infection ferret antisera raised against vaccine viruses and reference viruses in HI assays. For A(H3N2) viruses, the reactivity of test viruses in plaque reduction neutralisation assays (PNRA) is also summarised. Numbers and percentages of viruses showing ≤2-, 4- and ≥8-fold reductions in titres relative to the titres of the antisera for the respective homologous reference/vaccine viruses are shown.

Table 1. Summary of clinical samples and virus isolates received, with collection dates after 2016-08-31, by country*

MONTH	TOTAL RECEIVED	A		H1N1pdm09		H3N2			B		B Victoria lineage		B Yamagata lineage	
Country		Number received	Number propagated[1]	Number received	Number propagated[1]	Number received	Number propagated[2]		Number received	Number propagated[1]	Number received	Number propagated[1]	Number received	Number propagated[1]
2016														
SEPTEMBER														
Ghana	1			1	in process									
Iceland	12					12	0	9						
Madagascar	5	1	in process	2	in process						2	in process		
Oman	2			1	1	1	in process							
Saudi Arabia[3]	95			1	1	93	1	46					1	1
Slovenia	1			1	in process									
Turkey	1					1	in process							
United Kingdom	1			1	1									
OCTOBER														
Denmark	5					4	in process						1	1
Egypt	8	7	0	1	0									
Finland	2					2	0	2						
France	11					11	in process							
Germany	1					1	in process							
Ghana	16			1	in process	1	0	1			12	in process	2	in process
Iran	4					3	in process				1	1		
Jordan	1					1	0	1						
Kosovo	2					2	in process							
Kyrgyzstan	3					1	0	0			1	0	1	1
Madagascar	8			4	in process						4	in process		
Norway	17			1	1	16	5	9						
Oman	8					7	in process				1	1		
Portugal	2					2	in process							
Russia	1			1	1									
Spain	7					7	in process							
Switzerland	1					1	0	1						
Turkey	1					1	in process							
Ukraine	1					1	0	1						
United Kingdom	2					1	0	1					1	1
NOVEMBER														
Albania	3					3	0	3						
Algeria	3								3	in process				
Armenia	2					2	in process							
Austria	7			2	2	5	in process							
Azerbaijan	1					1	in process							
Belgium	3					3	in process							
Bosnia & Herzegovina	1					1	in process							
Croatia	1			1	1									
Denmark	5			1	1	4	in process							
Egypt	14	6	0	5	0				3	0				
Estonia	4					4	0	4						
Finland	4					3	0	3					1	1
France	20					19	in process				1	1		
Georgia	2					2	in process							
Germany	4					4	in process							
Ghana	10										4	3	6	4
Iran	3					3	0	3						
Ireland	3					3	in process							
Italy	7					7	0	6						
Jordan	5					5	0	4						
Kazakhstan	33					9	in process		24	in process				
Kosovo	28	1	in process			27	in process							
Kyrgyzstan	54	3	0			36	in process	19			14	in process	1	1
Latvia	2					2	in process							
Madagascar	1										1	1		
Montenegro	2					2	in process							
Morocco	7					6	in process		1	in process				
Netherlands	1					1	1	0						
Norway	38			4	2	28	1	26			2	1	4	3
Oman	7	1	in process			5	0	5			1	1		
Poland	1	1	in process											
Romania	4					4	in process							
Russia	6			1	in process	5	in process							
Slovenia	3					3	in process							
Spain	27					27	in process							
Sweden	5					5	in process							
Tunisia	1					1	0	1						
Turkey	2					2	in process							
Ukraine	20	1	in process	1	1	18	in process							
United Kingdom	9					9	2	7						
DECEMBER														
Albania	8					8	in process							
Algeria	10					5	in process		5	in process				
Armenia	28					28	in process							
Austria	10			1	1	9	in process							
Azerbaijan	8					8	in process							
Belgium	8			1	0	7	in process							
Bosnia & Herzegovina	7					6	in process				1	1		
Bulgaria	12			1	1	11	0	7						
Croatia	6			2	2	2	0	2					2	in process
Czech Republic	10					10	in process							
Denmark	3					3	0	3						
Egypt	3	3	0											
Estonia	16					16	0	15						
Finland	5					5	0	4						
France	26					26	1	25						
Georgia	9					8	in process				1	1		
Germany	15					15	in process							
Ghana	1										1	1		
Greece	5	1	in process			4	in process							
Hong Kong	1										1	1		
Hungary	13			1	1	12	in process							
Iceland	25					25	in process							
Iran	9					7	0	7			1	0	1	1

MONTH	TOTAL RECEIVED	A		H1N1pdm09		H3N2			B		B Victoria lineage		B Yamagata lineage	
Country		Number received	Number propagated[1]	Number received	Number propagated[1]	Number received	Number propagated[2]		Number received	Number propagated[1]	Number received	Number propagated[1]	Number received	Number propagated[1]
Ireland	17					17	in process							
Israel	4					4	in process							
Italy	10					10	0	10						
Kazakhstan	14					7	in process		5	in process	2	2		
Kosovo	9					9	in process							
Kyrgyzstan	2					1	0	1					1	1
Latvia	7					4	in process						3	3
Luxembourg	18					16	in process		1	in process			1	1
Macedonia	15					15	in process							
Madagascar	3								3	in process				
Moldova	4					4	in process							
Montenegro	28	19	in process			9	in process							
Morocco	5					5	in process							
Netherlands	11			1	1	10	2	7						
Norway	13					8	2	6			2	1	3	2
Poland	10					9	in process		1	in process				
Portugal	20			1	1	19	in process							
Romania	10					10	in process							
Russia	20					13	in process				7	in process		
Serbia	19					17	in process		2	0				
Slovenia	8					4	in process						4	4
Spain	9	1	0			8	1	7						
Sweden	11					11	in process							
Switzerland	19			2	2	15	0	15			1	1	1	1
Tunisia	3					1	0	1	2	in process				
Turkey	31					26	in process		5	in process				
Ukraine	41	4	in process			37	in process							
2017														
JANUARY														
Albania	9					9	in process							
Algeria	2								2	in process				
Belarus	10					10	in process							
Belgium	20					20	in process							
Bosnia & Herzegovina	12	1	0			11	in process							
Germany	20			2	2	14	2	12					4	4
Georgia	4					3	0	2			1	1		
Greece	35					27	in process		1	in process	7	in process		
Hong Kong	9			3	3	3	1	2			1	1	2	2
Ireland	3			1	1	2	in process							
Israel	13					13	in process							
Latvia	3					2	0	2					1	1
Luxembourg	2					2	0	2						
Macedonia	5					4	in process		1	in process				
Moldova	8					8	in process							
Morocco	1					1	in process							
Poland	11	2	in process			9	in process							
Portugal	1										1	in process		
Qatar	30			10	in process	10	in process		10	in process				
Romania	15					15	in process							
Slovenia	13					8	in process						5	5
Turkey	15					10	in process		5	in process				
	1412	52	0	56	27	1113	19	282	74	0	71	19	46	38
				4.0%			78.8%				5.0%		3.3%	
56 Countries		86.5%							13.5%					

1. Propagated to sufficient titre to perform HI assay (the totalled number does not include any from batches that are in process)
2. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir (the totalled number does not include any from batches that are in process)

Numbers in red indicate viruses recovered but with insufficient HA titre to permit HI assay

3. All were clinical specimens which were subjected to NGS - virus isolation was attempted only with those yielding complete genome sequences from the clinical specimen

* As of 2017-02-17

Table 2. Isolation rates for specimens collected after 2016-08-31

	Clinical sample			Virus isolate		
	Number processed	Number propagated[1]	Percent recovered	Number processed	Number propagated[1]	Percent recovered
H1	9	6	67%	26	21	81%
H3	246	8[2] 203[3]	3% 86% (4%)[4] 83%	330	48[2] 263[3]	15% 95% (20%)[4] 80%
B Victoria	26	18	69%	24	23(+1)[5]	100%
B Yamagata	13	9	69%	29	29	100%

1. Propagated to sufficient HA titre to perform HI assay
2. Propagated to sufficient HA titre in the presence of 20nM oseltamivir to allow HI assay
3. Viruses recovered (NA activity by MUNANA-based assay) but with insufficient HA titre to allow HI assay
4. The percentages for viruses recovered that could be assayed by HI are shown in parentheses
5. Virus from Oman with no collection date but from the 2016-17 season

Table 3. Summary of isolates with collection dates after 2016-08-31 showing reduction in reactivity with antisera raised against reference viruses in HI & plaque-reduction microneutralisation assays

Virus type/subtype	Number tested	≤2-fold[a]	4-fold[a]	≥8-fold[a]
H1[1]	27	26 (96.3%) 26 (96.3%)	0 (0%) 1(3.7%)	1 (3.7%) 0 (0%)
H3[2a]	56	53 (94.6%)	2 (3.6%)	1 (1.8%)
H3[2b]	56	21 (37.5%)	27 (48.2%)	8 (14.3%)
H3[2c]	56	29 (51.8%)	25 (44.6%)	2 (3.6%)
H3[3a]	144	111 (77.1%)	27 (18.8%)	6 (4.1%)
H3[3b]	144	29 (20.1%)	36 (25.0%)	79 (54.9%)
H3[3c]	144	133 (92.4%)	6 (4.2%)	5 (3.4%)
B/Victoria[4]	41	28 (68.3%)	9 (22%)	4 (9.7%)
B/Yamagata[5]	38	13 (34.2%)	12 (31.6%)	13 (34.2%)

a. Reduction in titre, compared to the homologous titre with post-infection ferret antisera indicated below:

1. Compared to homologous titres for ferret antisera raised against vaccine viruses A/California/7/2009 and A/Michigan/45/2015

2a. Compared to homologous titres for ferret antisera raised against tissue culture-propagated A/Hong Kong/5738/2014 (genetic subgroup 3C.2a)
2b. Compared to homologous titres for ferret antisera raised against egg-propagated A/Hong Kong/4801/2014 (genetic subgroup 3C.2a)
2c. Compared to homologous titres for ferret antisera raised against tissue culture-propagated A/Hong Kong/4801/2014 (genetic subgroup 3C.2a)
3a. Compared to homologous titres for ferret antisera raised against tisue culture-propagated A/Oman2585/2016 (genetic subgroup 3C.2a1) in plaque-reduction neutralisation assays
3b. Compared to homologous titres for ferret antisera raised against egg-propagated A/Hong Kong/4801/2014 (genetic subgroup 3C.2a [NYK]) in plaque-reduction neutralisation assays
3c. Compared to homologous titres for ferret antisera raised against tisue culture-propagated A/Hong Kong/7295/2014 (genetic subgroup 3C.2a) in plaque-reduction neutralisation assays

4. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture propagated B/Brisbane/60/2008-like equivalents (B/Hong Kong/514/2009, B/Ireland/3154/2016 & B/Nordrhein-Westfalen/1/2016)

5. Compared to homologous titres for ferret antisera raised against egg-propagated vaccine virus B/Phuket/3073/2013

Influenza A(H1N1)pdm09 virus analyses.

Influenza A(H1N1)pdm09 viruses have been received from NICs in 23 countries or administrative regions in four WHO Regions: African (2: Ghana and Madagascar), Eastern Mediterranean (4: Egypt, Oman, Qatar and Saudi Arabia), European (16: Austria, Belgium, Bulgaria, Croatia, Denmark, Germany, Hungary, Ireland, Netherlands, Norway, Portugal, Russia, Slovenia, Switzerland, Ukraine, United Kingdom) and Western Pacific (1: Hong Kong SAR). **Table 4** summarises the antigenic (HI) results for viruses, recovered from clinical samples with collection dates after 2016-08-31 that were propagated successfully at the Francis Crick Institute; the vast majority (>96%) were antigenically indistinguishable from egg-propagated viruses A/California/7/2009 and A/Michigan/45/2015.

Tables 5-1 to **5-4** show the results of HI assays of H1N1 viruses carried out since our September 2016 report. Test viruses in each table are sorted by date of collection and viruses with collection dates after 2016-08-31 are shown below red lines in the tables. A summary of the results for viruses with collection dates after 2016-08-31 is shown in **Table 5-5**.

Over 96% of the test viruses were recognised by the antisera raised against the A/California/7/2009 and A/Michigan/45/2015 vaccine viruses at titres within 2-fold of the titres of the antisera for the homologous viruses. With the exception of the antiserum raised against A/Lviv/N6/2009, over 90% of the test viruses were recognised at titres within 2-fold of the homologous titres of the antisera **(Table 5-5)**. The antiserum raised against A/Lviv/N6/2009 recognised 70% of test viruses at titres within 2-fold of the titre of the antiserum for the homologous virus but all test viruses were recognised at titres within 4-fold of the homologous titre. Of the test viruses for which the genetic group or subgroup were known, all viruses with collection dates after 2016-08-31 fell into genetic subgroup 6B.1.

Figure 1 shows a phylogenetic tree for the HA genes of representative H1N1 viruses. Over the last eight years the HA genes have evolved compared to A/California/7/2009 and eight genetic groups have been designated with viruses in group 6 forming clusters designated groups 6A, 6B and 6C. Viruses collected after 2016-08-31 fell into genetic group 6B and in two subgroups, 6B.1 and 6B.2 **(Figure 1).** Group 6B and subgroups 6B.1 and 6B.2 are defined by the following amino acid substitutions in **HA1** and **HA2**:

- **P83S, D97N**, **S185T, S203T K163Q, A256T, K283E** and **I321V** in **HA1**, and **E47K, S124N** and **E172K** in **HA2** compared with A/California/7/2009 define **genetic group 6B**, e.g. A/South Africa/3626/2013.
- **S84N, S162N (+CHO)** and **I216T** in **HA1** define **subgroup 6B.1** within group 6B, e.g. A/Slovenia/2903/2015
- **V152T** and **V173I** in **HA1** and **E164G** and **D174E** in **HA2** define **subgroup 6B.2** within group 6B, e.g. A/Israel/Q-504/2015.

An amino acid sequence alignment of the HAs for representative viruses is shown in **Figure 2**; the vaccine virus is shown in red, reference viruses are coloured blue, glycosylation motifs are highlighted, the genetic group/subgroup is shown in green and the HA1/HA2 proteolytic-processing site is highlighted.

Figure 3 illustrates the location of amino acid residues on an H1-HA structure defining genetic group 6B and the two major subgroups 6B.1 and 6B.2.

Figure 4 shows a phylogenetic tree for NA genes of representative H1N1 viruses. The HA and NA phylogenetic trees are generally congruent.

Figure 5 shows a NA amino acid alignment for a selected group of these viruses, annotated as **Figure 2** but lacking a proteolytic-processing site.

Antiviral resistance.

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 18 viruses with collection dates after 2016-08-31. All viruses tested showed normal inhibition (NI) to both neuraminidase inhibitors **(Table 14)**.

Conclusion.

The number of H1N1pdm09 viruses received in recent months has been small. However, the vast majority of A(H1N1)pdm09 viruses received were antigenically similar to the vaccine viruses A/California/7/2009 and A/Michigan/45/2015 and retained susceptibility to oseltamivir and zanamivir.

Table 4. Summary of influenza A(H1N1)pdm09 viruses analysed, collected after 2016-08-31

MONTH	H1N1pdm09				
			Fold reduction in HI titre		
Country	Number received[1]	Number propagated[2]	≤2 fold[3]	4 fold[3]	≥8 fold[3]
2016					
SEPTEMBER					
Ghana	1	in process			
Madagascar	2	in process			
Oman	1	1	1		
Saudi Arabia	1	1	1		
Slovenia	1	in process			
United Kingdom	1	1	1		
OCTOBER					
Egypt	1	0			
Madagascar	4	in process			
Norway	1	1	1		
Russia	1	1	1		
NOVEMBER					
Austria	2	2	2		
Croatia	1	1	1		
Denmark	1	1		1*	1
Egypt	5	0			
Norway	4	2	2		
Russia	1	in process			
Ukraine	1	1	1		
DECEMBER					
Austria	1	1	1		
Belgium	1	0			
Bulgaria	1	1	1		
Croatia	2	2	2		
Hungary	1	1	1		
Netherlands	1	1	1		
Portugal	1	1	1		
Switzerland	2	2	2		
2017					
JANUARY					
Germany	2	2	2		
Hong Kong	3	3	3		
Ireland	1	1	1		
Qatar	10	in process			
23 Countries	55	27	26 96.3%	0 0.0%	1 3.7%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared
2. Propagated to sufficient titre to perform HI assays
3. Reduction in HI titre with post-infection ferret antiserum raised against A/California/7/2009 (egg grown vaccine virus)

*. Reduction in HI titre with post-infection ferret antiserum raised against A/Michigan/45/2015 (egg grown vaccine virus) was the same as for A/California/7/2009 with the exception of this virus

Table 5-1. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-11-22 + 2016-11-29)

Viruses	Other information	Passage history / Ferret number / Genetic group	Collection date	Passage history	Haemagglutination inhibition titre – Post-infection ferret antisera: A/Mich 45/15 Egg NIB F42/16[*1] 6B.1	A/Mich 45/15 MDCK CDC F053/16 6B.1	A/Cal 7/09 Egg F06/16[*1]	A/Bayern 69/09 MDCK F09/15[*1]	A/Lviv N6/09 MDCK F14/13[*1]	A/Astrak 1/11 MDCK F22/13[*1] 5	A/St. P 27/11 Egg F26/14[*1] 6	A/St. P 100/11 Egg F24/11[*1] 7	A/HK 5659/12 MDCK F30/12[*1] 6A	A/Sth Afr 3626/13 Egg F03/14[*1] 6B	A/Slov 2903/2015 Egg F02/16[*2] 6B.1	A/Israel Q-504/15 MDCK F08/16[*2] 6B.2	
REFERENCE VIRUSES																	
A/Michigan/45/2015		6B.1	2015-09-07	E3/E2	1280	1280	1280	640	320	1280	640	2560	1280	1280	2560	1280	
A/Michigan/45/2015		6B.1	2015-09-07	MDCK1/C2/MDCK1	1280	1280	320	320	160	320	320	640	640	640	1280	1280	
A/California/7/2009	clone 38-32		2009-04-09	E3/E4	1280	1280	1280	640	320	1280	640	2560	1280	1280	2560	1280	
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	40	80	40	320	160	40	40	40	80	40	40	40	G155E
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	160	160	80	1280	640	80	160	80	160	80	160	160	G155E>G, D222G
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	1280	2560	1280	640	320	1280	640	2560	2560	1280	2560	1280	
A/St. Petersburg/27/2011		6	2011-02-14	E1/E4	2560	2560	1280	1280	640	1280	1280	5120	2560	1280	5120	2560	
A/St. Petersburg/100/2011		7	2011-03-14	E1/E4	1280	1280	640	640	320	640	640	1280	1280	640	1280	1280	
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK1	320	640	320	160	160	320	320	640	1280	640	640	320	
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	1280	1280	640	640	640	640	640	1280	1280	1280	1280	1280	
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E1	5120	5120	2560	1280	640	2560	1280	5120	2560	2560	5120	5120	
A/Israel/Q-504/2015		6B.2	2015-12-15	C1/MDCK2	2560	2560	1280	640	320	1280	640	2560	1280	1280	1280	2560	
TEST VIRUSES																	
A/California/7/2009			2009-04-09	C3/MDCK1	640	640	640	320	320	320	640	1280	1280	640	1280	640	
A/Singapore/GP1908/2015		6B.1	2015-04-11	MDCK3/MDCK1	1280	1280	320	320	160	320	320	640	640	640	1280	640	
A/Singapore/GP1908/2015		6B.1	2015-04-11	E3/E1	1280	ND	320	320	320	640	320	1280	640	640	25660	1280	
A/Singapore/GP1908/2015 (IVR-180)		6B.1	2015-04-11	E3/D7/E1	2560	ND	640	640	320	640	640	1280	1280	1280	2560	2560	
A/Sappro/18/2016		6B.2	2016-02-01	MDCK1/MDCK2/MDCK1	1280	2560	1280	640	320	1280	640	2560	1280	1280	2560	2560	
A/Saitama-C/10/2016		6B.1	2016-02-07	MDCK2/MDCK2/MDCK1	1280	1280	640	320	160	640	640	1280	1280	640	1280	1280	
A/Victoria/501/2016		6B.1	2016-05-23	SIAT1/MDCK1	2560	2560	1280	1280	640	1280	1280	2560	2560	2560	5120	5120	
A/Norway/4463/2016		6B.1	2016-10-27	MDCK1	2560	ND	2560	1280	1280	2560	1280	5120	5120	5120	5120	5120	
A/Norway/4462/2016		6B.1	2016-11-03	MDCK1	2560	ND	1280	1280	640	1280	1280	2560	2560	5120	5120	2560	
					Vaccine SH 2017		Vaccine NH 2016-17										

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND = Not Done

Sequences in phylogenetic trees

Table 5-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-12-20 + 2017-01-10)

Viruses	Other information		Collection date	Passage history	A/Mich	A/Mich	A/Mich	A/Cal	A/Bayern	A/Lviv	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel	
					Haemagglutination inhibition titre													
					Post-infection ferret antisera													
						NEW	NEW											
					45/15	45/15	45/15	7/09	69/09	N6/09	1/11	27/11	100/11	5659/12	3626/13	2903/2015	Q-504/15	
		Passage history			Egg	Egg	Egg	Egg	MDCK	MDCK	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	
		Ferret number			NIB F42/16[*1]	F31/16*[1]	F32/16*[1]	F06/16[*1]	F09/15[*1]	F14/13[*1]	F22/13[*1]	F26/14[*1]	F24/11[*1]	F30/12[*1]	F03/14[*1]	F02/16[*2]	F08/16[*2]	
		Genetic group			6B.1	6B.1	6B.1				5	6	7	6A	6B	6B.1	6B.2	
REFERENCE VIRUSES																		
A/Michigan/45/2015		6B.1		E3/E2	1280	2560	5120	1280	640	320	1280	640	2560	1280	1280	2560	1280	
A/California/7/2009	clone 38-32		2009-04-09	E3/E4	1280	2560	5120	640	640	320	640	640	1280	1280	640	1280	1280	
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	40	80	80	40	320	160	40	40	40	80	40	40	40	G155E
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	160	160	160	80	1280	640	80	160	80	160	80	160	160	G155E>G, D222G
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	1280	2560	2560	1280	640	320	1280	640	2560	2560	1280	2560	1280	
A/St. Petersburg/27/2011		6	2011-02-14	E1/E4	2560	2560	2560	1280	1280	640	1280	1280	5120	2560	1280	5120	2560	
A/St. Petersburg/100/2011		7	2011-03-14	E1/E4	1280	1280	1280	640	640	320	640	640	1280	1280	640	1280	1280	
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK1	320	640	640	320	160	160	320	320	640	1280	640	640	320	
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	1280	1280	1280	640	640	640	640	640	1280	1280	1280	1280	1280	
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E1	5120	5120	5120	2560	1280	640	2560	1280	5120	2560	2560	5120	5120	
A/Israel/Q-504/2015		6B.2	2015-12-15	C1/MDCK2	2560	2560	2560	1280	640	320	1280	640	2560	1280	1280	1280	2560	
TEST VIRUSES																		
A/Iowa/53/2015		6B.1	2015-11-04	E4/E1	640	640	1280	320	320	160	320	320	1280	640	640	1280	640	
A/Victoria/501/2016		6B.1	2016-05-23	E3/E1	1280	ND	2560	1280	320	320	640	640	1280	1280	640	1280	1280	
A/England/963/2016		6B.1	2016-09-27	MDCK1/MDCK1	1280	2560	5120	1280	640	320	1280	640	2560	1280	1280	2560	1280	
A/Norway/4555/2016		6B.1	2016-11-07	MDCK1/MDCK1	1280	ND	ND	640	320	160	640	320	1280	1280	640	1280	1280	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND = Not Done

Sequences in phylogenetic trees

Vaccine SH 2017

Vaccine NH 2016-17

Table 5-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2017-01-17 + 2017-01-24)

					Haemagglutination inhibition titre											
					Post-infection ferret antisera											
Viruses	Other information		Collection date	Passage history	A/Mich	A/Cal	A/Bayern	A/Lviv	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel	
					45/15	7/09	69/09	N6/09	1/11	27/11	100/11	5659/12	3626/13	2903/2015	Q-504/15	
		Passage history			Egg	Egg	MDCK	MDCK	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	
		Ferret number			NIB F42/16[*1]	F06/16[*1]	F09/15[*1]	F14/13[*1]	F22/13[*1]	F26/14[*1]	F24/11[*1]	F30/12[*1]	F03/14[*1]	F02/16[*2]	F08/16[*2]	
		Genetic group			6B.1				5	6	7	6A	6B	6B.1	6B.2	
REFERENCE VIRUSES																
A/Michigan/45/2015				E3/E2	1280	1280	640	320	1280	640	2560	1280	1280	2560	1280	
A/California/7/2009	clone 38-32		2009-04-09	E3/E4	1280	640	640	320	640	320	1280	1280	640	1280	640	
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	<	<	160	80	40	<	<	40	40	<	<	G155E
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	80	80	640	640	40	80	80	160	80	160	80	G155E>G, D222G
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	640	640	320	160	640	320	1280	640	640	1280	640	
A/St. Petersburg/27/2011		6	2011-02-14	E1/E4	1280	640	640	320	640	640	1280	1280	640	1280	640	
A/St. Petersburg/100/2011		7	2011-03-14	E1/E4	1280	640	640	320	1280	640	1280	1280	640	1280	640	
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	320	320	160	80	320	320	640	640	320	640	320	
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	1280	320	320	320	320	320	640	640	640	640	640	
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E1	2560	2560	1280	640	1280	640	5120	2560	1280	2560	2560	
A/Israel/Q-504/2015		6B.2	2015-12-15	C1/MDCK2	1280	640	320	160	640	320	1280	1280	640	1280	1280	
TEST VIRUSES																
A/Saudi Arabia/192788/2016		6B.1	2016-09-13	MDCK2	2560	1280	640	320	1280	640	1280	1280	1280	2560	1280	
A/Switzerland/19374003/2016		6B.1	2016-12-12	MDCK1	1280	640	640	320	640	320	1280	1280	1280	1280	1280	
A/Switzerland/19464226/2016		6B.1	2016-12-21	MDCK1	2560	640	640	320	1280	640	1280	1280	1280	1280	1280	
A/Bulgaria/074/2017		6B.1	2016-12-30	MDCK1	1280	640	320	160	640	320	1280	640	640	1280	640	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND = Not Done

Sequences in phylogenetic trees

Vaccine SH 2017 (A/Mich 45/15); Vaccine NH 2016-17 (A/Cal 7/09)

Table 5-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2017-02-07 + 2017-02-14)

Viruses	Other information		Collection date	Passage history	A/Mich	A/Cal	A/Bayern	A/Lviv	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel	
					Haemagglutination inhibition titre											
					Post-infection ferret antisera											
					45/15	7/09	69/09	N6/09	1/11	27/11	100/11	5659/12	3626/13	2903/2015	Q-504/15	
		Passage history			Egg	Egg	MDCK	MDCK	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	
		Ferret number			NIB F42/16*1	F06/16*1	F09/15*1	F14/13*1	F22/13*1	F26/14*1	F24/11*1	F30/12*1	F03/14*1	F02/16*2	F08/16*2	
		Genetic group			6B.1				5	6	7	6A	6B	6B.1	6B.2	
REFERENCE VIRUSES																
A/Michigan/45/2015				E3/E1	1280	1280	640	640	640	640	2560	1280	1280	2560	1280	
A/California/7/2009	clone 38-32		2009-04-09	E3/E4	2560	1280	1280	640	1280	1280	2560	2560	1280	2560	2560	
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	<	<	160	80	40	<	<	40	40	<	<	G155E
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	80	80	640	640	40	80	80	160	80	160	80	G155E>G, D222G
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	640	640	320	160	640	320	1280	640	640	1280	640	
A/St. Petersburg/27/2011		6	2011-02-14	E1/E4	1280	640	640	320	640	640	1280	1280	640	1280	640	
A/St. Petersburg/100/2011		7	2011-03-14	E1/E4	1280	640	640	320	1280	640	1280	1280	640	1280	640	
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	320	320	160	80	320	320	640	640	320	640	320	
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	1280	320	320	320	320	320	640	640	640	640	640	
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E1	2560	2560	1280	640	1280	640	5120	2560	1280	2560	2560	
A/Israel/Q-504/2015		6B.2	2015-12-15	C1/MDCK2	1280	640	320	160	640	320	1280	1280	640	1280	1280	
TEST VIRUSES																
A/Dnipro/580/2016		6B.2	2016-02-05	SIAT2/MDCK2	640	320	160	160	160	160	640	640	640	640	640	
A/Oman/6472/2016		6B.1	2016-09-17	MDCK1/MDCK1	2560	1280	640	320	1280	1280	2560	2560	2560	2560	2560	
A/Novosibirsk/121/2016		6B.1	2016-10-21	C2/MDCK2	2560	1280	640	640	1280	640	2560	2560	1280	2560	2560	
A/Denmark/53/2016		6B.1	2016-11-10	SIAT3/MDCK1	320	160	640	640	160	160	160	320	320	640	320	
A/Austria/953687/2016		6B.1	2016-11-13	C1/MDCK1	640	640	320	160	640	320	640	640	640	1280	640	
A/Austria/953688/2016		6B.1	2016-11-13	C1/MDCK1	2560	1280	640	320	1280	1280	2560	2560	1280	2560	2560	
A/Kharkiv/963/2016		6B.1	2016-11-21	SIAT1/MDCK1	2560	1280	640	640	1280	640	2560	2560	1280	2560	2560	
A/Croatia/3547/2016		6B.1	2016-11-28	MDCKx/MDCK1	640	640	320	320	640	320	1280	640	640	1280	640	
A/Croatia/3644/2016		6B.1	2016-12-09	MDCKx/MDCK1	1280	640	320	160	640	320	1280	1280	640	1280	1280	
A/Portugal/MS25/2016		6B.1	2016-12-12	SIAT1/MDCK1	2560	1280	640	320	1280	640	2560	2560	1280	2560	2560	
A/Austria/958726/2016		6B.1	2016-12-12	C1/MDCK1	2560	1280	640	320	1280	640	2560	1280	1280	2560	1280	
A/Croatia/3667/2016		6B.1	2016-12-12	MDCKx/MDCK1	640	640	320	160	320	320	1280	640	640	1280	640	
A/Hungary/187/2016		6B.1	2016-12-15	MDCK2/MDCK1	2560	1280	640	320	1280	640	2560	1280	1280	2560	2560	
A/Netherlands/3792/2016		6B.1	2016-12-29	MDCK2/MDCK1	1280	640	640	320	1280	640	2560	1280	1280	2560	1280	
A/Hong Kong/82/2017		P9	2017-01-02	MDCK1/MDCK1	1280	1280	320	160	1280	640	1280	1280	1280	2560	1280	
A/Nordrhein-Westfalen/12/2017		P10	2017-01-02	C1/MDCK1	1280	640	320	160	640	320	1280	640	640	1280	640	
A/Ireland/00509/2017		P9	2017-01-03	C1/MDCK1	1280	640	320	160	640	320	1280	640	640	1280	1280	
A/Hong Kong/217/2017		P9	2017-01-07	MDCK1/MDCK1	1280	640	320	320	640	640	1280	1280	1280	2560	640	
A/Hong Kong/219/2017		6B.1	2017-01-09	MDCK1/MDCK1	2560	1280	1280	320	1280	1280	2560	2560	1280	2560	2560	
A/Nordrhein-Westfalen/26/2017		P9	2017-01-16	C1/MDCK1	1280	640	320	320	640	640	1280	1280	1280	1280	1280	
					Vaccine SH 2017	Vaccine NH 2016-17										

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND = Not Done

Sequences in phylogenetic trees

Table 5-5. Antigenic analyses of influenza A(H1N1)pdm09 viruses with collection dates after 2016-08-31 - Summary

					Haemagglutination inhibition titre										
					Post-infection ferret antisera										
Viruses	Other information		Collection date	Passage history	A/Mich 45/15	A/Cal 7/09	A/Bayern 69/09	A/Lviv N6/09	A/Astrak 1/11	A/St. P 27/11	A/St. P 100/11	A/HK 5659/12	A/Sth Afr 3626/13	A/Slov 2903/2015	A/Israel Q-504/15
		Passage history			Egg	Egg	MDCK	MDCK	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK
		Ferret number			NIB F42/16[*1]	F06/16[*1]	F09/15[*1]	F14/13[*1]	F22/13[*1]	F26/14[*1]	F24/11[*1]	F30/12[*1]	F03/14[*1]	F02/16[*2]	F08/16[*2]
		Genetic group			6B.1				5	6	7	6A	6B	6B.1	6B.2
REFERENCE VIRUSES															
A/Michigan/45/2015				E3/E1	1280	1280	640	640	640	640	2560	1280	1280	2560	1280
A/California/7/2009	clone 38-32		2009-04-09	E3/E4	2560	1280	1280	640	1280	1280	2560	2560	1280	2560	2560
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	<	<	160	80	40	<	<	40	40	<	<
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	80	80	640	640	40	80	80	160	80	160	80
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	640	640	320	160	640	320	1280	640	640	1280	640
A/St. Petersburg/27/2011		6	2011-02-14	E1/E4	1280	640	640	320	640	640	1280	1280	640	1280	640
A/St. Petersburg/100/2011		7	2011-03-14	E1/E4	1280	640	640	320	1280	640	1280	1280	640	1280	640
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	320	320	160	80	320	320	640	640	320	640	320
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	1280	320	320	320	320	320	640	640	640	640	640
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E1	2560	2560	1280	640	1280	640	5120	2560	1280	2560	2560
A/Israel/Q-504/2015		6B.2	2015-12-15	C1/MDCK2	1280	640	320	160	640	320	1280	1280	640	1280	1280
TEST VIRUSES															
Number of viruses tested*					27	27	27	27	27	27	27	27	27	27	27
No with titre reduction ≤2-fold					26	26	27	19	26	25	26	27	27	25	25
%					96.3	96.3	100.0	70.4	96.3	92.6	96.3	100.0	100.0	92.6	92.6
No with titre reduction =4-fold					1			8	1	2				2	2
%					3.7			29.6	3.7	7.4				7.4	7.4
No with titre reduction ≥8-fold						1					1				
%						3.7					3.7				
					Vaccine SH 2017	Vaccine NH 2016-17									

* Of those with available HA sequence (22), all were clade 6B.1

Reference virus results are taken from individual tables as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Figure 1 . Phylogenetic comparison of A(H1N1)pdm09 HA genes

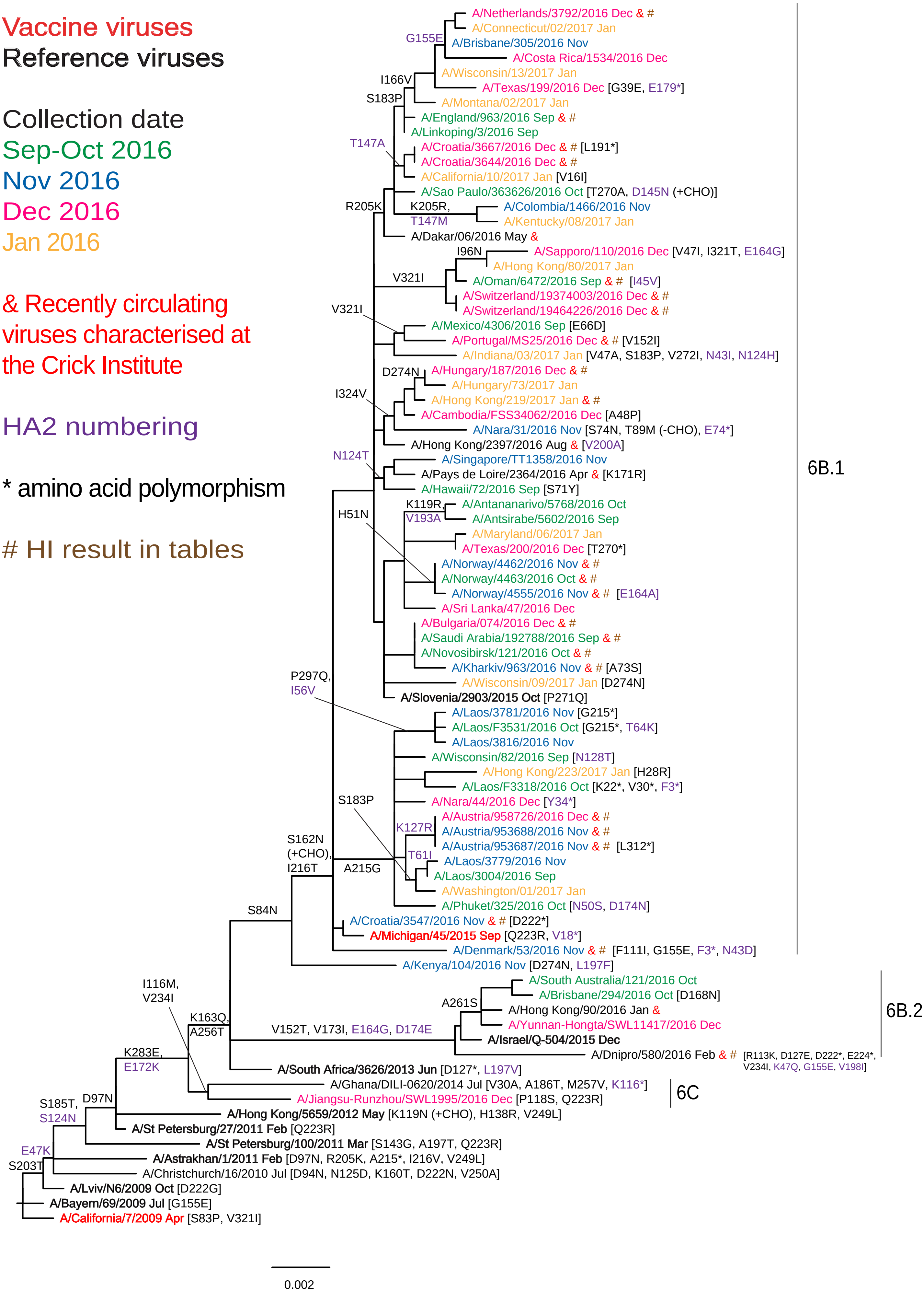


Figure 2. HA protein alignment for influenza A(H1N1)pdm09 viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

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                                    10        20        30        40        50        60        70        80        90       100
                               ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009            DTLCIGYHANNSTDTVDTVLEKNVTVTHSVNLLEDKHNGKLCKLRGVAPLHLGKCNIAGWILGNPECESLSTASSWSYIVETPSSDNGTCYPGDFIDYEE
A/Bayern/69/2009               ..................................................................................S.................
A/Lviv/N6/2009                 ..................................................................................S.................
A/Astrakhan/1/2011             ..................................................................................S.............N... 5
A/St Petersburg/100/2011       ..................................................................................S................. 7
A/St Petersburg/27/2011        ..................................................................................S.............N... 6
A/Hong Kong/5659/2012          ..................................................................................S.............N... 6A
A/Jiangsu-Runzhou/SWL1995/2016 ..................................................................................S.............N... 6C
A/Ghana/DILI-0620/2014         .............................A....................................................S.............N... 6C
A/South Africa/3626/2013       ..................................................................................S.............N... 6B
A/Israel/Q-504/2015            ..................................................................................S.............N... 6B.2
A/Yunnan-Hongta/SWL11417/2016  ..................................................................................S.............N... 6B.2
A/Brisbane/294/2016            ..................................................................................S.............N... 6B.2
A/Denmark/53/2016              ..................................................................................SN............N... 6B.1
A/Michigan/45/2015             ..................................................................................SN............N... 6B.1
A/Laos/3779/2016               ..................................................................................SN............N... 6B.1
A/Austria/958726/2016          ..................................................................................SN............N... 6B.1
A/Laos/3816/2016               ..................................................................................SN............N... 6B.1
A/Slovenia/2903/2015           ..................................................................................SN............N... 6B.1
A/Bulgaria/074/2016            ..................................................................................SN............N... 6B.1
A/Norway/4555/2016             ..................................................N...............................SN............N... 6B.1
A/Antananarivo/5768/2016       ..................................................................................SN............N... 6B.1
A/Singapore/TT1358/2016        ..................................................................................SN............N... 6B.1
A/Hong Kong/219/2017 Jan       ..................................................................................SN............N... 6B.1
A/Hungary/187/2016             ..............................................................X...................SN............N... 6B.1
A/Portugal/MS25/2016           ..................................................................................SN............N... 6B.1
A/Switzerland/19464226/2016    ..................................................................................SN............N... 6B.1
A/Oman/6472/2016               ..................................................................................SN............N... 6B.1
A/Croatia/3644/2016            ..................................................................................SN............N... 6B.1
A/England/963/2016             ..................................................................................SN............N... 6B.1
A/Wisconsin/13/2017 Jan        ..................................................................................SN............N... 6B.1
A/Netherlands/3792/2016        ..................................................................................SN............N... 6B.1

                                   110       120       130       140       150       160       170       180       190       200
                               ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009            LREQLSSVSSFERFEIFPKTSSWPNHDSNKGVTAACPHAGAKSFYKNLIWLVKKGNSYPKLSKSYINDKGKEVLVLWGIHHPSTSADQQSLYQNADAYVF
A/Bayern/69/2009               ......................................................E.............................................
A/Lviv/N6/2009                 ....................................................................................................
A/Astrakhan/1/2011             ....................................................................................................
A/St Petersburg/100/2011       ..........................................G.........................................T...........T...
A/St Petersburg/27/2011        ....................................................................................T...............
A/Hong Kong/5659/2012          ..................N..................R..............................................T...............
A/Jiangsu-Runzhou/SWL1995/2016 ...............M.S..................................................................T...............
A/Ghana/DILI-0620/2014         ...............M....................................................................TT..............
A/South Africa/3626/2013       ..........................X...................................Q.....................T...............
A/Israel/Q-504/2015            ...................................................T..........Q.........I...........T...............
A/Yunnan-Hongta/SWL11417/2016  ...................................................T..........Q.........I...........T...............
A/Brisbane/294/2016            ...................................................T..........Q....N....I...........T...............
A/Denmark/53/2016              ..........I...........................................E......NQ.....................T...............
A/Michigan/45/2015             .............................................................NQ.....................T...............
A/Laos/3779/2016               .............................................................NQ...................P.T...............
A/Austria/958726/2016          .............................................................NQ.....................T...............
A/Laos/3816/2016               .............................................................NQ.....................T...............
A/Slovenia/2903/2015           .............................................................NQ.....................T...............
A/Bulgaria/074/2016            .............................................................NQ.....................T...............
A/Norway/4555/2016             .............................................................NQ.....................T...............
A/Antananarivo/5768/2016       ..................R..........................................NQ.....................T...............
A/Singapore/TT1358/2016        .............................................................NQ.....................T...............
A/Hong Kong/219/2017 Jan       .............................................................NQ.....................T...............
A/Hungary/187/2016             .............................................................NQ.....................T...............
A/Portugal/MS25/2016           ...................................................I.........NQ.....................T...............
A/Switzerland/19464226/2016    .............................................................NQ.....................T...............
A/Oman/6472/2016               .............................................................NQ.....................T...............
A/Croatia/3644/2016            .............................................................NQ.....................T...............
A/England/963/2016             .............................................................NQ...................P.T...............
A/Wisconsin/13/2017 Jan        .............................................................NQ..V................P.T...............
A/Netherlands/3792/2016        .............................................................NQ..V................P.T...............

                                   210       220       230       240       250       260       270       280       290       300
                               ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009            VGSSRYSKKFKPEIAIRPKVRDQEGRMNYYWTLVEPGDKITFEATGNLVVPRYAFAMERNAGSGIIISDTPVHDCNTTCQTPKGAINTSLPFQNIHPITI
A/Bayern/69/2009               ....................................................................................................
A/Lviv/N6/2009                 ..T..................G..............................................................................
A/Astrakhan/1/2011             ..T.K.........XV................................L...................................................
A/St Petersburg/100/2011       ..T...................R.............................................................................
A/St Petersburg/27/2011        ..T...................R.............................................................................
A/Hong Kong/5659/2012          ..T.............................................L...................................................
A/Jiangsu-Runzhou/SWL1995/2016 ..T...................R..........I..................................................................
A/Ghana/DILI-0620/2014         ..T..............................I......................V.........................E.................
A/South Africa/3626/2013       ..T....................................................T..........................E.................
A/Israel/Q-504/2015            ..T....................................................T..........................E.................
A/Yunnan-Hongta/SWL11417/2016  ..T....................................................T....S.....................E.................
A/Brisbane/294/2016            ..T....................................................T....S.....................E.................
A/Denmark/53/2016              ..T............T.......................................T..........................E.................
A/Michigan/45/2015             ..T............T......R................................T..........................E.................
A/Laos/3779/2016               ..T...........GT.......................................T..........................E.................
A/Austria/958726/2016          ..T...........GT.......................................T..........................E.................
A/Laos/3816/2016               ..T...........GT.......................................T..........................E.............Q...
A/Slovenia/2903/2015           ..T............T.......................................T..............Q...........E.................
A/Bulgaria/074/2016            ..T............T.......................................T..........................E.................
A/Norway/4555/2016             ..T............T.......................................T..........................E.................
A/Antananarivo/5768/2016       ..T............T.......................................T..........................E.................
A/Singapore/TT1358/2016        ..T............T.......................................T..........................E.................
A/Hong Kong/219/2017 Jan       ..T............T.......................................T..........................E.................
A/Hungary/187/2016             ..T............T.......................................T.................N........E.................
A/Portugal/MS25/2016           ..T............T.......................................T..........................E.................
A/Switzerland/19464226/2016    ..T............T.......................................T..........................E.................
A/Oman/6472/2016               ..T............T.......................................T..........................E.................
A/Croatia/3644/2016            ..T.K..........T.......................................T..........................E.................
A/England/963/2016             ..T.K..........T.......................................T..........................E.................
A/Wisconsin/13/2017 Jan        ..T.K..........T.......................................T..........................E.................
A/Netherlands/3792/2016        ..T.K..........T.......................................T..........................E.................
```

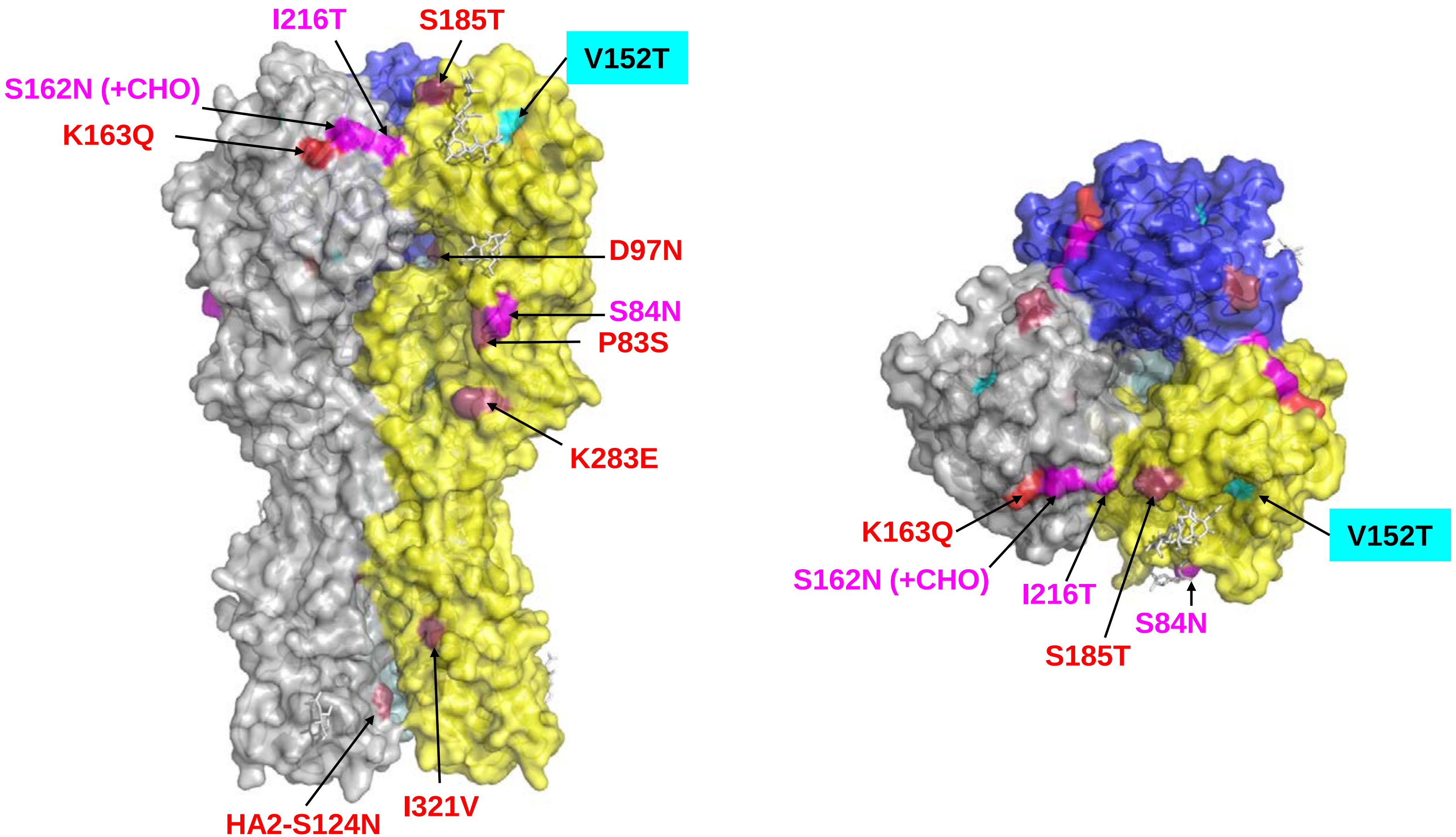
```
                                       310       320         3        13        23        33        43        53        63        73
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009             GKCPKYVKSTKLRLATGLRNIPSIQSRGLFGAIAGFIEGGWTGMVDGWYGYHHQNEQGSGYAADLKSTQNAIDEITNKVNSVIEKMNTQFTAVGKEFNHL
A/Bayern/69/2009                ....................V...............................................................................
A/Lviv/N6/2009                  ....................V...............................................................................
A/Astrakhan/1/2011              ....................V....................................................K..........................
A/St Petersburg/100/2011        ....................V....................................................K..........................
A/St Petersburg/27/2011         ....................V....................................................K..........................
A/Hong Kong/5659/2012           ....................V....................................................K..........................
A/Jiangsu-Runzhou/SWL1995/2016  ....................V....................................................K..........................
A/Ghana/DILI-0620/2014          ....................V....................................................K..........................
A/South Africa/3626/2013        ....................V....................................................K..........................
A/Israel/Q-504/2015             ....................V....................................................K..........................
A/Yunnan-Hongta/SWL11417/2016   ....................V....................................................K..........................
A/Brisbane/294/2016             ....................V....................................................K..........................
A/Denmark/53/2016               ....................V........X.......................................D...K..........................
A/Michigan/45/2015              ....................V.......................X............................K..........................
A/Laos/3779/2016                ....................V....................................................K.............I............
A/Austria/958726/2016           ....................V....................................................K..........................
A/Laos/3816/2016                ....................V....................................................K........V.................
A/Slovenia/2903/2015            ....................V....................................................K..........................
A/Bulgaria/074/2016             ....................V....................................................K..........................
A/Norway/4555/2016              ....................V....................................................K..........................
A/Antananarivo/5768/2016        ....................V....................................................K..........................
A/Singapore/TT1358/2016         ....................V....................................................K..........................
A/Hong Kong/219/2017 Jan        ....................V..V.................................................K..........................
A/Hungary/187/2016              ....................V..V.................................................K..........................
A/Portugal/MS25/2016            .........................................................................K..........................
A/Switzerland/19464226/2016     .........................................................................K..........................
A/Oman/6472/2016                .......................................................................V.K..........................
A/Croatia/3644/2016             ....................V....................................................K..........................
A/England/963/2016              ....................V....................................................K..........................
A/Wisconsin/13/2017 Jan         ....................V....................................................K..........................
A/Netherlands/3792/2016         ....................V....................................................K..........................

                                        83        93       103       113       123       133       143       153       163       173
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009             EKRIENLNKKVDDGFLDIWTYNAELLVLLENERTLDYHDSNVKNLYEKVRSQLKNNAKEIGNGCFEFYHKCDNTCMESVKNGTYDYPKYSEEAKLNREEI
A/Bayern/69/2009                ....................................................................................................
A/Lviv/N6/2009                  ....................................................................................................
A/Astrakhan/1/2011              ....................................................................................................
A/St Petersburg/100/2011        ..................................................N.................................................
A/St Petersburg/27/2011         ..................................................N.................................................
A/Hong Kong/5659/2012           ..................................................N.................................................
A/Jiangsu-Runzhou/SWL1995/2016  ..................................................N...............................................K.
A/Ghana/DILI-0620/2014          ..........................................X.......N...............................................K.
A/South Africa/3626/2013        ..................................................N...............................................K.
A/Israel/Q-504/2015             ..................................................N.......................................G.......K.
A/Yunnan-Hongta/SWL11417/2016   ..................................................N.......................................G.......K.
A/Brisbane/294/2016             ..................................................N.......................................G.......K.
A/Denmark/53/2016               ..................................................N...............................................K.
A/Michigan/45/2015              ..................................................N...............................................K.
A/Laos/3779/2016                ..................................................N...............................................K.
A/Austria/958726/2016           ..................................................N..R............................................K.
A/Laos/3816/2016                ..................................................N...............................................K.
A/Slovenia/2903/2015            ..................................................N...............................................K.
A/Bulgaria/074/2016             ..................................................N...............................................K.
A/Norway/4555/2016              ..................................................N.......................................A.......K.
A/Antananarivo/5768/2016        ..................................................N...............................................K.
A/Singapore/TT1358/2016         ..................................................T...............................................K.
A/Hong Kong/219/2017 Jan        ..................................................N...............................................K.
A/Hungary/187/2016              ..................................................N...............................................K.
A/Portugal/MS25/2016            ..................................................N...............................................K.
A/Switzerland/19464226/2016     ..................................................N...............................................K.
A/Oman/6472/2016                ..................................................N...............................................K.
A/Croatia/3644/2016             ..................................................N.......................A........................K.
A/England/963/2016              ..................................................N...............................................K.
A/Wisconsin/13/2017 Jan         ..................................................N...............................................K.
A/Netherlands/3792/2016         ..................................................N..............................E................K.

                                       183       193       203       213
                                ....|....|....|....|....|....|....|....|....|....
A/California/7/2009             DGVKLESTRIYQILAIYSTVASSLVLVVSLGAISFWMCSNGSLQCRICI
A/Bayern/69/2009                .................................................
A/Lviv/N6/2009                  .................................................
A/Astrakhan/1/2011              .................................................
A/St Petersburg/100/2011        .................................................
A/St Petersburg/27/2011         .................................................
A/Hong Kong/5659/2012           .................................................
A/Jiangsu-Runzhou/SWL1995/2016  .................................................
A/Ghana/DILI-0620/2014          .................................................
A/South Africa/3626/2013        .......................V.........................
A/Israel/Q-504/2015             E................................................
A/Yunnan-Hongta/SWL11417/2016   E................................................
A/Brisbane/294/2016             E................................................
A/Denmark/53/2016               .................................................
A/Michigan/45/2015              .................................................
A/Laos/3779/2016                .................................................
A/Austria/958726/2016           .................................................
A/Laos/3816/2016                .................................................
A/Slovenia/2903/2015            .................................................
A/Bulgaria/074/2016             .................................................
A/Norway/4555/2016              .................................................
A/Antananarivo/5768/2016        ...................A.............................
A/Singapore/TT1358/2016         .................................................
A/Hong Kong/219/2017 Jan        .................................................
A/Hungary/187/2016              .................................................
A/Portugal/MS25/2016            .................................................
A/Switzerland/19464226/2016     .................................................
A/Oman/6472/2016                .................................................
A/Croatia/3644/2016             .................................................
A/England/963/2016              .................................................
A/Wisconsin/13/2017 Jan         .................................................
A/Netherlands/3792/2016         .................................................
```

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 3. H1 HA locations of amino acids substitutions in genetic subgroups 6B.1 and 6B.2



Amino acid substitutions compared with A/California/7/2009 that define genetic group 6B are in red (S203T, A256T in HA1 with E47K, E172K in HA2 not visible on this structure), those that define subgroup 6B.1 in magenta, those that define subgroup 6B.2 in cyan (V173I in HA1 with E164G, D174E in HA2 not visible on this structure). A number of viruses in subgroup 6B.2 carry additional substitutions (e.g. R113K, D127E in HA1 with K47Q in HA2).

Figure 4. Phylogenetic comparison of A(H1N1)pdm09 NA genes

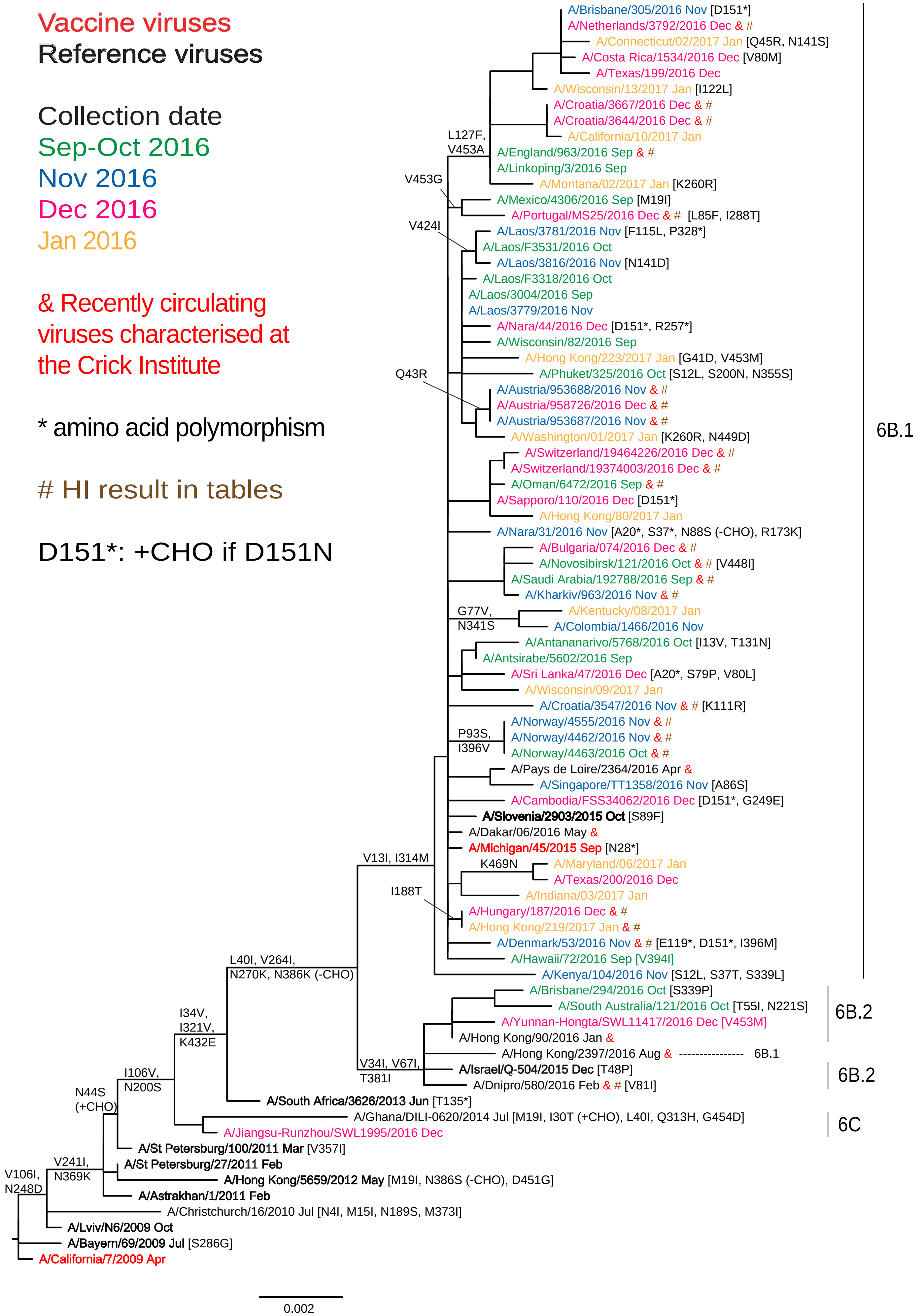


Figure 5. NA protein alignment for influenza A(H1N1)pdm09 viruses

Vaccine virus: Reference virus: Genetic group: **Potential N-linked glycosylation site**

```
                                          10        20        30        40        50        60        70        80        90       100
                                 ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009              MNPNQKIITIGSVCMTIGMANLILQIGNIISIWISHSIQLGNQNQIETCNQSVITYENNTWVNQTYVNISNTNFAAGQSVVSVKLAGNSSLCPVSGWAIY
A/Bayern/69/2009                 ....................................................................................................
A/Lviv/N6/2009                   ....................................................................................................
A/Astrakhan/1/2011               .................................................................................................... 5
A/St Petersburg/100/2011         ...........................................S........................................................ 7
A/St Petersburg/27/2011          .................................................................................................... 6
A/Hong Kong/5659/2012            ..................I................................................................................. 6A
A/Jiangsu-Runzhou/SWL1995/2016   ...........................................S........................................................ 6C
A/Ghana/DILI-0620/2014           ..................I..........T.........I...S........................................................ 6C
A/South Africa/3626/2013         .................................V.........S........................................................ 6B
A/Israel/Q-504/2015              .......................................I...S...P..................I................................. 6B.2
A/Yunnan-Hongta/SWL11417/2016    .......................................I...S......................I................................. 6B.2
A/Brisbane/294/2016              .......................................I...S......................I................................. 6B.2
A/Denmark/53/2016                ............I....................V.....I...S........................................................ 6B.1
A/Michigan/45/2015               ............I..............X.....V.....I...S........................................................ 6B.1
A/Laos/3779/2016                 ............I....................V.....I...S........................................................ 6B.1
A/Austria/958726/2016            ............I....................V.....I..RS........................................................ 6B.1
A/Laos/3816/2016                 ............I....................V.....I...S........................................................ 6B.1
A/Slovenia/2903/2015             ............I....................V.....I...S............................................F........... 6B.1
A/Bulgaria/074/2016              ............I....................V.....I...S........................................................ 6B.1
A/Norway/4555/2016               ............I....................V.....I...S................................................S....... 6B.1
A/Antananarivo/5768/2016         .................................V.....I...S........................................................ 6B.1
A/Singapore/TT1358/2016          ............I....................V.....I...S.........................................S.............. 6B.1
A/Hong Kong/219/2017 Jan         ............I....................V.....I...S........................................................ 6B.1
A/Hungary/187/2016               ............I....................V.....I...S........................................................ 6B.1
A/Portugal/MS25/2016             ............I....................V.....I...S........................................F............... 6B.1
A/Switzerland/19464226/2016      ............I....................V.....I...S........................................................ 6B.1
A/Oman/6472/2016                 ............I....................V.....I...S........................................................ 6B.1
A/Croatia/3644/2016              ............I....................V.....I...S........................................................ 6B.1
A/England/963/2016               ............I....................V.....I...S........................................................ 6B.1
A/Wisconsin/13/2017 Jan          ............I....................V.....I...S........................................................ 6B.1
A/Netherlands/3792/2016          ............I....................V.....I...S........................................................ 6B.1

                                         110       120       130       140       150       160       170       180       190       200
                                 ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009              SKDNSVRIGSKGDVFVIREPFISCSPLECRTFFLTQGALLNDKHSNGTIKDRSPYRTLMSCPIGEVPSPYNSRFESVAWSASACHDGINWLTIGISGPDN
A/Bayern/69/2009                 ....................................................................................................
A/Lviv/N6/2009                   .....I..............................................................................................
A/Astrakhan/1/2011               .....I..............................................................................................
A/St Petersburg/100/2011         .....I..............................................................................................
A/St Petersburg/27/2011          .....I..............................................................................................
A/Hong Kong/5659/2012            .....I..............................................................................................
A/Jiangsu-Runzhou/SWL1995/2016   ...................................................................................................S
A/Ghana/DILI-0620/2014           ...................................................................................................S
A/South Africa/3626/2013         ..................................X................................................................S
A/Israel/Q-504/2015              ...................................................................................................S
A/Yunnan-Hongta/SWL11417/2016    ...................................................................................................S
A/Brisbane/294/2016              ...................................................................................................S
A/Denmark/53/2016                ..................X...............................X................................................S
A/Michigan/45/2015               ...................................................................................................S
A/Laos/3779/2016                 ...................................................................................................S
A/Austria/958726/2016            ...................................................................................................S
A/Laos/3816/2016                 ........................................D..........................................................S
A/Slovenia/2903/2015             ...................................................................................................S
A/Bulgaria/074/2016              ...................................................................................................S
A/Norway/4555/2016               ...................................................................................................S
A/Antananarivo/5768/2016         ..............................N....................................................................S
A/Singapore/TT1358/2016          ...................................................................................................S
A/Hong Kong/219/2017 Jan         .......................................................................................T...........S
A/Hungary/187/2016               .......................................................................................T...........S
A/Portugal/MS25/2016             ...................................................................................................S
A/Switzerland/19464226/2016      ...................................................................................................S
A/Oman/6472/2016                 ...................................................................................................S
A/Croatia/3644/2016              ..........................F........................................................................S
A/England/963/2016               ..........................F........................................................................S
A/Wisconsin/13/2017 Jan          .....................L....F........................................................................S
A/Netherlands/3792/2016          ..........................F........................................................................S

                                         210       220       230       240       250       260       270       280       290       300
                                 ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009              GAVAVLKYNGIITDTIKSWRNNILRTQESECACVNGSCFTVMTDGPSNGQASYKIFRIEKGKIVKSVEMNAPNYHYEECSCYPDSSEITCVCRDNWHGSN
A/Bayern/69/2009                 .....................................................................................G..............
A/Lviv/N6/2009                   ...............................................D....................................................
A/Astrakhan/1/2011               ........................................I......D....................................................
A/St Petersburg/100/2011         ........................................I......D....................................................
A/St Petersburg/27/2011          ........................................I......D....................................................
A/Hong Kong/5659/2012            ........................................I......D....................................................
A/Jiangsu-Runzhou/SWL1995/2016   ........................................I......D....................................................
A/Ghana/DILI-0620/2014           ........................................I......D....................................................
A/South Africa/3626/2013         ........................................I......D....................................................
A/Israel/Q-504/2015              ........................................I......D...............I.....K..............................
A/Yunnan-Hongta/SWL11417/2016    ........................................I......D...............I.....K..............................
A/Brisbane/294/2016              ........................................I......D...............I.....K..............................
A/Denmark/53/2016                ........................................I......D...............I.....K..............................
A/Michigan/45/2015               ........................................I......D...............I.....K..............................
A/Laos/3779/2016                 ........................................I......D...............I.....K..............................
A/Austria/958726/2016            ........................................I......D...............I.....K..............................
A/Laos/3816/2016                 ........................................I......D...............I.....K..............................
A/Slovenia/2903/2015             ........................................I......D...............I.....K..............................
A/Bulgaria/074/2016              ........................................I......D...............I.....K..............................
A/Norway/4555/2016               ........................................I......D...............I.....K..............................
A/Antananarivo/5768/2016         ........................................I......D...............I.....K..............................
A/Singapore/TT1358/2016          ........................................I......D...............I.....K..............................
A/Hong Kong/219/2017 Jan         ........................................I......D...............I.....K..............................
A/Hungary/187/2016               ........................................I......D...............I.....K..............................
A/Portugal/MS25/2016             ........................................I......D...............I.....K.................T............
A/Switzerland/19464226/2016      ........................................I......D...............I.....K..............................
A/Oman/6472/2016                 ........................................I......D...............I.....K..............................
A/Croatia/3644/2016              ........................................I......D...............I.....K..............................
A/England/963/2016               ........................................I......D...............I.....K..............................
A/Wisconsin/13/2017 Jan          ........................................I......D...............I.....K..............................
A/Netherlands/3792/2016          ........................................I......D...............I.....K..............................
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                                           310       320       330       340       350       360       370       380       390       400
                                  ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009               RPWVSFNQNLEYQIGYICSGIFGDNPRPNDKTGSCGPVSSNGANGVKGFSFKYGNGVWIGRTKSISSRNGFEMIWDPNGWTGTDNNFSIKQDIVGINEWS
A/Bayern/69/2009                  ....................................................................................................
A/Lviv/N6/2009                    ....................................................................................................
A/Astrakhan/1/2011                ....................................................................K...............................
A/St Petersburg/100/2011          ........................................................I...........K...............................
A/St Petersburg/27/2011           ....................................................................K...............................
A/Hong Kong/5659/2012             ....................................................................K................S..............
A/Jiangsu-Runzhou/SWL1995/2016    ....................................................................K...............................
A/Ghana/DILI-0620/2014            ............H.......................................................K...............................
A/South Africa/3626/2013          ....................V...............................................K...............................
A/Israel/Q-504/2015               ....................V...............................................K...........I....K..............
A/Yunnan-Hongta/SWL11417/2016     ....................V...............................................K...........I....K..............
A/Brisbane/294/2016               ....................V.................P.............................K...........I....K..............
A/Denmark/53/2016                 .............M......V...............................................K................K.........M....
A/Michigan/45/2015                .............M......V...............................................K................K..............
A/Laos/3779/2016                  .............M......V...............................................K................K..............
A/Austria/958726/2016             .............M......V...............................................K................K..............
A/Laos/3816/2016                  .............M......V...............................................K................K..............
A/Slovenia/2903/2015              .............M......V...............................................K................K..............
A/Bulgaria/074/2016               .............M......V...............................................K................K..............
A/Norway/4555/2016                .............M......V...............................................K................K.........V....
A/Antananarivo/5768/2016          .............M......V...............................................K................K..............
A/Singapore/TT1358/2016           .............M......V...............................................K................K..............
A/Hong Kong/219/2017 Jan          .............M......V...............................................K................K..............
A/Hungary/187/2016                .............M.X....V...............................................K................K..............
A/Portugal/MS25/2016              .............M......V...............................................K................K..............
A/Switzerland/19464226/2016       .............M......V...............................................K................K..............
A/Oman/6472/2016                  .............M......V...............................................K................K..............
A/Croatia/3644/2016               .............M......V...............................................K................K..............
A/England/963/2016                .............M......V...............................................K................K..............
A/Wisconsin/13/2017 Jan           .............M......V...............................................K................K..............
A/Netherlands/3792/2016           .............M......V...............................................K................K..............

                                           410       420       430       440       450       460
                                  ....|....|....|....|....|....|....|....|....|....|....|....|....|....
A/California/7/2009               GYSGSFVQHPELTGLDCIRPCFWVELIRGRPKENTIWTSGSSISFCGVNSDTVGWSWPDGAELPFTIDK
A/Bayern/69/2009                  .....................................................................
A/Lviv/N6/2009                    .....................................................................
A/Astrakhan/1/2011                .....................................................................
A/St Petersburg/100/2011          .....................................................................
A/St Petersburg/27/2011           .....................................................................
A/Hong Kong/5659/2012             ..................................................G..................
A/Jiangsu-Runzhou/SWL1995/2016    .....................................................................
A/Ghana/DILI-0620/2014            .....................................................D...............
A/South Africa/3626/2013          ...............................E.....................................
A/Israel/Q-504/2015               ...............................E.....................................
A/Yunnan-Hongta/SWL11417/2016     ...............................E....................M................
A/Brisbane/294/2016               ...............................E.....................................
A/Denmark/53/2016                 ...............................E.....................................
A/Michigan/45/2015                ...............................E.....................................
A/Laos/3779/2016                  ...............................E.....................................
A/Austria/958726/2016             ...............................E.....................................
A/Laos/3816/2016                  .......................I.......E.....................................
A/Slovenia/2903/2015              ...............................E.....................................
A/Bulgaria/074/2016               ...............................E.....................................
A/Norway/4555/2016                ...............................E.....................................
A/Antananarivo/5768/2016          ...............................E.....................................
A/Singapore/TT1358/2016           ...............................E.....................................
A/Hong Kong/219/2017 Jan          ...............................E.....................................
A/Hungary/187/2016                ...............................E.....................................
A/Portugal/MS25/2016              ...............................E....................G................
A/Switzerland/19464226/2016       ...............................E.....................................
A/Oman/6472/2016                  ...............................E.....................................
A/Croatia/3644/2016               ...............................E....................A................
A/England/963/2016                ...............................E....................A................
A/Wisconsin/13/2017 Jan           ...............................E....................A................
A/Netherlands/3792/2016           ...............................E....................A................
```

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Influenza A(H3N2) virus analyses.

Influenza A(H3N2) viruses collected since 2016-08-31 have been received from NICs in 56 countries or administrative regions in four WHO Regions: African (3: Algeria, Ghana and Madagascar), Eastern Mediterranean (8: Egypt, Iran, Jordan, Morocco, Oman, Qatar, Saudi Arabia and Tunisia), European (44 countries) and Western Pacific (1: Hong Kong SAR).

Influenza A(H3N2) viruses have become increasingly difficult to characterise antigenically by HI assay due to either the variable agglutination of red blood cells (RBCs) from guinea pig, turkey and humans or the loss of the ability of viruses to agglutinate any of these RBCs. Of the successfully propagated viruses collected since 2016-08-31, just over 10% had sufficient HA titre to be analysed by HI assays conducted using guinea pig RBCs in the presence of 20nM oseltamivir, added to circumvent NA-mediated binding to the RBCs **(Tables 1, 2, 6 & 7, Figure 6)**. The remainder were unable to agglutinate guinea pig RBCs at all **(Tables 2 & 6)**, or were unable to agglutinate RBCs in the presence of 20nM oseltamivir **(Figure 6)**. Of the viruses in these last two categories that were sequenced, all but one belonged to subclades 3C.2a and 3C.2a1 **(Table 6)**. Of the 41 viruses with collection dates after 2016-08-31 that were able to bind guinea pig RBCs in the presence of oseltamivir, 31 (75%) showed either loss of or polymorphism at the HA1 158-160 glycosylation motif acquired by viruses in this subclade (**Tables 8-1 to 8-6**).

HI analyses.

A total of 56 viruses with collection dates after 2016-08-31 could be analysed by HI assay and summaries of the HI results are shown in **Tables 2 and 7**. Individual HI results are shown in **Tables 8-1** to **8-6**. Test viruses in each table are sorted by genetic subclade and by date of collection and those below the red horizontal lines in the tables are viruses with collection dates after 2016-08-31. The HA genetic group is indicated for those viruses that have been sequenced and those included in phylogenetic trees are highlighted. Viruses falling above the blue line in **Tables 8-1 to 8-4** are viruses from genetic subclade 3C.2a and below the blue lines from genetic subclade 3C.2a1. A/Valladolid/290/2016 **(Table 8-1)** falls into genetic subclade 3C.3a.

A summary of the HI results for viruses collected since 2016-08-31 with antisera raised against the cell culture-propagated cultivars of A/Hong Kong/5738/2014 and A/Hong Kong/4801/2014 and the egg-propagated cultivar of A/Hong Kong/4801/2014 is shown in **Table 7**. A more detailed summary of the recognition of the viruses by the full panel of antisera is shown in **Table 8-7**, together with a breakdown based on gene sequencing for viruses in subclades 3C.2a and 3C.2a1.The antiserum raised against A/Hong Kong/5738/2014 recognised 98% of the test viruses at titres within 4-fold of the titre of the antiserum with the homologous virus, 95% within 2-fold of the homologous titre. Similarly, the antiserum raised against the cell culture-propagated A/Hong Kong/4801/2014 recognised 96% of test viruses with collection dates after 2016-08-31 at titres within 4-fold of the titre of the antiserum with the homologous virus and 52% at titres within 2-fold of the titre with the homologous virus. However, although the antiserum raised against egg-propagated A/Hong Kong/4801/2014 recognised 86% of test viruses with collection dates after 2016-08-31 at titres within 4-fold of the titre of the antiserum with the homologous virus, only 37.5% of the test viruses were recognised at titres within 2-fold of the titre with the homologous virus **(Table 7)**.

For test viruses with collection dates after 2016-08-31 characterised genetically and antigenically by HI since our September 2016 report **(Table 8-7)**:

- an antiserum raised against the egg-propagated cultivar of A/Hong Kong/4801/2014 (3C.2a) recognised 78% of those viruses in subclade 3C.2a and 87% of the viruses in subclade 3C.2a1 at titres within 4-fold of the homologous titre of the antiserum;
- an antiserum raised against the cell culture-propagated cultivar of A/Hong Kong/4801/2014 (3C.2a) recognised all of the test viruses in subclade 3C.2a and all but one (96%) of the viruses in subclade 3C.2a1 at titres within 4-fold of the homologous titre of the antiserum;
- an antiserum raised against the cell culture-propagated cultivar of A/Hong Kong/5738/2014 (3C.2a) recognised all of the test viruses in subclade 3C.2a and all but one (96%) of the test viruses in subclade 3C.2a1 at titres within 4-fold of the homologous titre of the antiserum;
- an antiserum raised against a cell culture-propagated cultivar of A/Georgia/532/2015 (3C.2a), a reference virus that carries the HA1 amino acid substitutions R142K and Q197R and shows polymorphism at the 158-160 glycosylation motif, recognised only 16.7% of those viruses in subclade 3C.2a and only one (4%) of the viruses in subclade 3C.2a1 at titres within 4-fold of the homologous titre of the antiserum;
- an antiserum raised against the previously recommended vaccine virus, egg-propagated A/Switzerland/9715293/2013 (3C.3a), recognised none of the test viruses at a titre within 4-fold of the titre of the antiserum with the homologous virus;
- the antiserum raised against a cell culture-propagated cultivar of A/Switzerland/9715293/2013 (3C.3a) had an homologous titre of 160 and, in contrast, recognised 73% of all the test viruses, 72% in subclade 3C.2a and 78% in subcalde 3C.2a1, at titres within 4-fold of the homologous titre;
- an antiserum raised against a cell culture-propagated cultivar of A/Stockholm/6/2014 (3C.3a) had a higher homologous titre of 320 or 640 and recognised 78% of test viruses in subclade 3C.2a and 77% of test viruses in subclade 3C.2a1 at titres within 4-fold of the homologous titre of the antiserum;

- an antiserum raised against an egg-propagated cultivar of A/Stockholm/6/2014 (3C.3a) had a homologous titre of 320 and recognised only 62.5% of all the test viruses, 61% of viruses in each of subclades 3C.2a and 3C.2a1, at titres within 4-fold of the homologous titre of the antiserum.

Antisera raised against viruses in other genetic groups generally recognised the test viruses poorly; these were raised against A/Texas/50/2012 (a former vaccine virus in genetic group 3C.1) and reference viruses in other genetic groups - A/Hong Kong/146/2013 (3C.2), A/Samara/73/2013 (3C.3) and A/Netherlands/525/2014 (3C.3b).

Virus Neutralisation.

Plaque reduction neutralisation assays (PRNA) were used to complement HI tests and allowed antigenic characterisation of representative viruses of the many that could not be assayed by HI. The results of virus neutralisation assays are shown in **Tables 9-1 to 9-8**.

Selected antisera were raised against the currently recommended vaccine virus, egg-propagated A/Hong Kong/4801/2014 (subclade 3C.2a), and four cell culture-propagated viruses from genetic subclades 3C.2a or 3C.2a1: A/Hong Kong/7295/2014 (3C.2a), A/Cote d'Ivoire/544/2016 (3C.2a), A/Oman/2585/2016 (3C.2a1), were used in all assays. An antiserum raised against egg-propagated A/Norway/3806/2016 (3C.2a1) was used in **Tables 9-1 to 9-4** and an antiserum raised against cell culture-propagated A/Norway/4436/2016 was used in **Tables 9-5 to 9-8**. All four cell culture-propagated reference viruses retained the HA1 158-160 glycosylation motif. A/Hong Kong/7295/2014 represented a non-polymorphic virus genetically very closely related to the prototype A/Hong Kong/4801/2014; A/Cote d'Ivoire/544/2016 represented a group that carries **HA1 N121K, S144K** and **S198P** amino acid substitutions. The reference virus A/Oman/2585/2016 was chosen as a representative of subclade 3C.2a1 viruses, 3C.2a1 viruses defined by an HA gene that encodes **N171K** in **HA1** and **I77V** and **G155E** in **HA2**, with A/Oman/2585/2016 also falling into a genetic subgroup of 3C.2a1 with an HA gene also encoding **N121K** in **HA1**. A/Norway/4436/2016 also falls in genetic subgroup 3C.2a1 but it also falls into a further cluster of viruses defined by the amino acid substitutions **I140M** in **HA1** and **G150E** in **HA2**. The egg-propagated A/Norway/3806/2016 falls into the same genetic subgroup as A/Norway/4436/2016 and has the egg adaptive amino acid substitutions **T160K** (resulting in the loss of the glycosylation motif at residues 158-160 in HA1) and **L194P** in **HA1**.

Test viruses were selected to represent genetic groups within subclades 3C.2a and 3C.2a1. Of the 144 test viruses examined, 143 have been genetically characterised to date: 64 (45%) were from genetic groups in subclade 3C.2a and 79 (55%) were from genetic groups within subclade 3C.2a1. A summary of the results is shown in **Table 9-9**.

The results in **Tables 9-1 to 9-8** show the neutralisation titres as estimated by the software used to score the neutralisation titres and these numbers are shown 'rounded' to the nearest value in a 2-fold dilution series, analogous to the HI assay. To summarise the results shown in **Table 9-9**:

The antiserum raised against the egg-propagated vaccine virus A/Hong Kong/4801/2014 (3C.2a) recognised:

- 68.8% of the test viruses in subclade 3C.2a at titres within 4-fold of the homologous titre of the antiserum, which were
  - 56% of those with substitutions N121K and S144 K in HA1,
  - 77% of those with substitutions T131K, R142K and R261Q in HA1,
  - 85% of those with substitutions N121K, S144K and S219Y in HA1,
  - 81.3% of those with substitutions N121K, N122D (resulting in the loss of the glycosylation site at residue 122) and S144K in HA1;
- 26.6% of the test viruses in subclade 3C.2a1 at titres within 4-fold of the homologous titre of the antiserum, which were
  - 44% of those with only the N171K substitution in HA1,
  - 35.7% of those with substitutions N121K and N171K in HA1,
  - 25% of those with substitutions N121K, I140M and N171K in HA1,
  - a single virus of eight (12.5%) with substitutions N121K, R142G and N171K in HA1,
  - none of the viruses with HA1 substitutions R142G and N171K (n=7) or with substitutions N121K, T135K (resulting in the loss of the glycosylation motif at residue 133-135 of HA1) and N171K (n=4).

The antiserum raised against cell culture-propagated A/Hong Kong/7295/2014 (3C.2a) recognised:

- 96.9% of the test viruses in subclade 3C.2a at titres within 4-fold of the homologous titre of the antiserum, which were
  - all of those with substitutions N121K and S144 K in HA1,
  - all of those with substitutions T131K, R142K and R261Q in HA1,
  - all of those with substitutions N121K, N122D (resulting in the loss of the glycosylation site at residue 122) and S144K in HA1,
  - 85% of those with substitutions N121K, S144K and S219Y in HA1;
- 96.2% of the test viruses in subclade 3C.2a1 at titres within 4-fold of the homologous titre of the antiserum, which were
  - 96% of those with only the N171K substitution in HA1,
  - all of those with substitutions N121K and N171K in HA1,
  - 85.7% of those with substitutions R142G and N171K,
  - all with HA1 substitutions N121K, T135K (resulting in the loss of the glycosylation motif at residue 133-135) and N171K (n=4),
  - all of those with substitutions N121K, R142G and N171K (n=8) in HA1,
  - 95% of those with substitutions N121K, I140M and N171K in HA1.

The antiserum raised against the cell culture-propagated A/Cote d'Ivoire/544/2016 (3C.2a) recognised:

- 76.6% of the test viruses in subclade 3C.2a at titres within 4-fold of the homologous titre of the antiserum, which were
  - all of those with substitutions N121K and S144 K in HA1,
  - only 15.4% of those with substitutions T131K, R142K and R261Q in HA1,
  - 77% of those with substitutions N121K, S144K and S219Y in HA1,
  - all of those with substitutions N121K, N122D (resulting in the loss of the glycosylation site at residue 122) and S144K in HA1;
- 50.6% of the test viruses in subclade 3C.2a1 at titres within 4-fold of the homologous titre of the antiserum, which were
  - 68% of those with only the N171K substitution in HA1,
  - 57% of those with substitutions N121K and N171K in HA1,
  - 28.6% of those with substitutions R142G and N171K,
  - 75% of those with HA1 substitutions N121K, T135K (resulting in the loss of the glycosylation motif at residue 133-135) and N171K (n=4),
  - 37.5% of those with HA1 substitutions N121K, R142G and N171K (n=8),
  - 45% of those with substitutions N121K, I140M and N171K in HA1.

The antiserum raised against cell culture-propagated A/Oman/2585/2016 (3C.2a1) recognised:

- 95.3% of the test viruses in subclade 3C.2a at titres within 4-fold of the homologous titre of the antiserum, which were
  - all of those with substitutions N121K and S144 K in HA1,
  - all of those with substitutions T131K, R142K and R261Q in HA1,
  - 7 of 8 (87.5%) with substitutions N121K, S144K and S219Y in HA1,
  - all of those with substitutions N121K, N122D (resulting in the loss of the glycosylation site at residue 122) and S144K in HA1;
- 96.2% of the test viruses in subclade 3C.2a1 at titres within 4-fold of the homologous titre of the antiserum, which were
  - 96% of those with only the N171K substitution in HA1,
  - all of those with substitutions N121K and N171K in HA1,
  - 6 of 7 (85.7%) with the substitutions R142G and N171K,
  - all four with HA1 substitutions N121K, T135K (resulting in the loss of the glycosylation motif at residue 133-135) and N171K,
  - all eight of those with substitutions N121K, R142G and N171K in HA1,
  - 19 of 20 (95%) with substitutions N121K, I140M and N171K in HA1.

The antiserum raised against cell culture-propagated A/Norway/4436/2016 (3C.2a1) was used with a total of 94 viruses in **Tables 9-5 to 9-8** and recognised:

- 96.9% of the test viruses in subclade 3C.2a at titres within 4-fold of the homologous titre of the antiserum, which were
    - all of those with substitutions N121K and S144 K in HA1,
    - all of those with substitutions T131K, R142K and R261Q in HA1,
    - 7 of 8 (87.5%) with the substitutions N121K, S144K and S219Y in HA1,
    - all five of those with substitutions N121K, N122D (resulting in the loss of the glycosylation site at residue 122) and S144K in HA1;
- 98.4% of the test viruses in subclade 3C.2a1 at titres within 4-fold of the homologous titre of the antiserum, which were
    - 95.5% of those with only the N171K substitution in HA1,
    - all 12 of those with substitutions N121K and N171K in HA1,
    - all six of those with substitutions R142G and N171K in HA1,
    - all four with HA1 substitutions N121K, T135K (resulting in the loss of the glycosylation motif at residue 133-135) and N171K,
    - all six with HA1 substitutions N121K, R142G and N171K,
    - all 11 of those with substitutions N121K, I140M and N171K in HA1.

The antiserum raised against egg-propagated A/Norway/3806/2016 (3C.2a1) was used with a total of 39 test viruses in **Tables 9-1 to 9-4**, for 38 of which the HA gene sequence has been determined, and recognised:

- 69.4% of the test viruses in subclade 3C.2a at titres within 4-fold of the homologous titre of the antiserum, which were
    - Seven of ten of those with substitutions N121K and S144 K in HA1,
    - none of those with HA1 substitutions T131K, R142K and R261Q (n=7),
    - all five of those with substitutions N121K, S144K and S219Y in HA1,
    - all 11 of those with substitutions N121K, N122D (resulting in the loss of the glycosylation site at residue 122) and S144K in HA1;
- 52% of the test viruses in subclade 3C.2a1 at titres within 4-fold of the homologous titre of the antiserum, which were
    - three of four (75%) with only the N171K substitution in HA1,
    - all three of those with substitutions N121K and N171K in HA1,
    - none of three with HA1 substitutions R142G and N171K,
    - none of four with HA1 substitutions N121K, R142G and N171K,
    - 9 of 13 (69.2%) with HA1 substitutions N121K, I140M and N171K.

Genetic analyses.

Phylogenetic analysis of the HA genes of representative recently circulating H3N2 viruses is shown in **Figure 7**. The HA genes fell within genetic group 3C. This group has three subdivisions: 3C.1, 3C.2 and 3C.3; subdivisions 3C.2 and 3C.3 have undergone further evolution to yield subclades designated as 3C.2a, 3C.3a and 3C.3b respectively **(Figure 7)**.

The previously used vaccine virus A/Texas/50/2012 belongs to genetic subdivision 3C.1. Viruses received, collected during the course of the 2016-2017 season, that have been sequenced fell into subclades 3C.2a (440: 99.8% with 251 [56.9%] falling in 3C.2a1) and 3C.3a (1: 0.2%). Amino acid substitutions that define these subclades compared with A/Texas/50/2012 are:

- (**3C.3a**) **T128A** (resulting in the loss of a potential glycosylation site), **A138S, R142G, N145S, F159S, V186G**[@], **P198S**, **F219S** and **N225D** in **HA1**, e.g. A/Switzerland/9715293/2013. More recent viruses have fallen into two groups, one group carrying HA genes that encode the amino acid substitutions **K92R, Q197K, S198P, S312N** in **HA1** and the other group carrying HA genes that encode the substitutions **L3I, S91N, N144K, F193S** in **HA1** and **D160N** in **HA2**.
- (**3C.2a**) **L3I**, **N128T** (resulting in the gain of a potential glycosylation site), **N144S** (resulting in the loss of a potential glycosylation site), **N145S, F159Y**, **K160T** (resulting in the gain of a potential glycosylation site), **V186G**[@], **P198S**, **F219S**, **N225D** and **Q311H** in **HA1**, e.g. A/Hong Kong/5738/2014. Several subgroups have emerged within 3C.2a and 3C.2a1 subclades:
  Within 3C.2a viruses with HA substitutions:
    - **N121K** and **S144K** in **HA**1, with further subgroups of viruses that carry HA genes that encode **S219Y** or **N122D** (resulting in the loss of a glycosylation motif at residues **122-124**) in **HA1**, or the substitutions **Y94H, I140T, R269K** in **HA1** and **A166S** and **Q172H** in **HA2**,
    - **T131K** and **R142K** in **HA1** many also having HA genes encoding **R261Q** in **HA1.**

  Within 3C.2a1 with HA substitutions **N171K** in **HA1** and **I77V** and **G155E** in **HA2**:
    - **N121K** and **R142G** in **HA1**,
    - **N121K** and **R142G** in **HA1** and **G150E** in **HA2**,
    - **N121K** and **I140M** in **HA1** and **G150E** in **HA2**.

[@]Note: the **G186V** substitution in HA1 occurred during adaptation of A/Texas/50/2012 to propagation in hens' eggs.

Figure 8 shows an HA amino acid sequence alignment for a representative selection of the viruses used to generate the phylogeny (**Figure 7**). The vaccine virus A/Hong

Kong/4801/2014 is shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the HA1/HA2 proteolytic cleavage site is highlighted. The group/subclade to which each of the sequences belongs is also shown.

Figure 9 shows the location on a 2005 H3 HA structure of amino acid substitutions that define the main genetic subgroups of 3C.2a and 3C.2a1 subclade viruses defined in **Table 9-9,** each compared with the vaccine virus A/Hong Kong/4801/2014.

Figure 10 shows a NA phylogenetic tree for representative viruses with their HA genetic groups indicated. **Figure 11** shows a NA amino acid sequence alignment for a representative selection of these viruses. As above, the vaccine virus is shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the group/subclade to which the corresponding HA gene sequences belong is also shown.

The NA gene sequences of recent viruses with HA genes in genetic groups within 3C.2a (including 3C.2a1) generally cluster similarly to the HA sequences. Most viruses fall into genetic groups that differ from the NA gene sequence of A/Hong Kong/4801/2014 at several positions: **S245N**, resulting in the formation of a glycosylation motif at residues **245-247**, **S247T** maintaining the glycosylation motif, **T267K, I380V and T392I.** There are several subgroups defined by amino acid substitutions:

- **S335G**, associated with 3C.2a viruses,
- **G93D, D339N, P468L**, also associated with 3C.2a viruses,
- **P468H**, many with the additional substitution **D339N** and many of these have the substitution **N329S**, resulting in the loss of a glycosylation motif, associated with viruses in both 3C.2a and 3C.2a1 subclades.

Figure 12 illustrates these amino acid substitutions on an NA tetramer.

Antiviral resistance

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 450 viruses with collection dates after 2016-08-31. All showed normal inhibition (NI) by both neuraminidase inhibitors **(Table 14)**.

Conclusion.

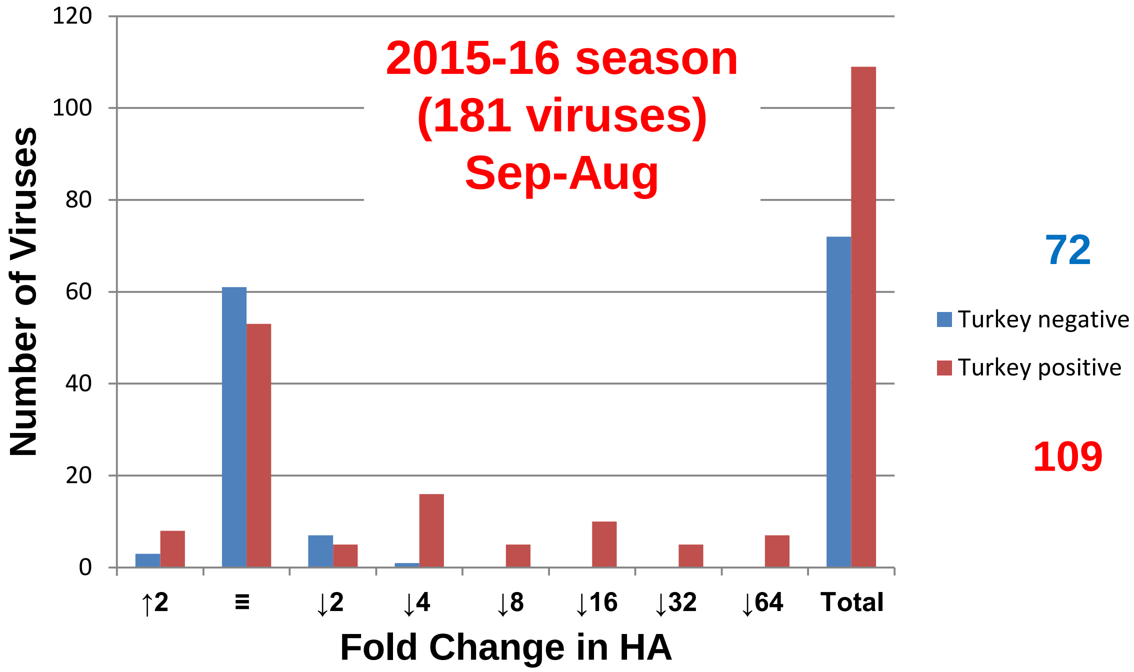
Viruses in subclades 3C.2a and 3C.2a1 were by far the most common but 90% of viruses in these subclades did not agglutinate RBCs and could not therefore be characterised by HI assay.

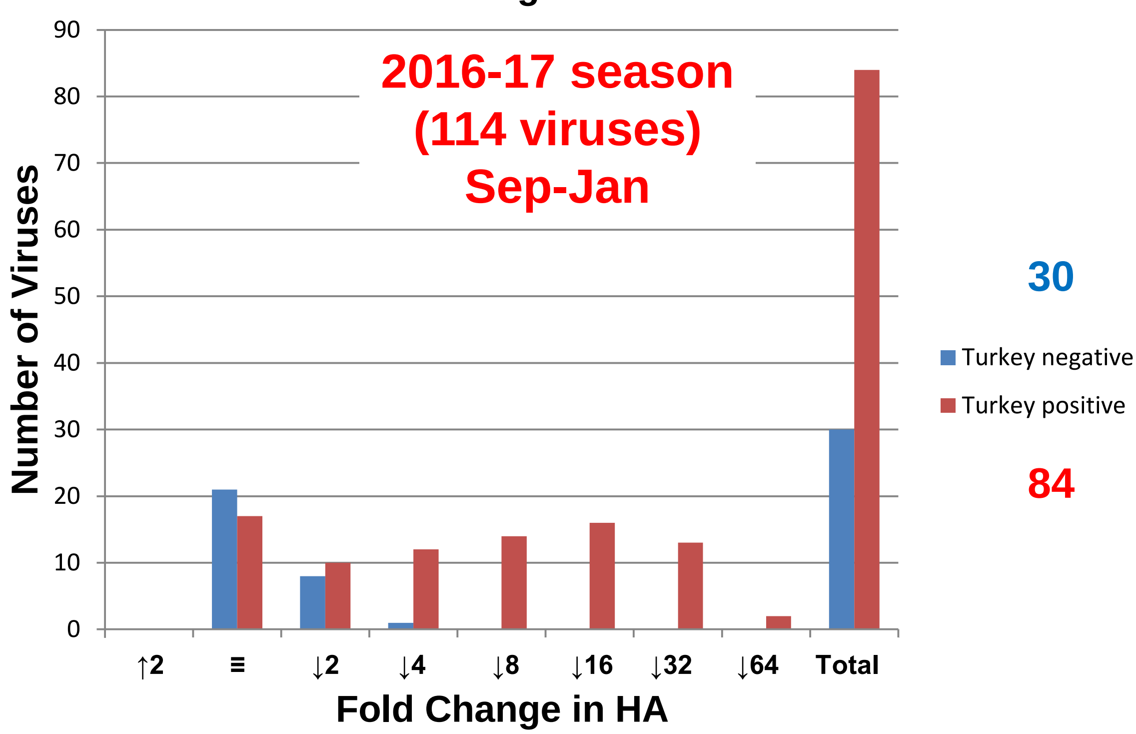
In HI assays that could be carried out the antiserum raised against a cell culture-propagated 3C.2a virus (genetically similar to the vaccine virus A/Hong Kong/4801/2014) recognised test viruses in subclades 3C.2a and 3C.2a1 very well.

An antiserum raised against the cell culture-propagated cultivar of A/Hong Kong/4801/2014, polymorphic at the 158-160 glycosylation motif, also recognised the test viruses well. An antiserum raised against the egg-propagated vaccine virus A/Hong Kong/4801/2014 recognised the majority (85%) of viruses at titres within 4-fold of its homologous titre.

In PRNA, subclade 3C.2a and 3C.2a1 viruses were recognised very well by antisera raised against three cell culture-propagated viruses in genetic group 3C.2a: these were antisera raised against a cell culture-propagated virus genetically similar to the vaccine virus A/Hong Kong/4801/2014 and two recent cell culture-propagated reference viruses in subclade 3C.2a1 (A/Oman/2585/2016 and A/Norway/4436/2016). An antiserum raised against another recent 3C.2a virus, A/Cote d'Ivoire/544/2016, showed more variable recognition of the test viruses in each genetic subclade. An antiserum raised against the egg-propagated vaccine virus A/Hong Kong/4801/2014 (3C.2a) recognised approximately 66% of test viruses in subclade 3C.2a but only 26% of test viruses in subclade 3C.2a1 at titres within 4-fold of the titre of the antiserum for the homologous virus.

Figure 6. Analysis of H3N2 Influenza Virus Agglutination of Guinea Pig RBCs and the Effect of 20nM Oseltamivir





Fold shift in HA activity in the presence of 20nM oseltamivir: [Enhancement (↑2) Equivalence (≡) Reduction (↓2-↓64)]

Table 6. A(H3N2) viruses recovered from specimens collected during the 2016 - 2017 influenza season (after 2016-08-31), as assessed by NA activity (MUNANA-based assay)

Situation at the time of the February 2017 VCM (viruses with collection dates Sep 2016-Jan 2017)

RBC binding phenotype[1]			HI assay possible	Number of isolates	Number sequenced[2]	Number in clade 3.3	Number in clade 3C.3b	Number in clade 3C.3a	Number in clade 3C.2a	Number in clade 3C.2a1
Turkey	Guinea pig	Guinea pig + 20nM oseltamivir								
+	+	+	Yes	33	24 (73%)	0	0	1	11	12
-	+	+	Yes	23	18 (78%)	0	0	0	7	11
-	-	+	Yes	0	0	0	0	0	0	0
+	-	+	Yes	0	0	0	0	0	0	0
			Totals	56	42 (75%)	0	0	1	18	23
+	+	-	No	52	46 (88%)	0	0	0	19	27
+	-	-	No	0	0	0	0	0	0	0
-	+	-	No	7	5 (71%)	0	0	0	4	1
-	-	-	No	407	324 (80%)	0	0	0	144	180
			Totals	466	375 (80%)	0	0	0	167	208
			Totals	522	417 (80%)	0	0	1	185	231

1. Negative (-) corresponds to HA titres of <4, while positive (+) corresponds to HA titres of ≥4
2. As of 2017-02-17: a number still in process

Situation at the time of the September 2016 VCM (viruses with collection dates Feb-Aug 2016)

RBC binding phenotype[1]			HI assay possible	Number of isolates	Number sequenced[2]	Number in clade 3.3	Number in clade 3C.3b	Number in clade 3C.3a	Number in clade 3C.2a
Turkey	Guinea pig	Guinea pig + 20nM oseltamivir							
+	+	+	Yes	88	84 (95%)	0	0	59	25
-	+	+	Yes	72	67 (93%)	1	0	10	56
-	-	+	Yes	0	0	0	0	0	0
+	-	+	Yes	0	0	0	0	0	0
			Totals	160	151 (94%)	1	0	69	81
+	+	-	No	21	15 (71%)	0	0	0	15
+	-	-	No	0	0	0	0	0	0
-	+	-	No	28	25 (89%)	0	0	0	25
-	-	-	No	187	147 (79%)	0	0	1	146
			Totals	236	187 (79%)	0	0	0	186
			Totals	396	338 (85%)	1	0	70	267

1. Negative (-) corresponds to HA titres of <4, while positive (+) corresponds to HA titres of ≥4
2. As of 2016-09-16: a number still in process

Table 7. Summary of influenza A(H3N2) viruses analysed, collected after 2016-08-31

MONTH	H3N2										
Country	Number received	Number propagated[1]	≤2 fold[2]	4 fold[2]	≥8 fold[2]	≤2 fold[3]	4 fold[3]	≥8 fold[3]	≤2 fold[4]	4 fold[4]	≥8 fold[4]
2016											
SEPTEMBER											
Iceland	12	0									
Oman	1	in process									
Saudi Arabia	93	2	2			1	1		2		
Turkey	1	in process									
OCTOBER											
Denmark	4	in process									
Finland	2	0									
France	11	in process	2				2		1	1	
Germany	1	in process									
Ghana	1	0									
Iran	3	in process									
Jordan	1	0									
Kosovo	2	in process									
Kyrgyzstan	1	0									
Norway	16	5	5				5		5		
Oman	7	in process									
Portugal	2	in process									
Spain	7	in process									
Switzerland	1	0									
Turkey	1	in process									
Ukraine	1	0									
United Kingdom	1	0									
NOVEMBER											
Albania	3	0									
Armenia	2	in process									
Austria	5	in process									
Azerbaijan	1	in process									
Belgium	3	in process									
Bosnia & Herzegovina	1	in process									
Denmark	4	in process									
Estonia	4	0									
Finland	3	0									
France	19	in process	1			1			1		
Georgia	2	in process									
Germany	4	in process									
Iran	3	0									
Italy	7	0									
Ireland	3	in process									
Jordan	5	0									
Kazakhstan	9	in process									
Kosovo	27	in process									
Kyrgyzstan	36	in process	3			3			1	2	
Latvia	2	in process									
Montenegro	2	in process									
Morocco	6	in process									
Netherlands	1	1	1			1			1		
Norway	28	1	1				1				1
Oman	5	0									
Romania	4	in process									
Russia	5	in process									
Slovenia	3	in process									
Spain	27	in process									
Sweden	5	in process	2			2			1	1	
Tunisia	1	0									
Turkey	2	in process									
Ukraine	18	in process									
United Kingdom	9	2	2				2			2	
DECEMBER											
Albania	8	in process									
Algeria	5	in process									
Armenia	28	in process									
Austria	9	in process	2			2				1	1
Azerbaijan	8	in process									
Belgium	7	in process									
Bosnia & Herzegovina	6	in process									
Bulgaria	11	0									
Croatia	2	0									
Czech Republic	10	in process									
Denmark	3	0									
Estonia	16	0									
Finland	5	0									
France	26	1	1			1			1		
Georgia	8	in process	1			1				1	
Germany	15	in process									
Greece	4	in process									
Hungary	12	in process	2	2	1	2	2	1	1	1	3
Iceland	25	in process	2			2			1	1	
Iran	7	0									
Ireland	17	in process									
Israel	4	in process									
Italy	10	0									
Kazakhstan	7	in process									
Kosovo	9	in process									
Kyrgyzstan	1	0									
Latvia	4	in process	1			1			1		

Luxembourg	16	in process									
Macedonia	15	in process									
Moldova	4	in process									
Montenegro	9	in process	1			1				1	
Morocco	5	in process									
Netherlands	10	2	2			2			2		
Norway	8	2	2				2			2	
Poland	9	in process									
Portugal	19	in process									
Romania	10	in process									
Russia	13	in process	3			3				3	
Serbia	17	in process									
Slovenia	4	in process									
Spain	8	1	1				1			1	
Sweden	11	in process	6			3	3		1	3	2
Switzerland	15	0									
Tunisia	1	0									
Turkey	26	in process									
Ukraine	37	in process	2			1	1			2	
2017											
JANUARY											
Albania	9	in process									
Belarus	10	in process									
Belgium	20	in process									
Bosnia & Herzegovina	11	in process	1			1				1	
Germany	14	2	2				2			2	
Georgia	3	0									
Greece	27	in process	1			1			1		
Hong Kong	3	1	1				1			1	
Ireland	2	in process									
Israel	13	in process	2				1	1		1	1
Latvia	2	0									
Luxembourg	2	0									
Macedonia	4	in process									
Moldova	8	in process									
Morocco	1	in process									
Poland	9	in process	1				1		1		
Qatar	10	in process									
Romania	15	in process									
Slovenia	8	in process									
Turkey	10	in process									
	1113	20	53	2	1	29	25	2	21	27	8
54 Countries			94.6%	3.6%	1.8%	51.8%	44.6%	3.6%	37.5%	48.2%	14.3%

1. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir
2. Compared to homologous titres for ferret antisera raised against tissue culture-propagated virus A/Hong Kong/5738/2014 (3C.2a)
3. Compared to homologous titres for ferret antisera raised against tissue culture-propagated virus A/Hong Kong/4801/2014 (3C.2a)
4. Compared to homologous titres for ferret antisera raised against egg-propagated virus A/Hong Kong/4801/2014 (3C.2a)

Table 8-1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2016-11-25 + 2016-12-02 + 2016-12-09 + 2016-12-16 + 2016-12-20 + 2017-01-10

Haemagglutination inhibition titre — Post-infection ferret antisera

Viruses	Other information	Genetic group	Collection date	Passage history	A/Texas 50/12 Egg F36/12[*1] 3C.1	A/Samara 73/13 SIAT F35/15[*1] 3C.3	A/Stock 6/14 SIAT F14/14[*1] 3C.3a	A/Stock 6/14 Egg F20/14[*1] 3C.3a	A/Switz 9715293/13 SIAT F18/15[1] 3C.3a	A/Switz 9715293/13 Egg F32/14[*1] 3C.3a	A/Neth 525/14 SIAT F23/15[*1] 3C.3b	A/HK 146/13 Egg F10/15[*1] 3C.2	A/HK 5738/14 MDCK F30/14[*1] 3C.2a	A/HK 4801/14 MDCK F43/15[*1] 3C.2a	A/HK 4801/14 Egg F12/15[*1] 3C.2a	A/Georgia 532/15 SIAT F33/15[*1] 3C.2a	HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
REFERENCE VIRUSES																	
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	5120	1280	320	1280	80	1280	640	640	320	640	80	320	
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT2	1280	640	640	640	160	640	640	640	320	1280	320	640	
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT3	80	40	640	160	160	160	80	80	160	320	80	160	
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	640	160	160	320	80	640	80	80	160	640	40	80	
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT2	40	40	320	160	160	160	40	80	160	160	80	160	
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	640	320	160	320	80	1280	80	80	320	320	80	320	
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT4	640	320	320	320	80	320	1280	160	160	320	80	320	
A/Hong Kong/146/2013		3C.2	2013-01-11	E6	2560	640	80	640	40	640	320	1280	320	640	160	640	
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	80	80	320	80	80	80	40	80	160	320	160	320	[N158n/k, T160t/r] (-cho), S312N
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK2	80	80	320	160	80	80	40	80	160	640	160	160	Parent
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	40	80	80	80	40	40	40	<	320	640	160	640	N96S, T160K (-CHO), L194P
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT3	160	80	320	160	80	160	40	160	320	640	160	1280	R142K, N158H (-CHO), Q197R
TEST VIRUSES																	
A/Fiji/60/2016		3C.2a	2016-05-27	E4/E1	<	40	<	40	<	<	<	<	80	320	40	160	N121T, T160K (-CHO), L194P, D225G
A/Alaska/232/2015		3C.2a1	2015-09-09	E6/E1	<	<	40	80	40	40	<	40	160	160	80	160	N171K
A/Sri Lanka/61/2015		3C.2a1	2015-11-29	E5/E1	<	<	40	40	40	<	<	<	160	640	80	160	A138S, T160K (-CHO), N171K, L194P, T203I
A/Nagano-C/21/2016		3C.2a1	2016-02-20	MDCK2/SIAT2/SIAT1	40	40	160	80	80	40	<	40	80	320	80	80	T160K (-CHO), N171K
A/Newcastle/30/2016		3C.2a1	2016-03-29	SIAT3/SIAT1	40	40	160	80	80	40	<	80	160	320	80	160	N6n/s, N121K, N122D (-CHO), T160t/k (-cho), N171K
A/Yokohama/120/2016		3C.2a1	2016-04-12	MDCK2/SIAT2/SIAT1	80	40	320	80	80	40	<	80	160	320	80	160	N158n/k (-cho), N171K
A/Sydney/142/2016		3C.3a	2016-06-30	E4/E1	160	80	320	320	80	160	80	40	80	160	40	80	
A/Sydney/142/2016		3C.3a	2016-06-30	SIAT1/SIAT1	80	80	640	320	160	160	80	80	160	160	160	160	
A/South Australia/54/2016		3C.3a	2016-07-12	E4/E1	320	80	320	320	80	160	80	40	80	160	40	80	
A/South Auckland/23/2016		3C.3a	2016-07-13	MDCKx/SIAT1/SIAT1	80	80	640	320	160	160	80	80	160	160	80	160	
A/England/962/2016		3C.3a	2016-08-30	SIAT1/SIAT1	40	40	320	160	80	80	<	40	80	160	80	80	
A/England/64520167/2016		3C.2a	2016-11-02	SIAT1/SIAT1	40	40	80	40	40	40	<	40	80	160	40	320	T131K, R142K, N158H (-CHO), R261Q
A/England/64580215/2016		3C.2a	2016-11-08	SIAT1/SIAT1	40	<	160	80	40	<	<	40	80	160	40	40	N121K, S144K
A/Norway/4849/2016		3C.2a	2016-12-02	SIAT1	<	<	80	80	<	<	<	40	80	160	40	80	N121K, S144K
A/Saudi Arabia/192840/2016		3C.2a1	2016-09-02	SIAT2	40	80	160	80	40	40	<	80	320	320	80	160	R142G, N171K
A/Saudi Arabia/192838/2016		3C.2a1	2016-09-13	SIAT1	40	40	80	80	40	40	<	40	80	160	80	80	T160K (-CHO), N171K
A/Bretagne/2836/2016		3C.2a1	2016-10-03	MDCK1/SIAT1	40	<	80	80	40	40	<	40	80	160	80	80	S47T, G78S, T160K (-CHO), N171K
A/Norway/4238/2016		3C.2a1	2016-10-12	SIAT1/SIAT1	40	<	80	40	40	40	<	<	80	160	80	40	N121K, S124G (-CHO), N126D (-CHO), T160t/k (-cho), N171K
A/Norway/4256/2016		3C.2a1	2016-10-13	SIAT1/SIAT1	80	40	160	80	80	40	<	80	80	160	80	80	N121K, I140M, T160t/k (-cho), N171K
A/Bretagne/2845/2016		3C.2a1	2016-10-13	MDCK1/SIAT1	40	<	80	80	40	40	<	40	80	160	40	80	S47T, G78S, T160t/k (-cho), N171K
A/Norway/4297/2016		3C.2a1	2016-10-17	SIAT1/SIAT3	80	40	160	80	80	40	<	80	80	160	80	80	N121K, I140M, N171K
A/Norway/4395/2016		3C.2a1	2016-10-20	SIAT1	80	40	160	80	80	40	<	80	160	160	160	80	N121K, T135K (-CHO), T160t/k (-cho), N171K
A/Norway/4497/2016		3C.2a1	2016-10-30	SIAT1	40	40	160	80	80	40	<	80	80	160	80	80	N121K, I140M, T160t/k/i (-cho) N171K
A/Valladolid/290/2016		3C.3a	2016-12-03	MDCK1/SIAT1	40	40	320	160	80	80	40	40	80	160	40	80	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) [1] < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-01-13 + 2017-01-17

Haemagglutination inhibition titre — Post-infection ferret antisera

Viruses	Other information	Passage history / Ferret number / Genetic group	Collection date	Passage history	A/Texas 50/12 Egg F36/12^{*1} 3C.1	A/Samara 73/13 SIAT F35/15^{*1} 3C.3	A/Stock 6/14 SIAT F14/14^{*1} 3C.3a	A/Stock 6/14 Egg F20/14^{*1} 3C.3a	A/Switz 9715293/13 SIAT F18/15^{1} 3C.3a	A/Switz 9715293/13 Egg F32/14^{*1} 3C.3a	A/Neth 525/14 SIAT F23/15^{*1} 3C.3b	A/HK 146/13 Egg F10/15^{*1} 3C.2	A/HK 5738/14 MDCK F30/14^{*1} 3C.2a	A/HK 4801/14 MDCK F43/15^{*1} 3C.2a	A/HK 4801/14 Egg F12/15^{*1} 3C.2a	A/Georgia 532/15 SIAT F33/15^{*1} 3C.2a	HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
REFERENCE VIRUSES																	
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	5120	1280	320	1280	80	1280	640	640	320	640	80	320	
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT2	1280	640	640	640	160	640	640	640	320	1280	320	640	
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT3	80	40	640	160	160	160	80	80	160	320	80	160	
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	640	160	160	320	80	640	80	80	160	640	40	80	
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT2	40	40	320	160	160	160	40	80	160	160	80	160	
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	320	160	160	320	40	1280	40	80	320	320	40	160	
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT4	640	320	320	320	80	320	1280	160	160	320	80	320	
A/Hong Kong/146/2013		3C.2	2013-01-11	E6	2560	640	80	640	40	640	320	1280	320	640	160	640	
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	80	80	320	80	80	80	40	80	160	320	160	320	[N158n/k, T160t/r] (-cho), S312N
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK2	80	80	320	160	80	80	40	80	160	640	160	160	Parent
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	40	160	80	80	40	40	40	<	320	640	320	640	N96S, T160K (-CHO), L194P
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT3	160	80	320	160	80	160	40	160	320	640	160	1280	R142K, N158H (-CHO), Q197R
TEST VIRUSES																	
A/Norway/4676/2016		3C.2a	2016-11-23	SIAT1/SIAT2	<	<	80	40	40	<	<	<	80	160	40	160	T131K, R142K, R261Q
A/Norway/4858/2016		3C.2a	2016-12-02	SIAT1/SIAT2	40	40	160	80	80	40	<	40	80	160	80	160	T131K, R142K, R261Q
A/Lyon/CHU/23.11.R8/2016		3C.2a1	2016-11-23	MDCK2/SIAT1	80	40	160	80	80	80	<	80	160	320	160	80	N31S, T160k/i (-CHO)N171K
A/Alsace/3215/2016		3C.2a1	2016-12-09	MDCK1/SIAT1	80	40	160	80	80	80	<	80	160	320	160	80	N121K, I140M, T160t/I (-cho), N171K

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) [1] < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-01-27 + 2017-01-31

					Haemagglutination inhibition titre												
					Post-infection ferret antisera												
Viruses	Other information		Collection date	Passage history	A/Texas 50/12	A/Samara 73/13	A/Stock 6/14	A/Stock 6/14	A/Switz 9715293/13	A/Switz 9715293/13	A/Neth 525/14	A/HK 146/13	A/HK 5738/14	A/HK 4801/14	A/HK 4801/14	A/Georgia 532/15	HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
		Passage history			Egg	SIAT	SIAT	Egg	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT	
		Ferret number			F36/12[*1]	F35/15[*1]	F14/14[*1]	F20/14[*1]	F18/15[1]	F32/14[*1]	F23/15[*1]	F10/15[*1]	F30/14[*1]	F43/15[*1]	F12/15[*1]	F33/15[*1]	
		Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.2a	
REFERENCE VIRUSES																	
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	2560	640	160	640	80	640	320	320	160	640	80	160	
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT2	1280	640	320	320	160	640	320	640	320	640	320	320	
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT3	80	40	320	160	160	80	40	80	160	320	80	80	
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	320	80	80	320	40	640	40	40	160	320	40	80	
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT3	40	40	320	160	160	160	40	40	160	160	80	160	
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	320	160	160	320	40	640	40	80	160	320	80	160	
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT4	640	320	320	160	80	160	1280	160	160	320	80	160	
A/Hong Kong/146/2013		3C.2	2013-01-11	E6	2560	640	40	320	40	640	320	640	320	640	80	320	
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	40	40	160	80	40	40	<	40	160	320	80	160	[N158n/k, T160t/r] (-cho), S312N
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK2	80	40	160	80	40	40	<	40	80	320	80	80	Parent
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	40	80	80	40	40	40	40	<	320	640	160	640	N96S, T160K (-CHO), L194P
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT3	80	80	160	160	80	80	<	80	320	320	160	640	R142K, N158H (-CHO), Q197R
TEST VIRUSES																	
A/Belgorod/2/2016		3C.2a	2016-12-13	MDCK1/SIAT1	40	40	160	80	40	40	<	40	80	160	40	160	N121K, S144K, T160t/k (-cho)
A/Montenegro/110/2016		3C.2a	2016-12-15	MDCK1/SIAT1	40	40	80	80	40	<	<	40	80	320	40	80	I58V, N121K, S144k, T160K (-CHO), D188G, S219Y
A/Latvia/12-064825/2016		3C.2a	2016-12-27	C2/SIAT1	<	40	160	40	40	<	<	40	160	160	80	160	N121K, S144K
A/Austria/960696/2016		3C.2a1	2016-12-22	C1/SIAT1	40	<	160	40	40	<	<	<	80	160	40	80	N121K, T135K (-CHO), T160t/k (-cho), N171K
A/Georgia/1819/2016		3C.2a1	2016-12-23	SIAT1	40	<	160	80	40	<	<	40	80	160	40	40	R142G, T160t/k (-cho), N171K
A/Iceland/81/2016		3C.2a1	2016-12-28	MDCK1/SIAT1	40	<	80	40	40	40	<	40	80	160	40	80	A96V, N121K, T135K (-CHO), N158n/k (-cho), N171K
A/Iceland/84/2016		3C.2a1	2016-12-29	MDCK1/SIAT1	80	80	160	160	80	80	40	80	160	640	80	160	A96V, N121K, T135K (-CHO), T160t/k (-cho), N171K

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) [1] < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-02-03

Viruses	Other information		Collection date	Passage history	Haemagglutination inhibition titre – Post-infection ferret antisera: A/Texas 50/12	A/Samara 73/13	A/Stock 6/14	A/Stock 6/14	A/Switz 9715293/13	A/Switz 9715293/13	A/Neth 525/14	A/HK 146/13	A/HK 5738/14	A/HK 4801/14	A/HK 4801/14	A/Georgia 532/15	HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
		Passage history			Egg	SIAT	SIAT	Egg	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT	
		Ferret number			F36/12[*1]	F35/15[*1]	F14/14[*1]	F20/14[*1]	F18/15[1]	F32/14[*1]	F23/15[*1]	F10/15[*1]	F30/14[*1]	F43/15[*1]	F12/15[*1]	F33/15[*1]	
		Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.2a	
REFERENCE VIRUSES																	
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	5120	640	160	640	80	640	320	640	160	640	80	320	
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT2	1280	640	640	640	160	640	320	640	320	640	160	320	
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT3	80	40	320	160	160	160	40	80	160	160	80	160	
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	640	80	160	320	80	640	80	80	160	640	40	80	
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT3	40	40	320	160	160	160	40	40	160	160	40	160	
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	320	160	160	320	80	640	80	80	160	320	40	160	
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT3	640	320	320	160	80	160	1280	160	80	160	80	160	
A/Hong Kong/146/2013		3C.2	2013-01-11	E6	2560	640	80	640	40	640	320	640	320	640	80	320	
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	40	<	160	80	40	40	<	40	160	160	80	80	[N158n/k, T160t/r] (-cho), S312N
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK2	80	80	160	160	80	80	40	80	160	320	80	160	Parent
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	40	80	80	80	40	40	40	<	320	320	160	320	N96S, T160K (-CHO), L194P
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT5	80	80	160	160	80	80	<	80	320	320	160	640	R142K, N158H (-CHO), Q197R
TEST VIRUSES																	
A/Kyrgyzstan/15/2017		3C.2a	2016-11-28	MDCKx/SIAT1	40	40	160	80	40	40	<	40	80	160	40	80	N121K, N122D (-CHO), S144K, T160t/k (-cho), G240R, S262N
A/Kyrgyzstan/11/2017		3C.2a	2016-11-30	MDCKx/SIAT1	40	40	160	80	40	40	<	40	160	160	40	80	N121K, N122D (-CHO), S144K, T160t/k (-cho), S262N
A/Hungary/195/2016		3C.2a	2016-12-15	MDCK2/SIAT1	40	40	80	80	40	40	<	40	80	160	40	80	N121K, S144K, T160t/k (-cho)
A/Hungary/203/2016		3C.2a	2016-12-19	MDCK2/SIAT1	<	<	80	40	<	<	<	<	40	80	<	<	N121K, S144K, T160t/k (-cho)
A/Sweden/152/2016		3C.2a1	2016-12-17	MDCK1/SIAT1	40	<	80	80	<	<	<	<	80	160	40	40	N121K, T135K (-CHO), N171K
A/Sweden/154/2016		3C.2a1	2016-12-17	MDCK1/SIAT1	40	<	80	80	40	40	<	80	160	160	80	40	N121K, T135K (-CHO), N171K
A/Sweden/155/2016		3C.2a1	2016-12-17	MDCK1/SIAT1	40	<	40	40	<	<	<	<	80	80	<	<	N121K, T135K (-CHO), N171K
A/Sweden/159/2016		3C.2a1	2016-12-18	MDCK1/SIAT1	<	<	80	40	<	<	<	<	80	80	<	40	N121K, T135K (-CHO), N171K
A/Khmelnitsky/2245/2016		3C.2a1	2016-12-16	MDCK1/SIAT2	40	<	80	40	40	40	<	40	80	80	40	40	N121K, T135K (-CHO), N171K, S279Y
A/Hungary/198/2016		3C.2a1	2016-12-19	MDCK2/SIAT1	40	<	80	40	<	<	<	<	40	80	<	40	S91N, N121K, I140M, T160t/k (-cho), N171K
A/Hungary/197/2016		3C.2a1	2016-12-19	MDCK2/SIAT1	<	<	80	40	<	<	<	<	<	40	<	<	N121K, T135K (-CHO), N171K, E280G
A/Austria/960579/2016		3C.2a1	2016-12-20	C1/SIAT2	<	<	80	40	<	<	<	<	80	160	<	40	G78S, N121KR142G, N171K
A/Khmelnitsky/2479/2016		3C.2a1	2016-12-21	MDCK1/SIAT2	40	<	80	40	<	<	<	<	80	160	40	40	N121K, T135K (-CHO), T160t/k (-cho), N171K, S279Y
A/Bosnia & Herzegovina/40-G/2017		3C.2a1	2017-01-05	SIAT2	40	<	160	80	40	40	<	40	80	160	40	80	N121K, R142G, T160t/k (-ch0), N171K

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) [1] < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-02-07

Viruses	Other information	Passage history / Ferret number / Genetic group	Collection date	Passage history	A/Texas 50/12 Egg F36/12*1 3C.1	A/Samara 73/13 SIAT F35/15*1 3C.3	A/Stock 6/14 SIAT F14/14*1 3C.3a	A/Stock 6/14 Egg F20/14*1 3C.3a	A/Switz 9715293/13 SIAT F18/15[1] 3C.3a	A/Switz 9715293/13 Egg F32/14*1 3C.3a	A/Neth 525/14 SIAT F23/15*1 3C.3b	A/HK 146/13 Egg F10/15*1 3C.2	A/HK 5738/14 MDCK F30/14*1 3C.2a	A/HK 4801/14 MDCK F43/15*1 3C.2a	A/HK 4801/14 Egg F12/15*1 3C.2a	A/Georgia 532/15 SIAT F33/15*1 3C.2a	HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
					Haemagglutination inhibition titre												
					Post-infection ferret antisera												
REFERENCE VIRUSES																	
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	5120	640	160	640	80	640	320	640	160	640	80	320	
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT2	1280	640	640	640	160	640	320	640	320	640	160	320	
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT3	80	40	320	160	160	160	40	80	160	160	80	160	
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	640	80	160	320	80	640	80	80	160	640	40	80	
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT3	40	40	320	160	160	160	40	40	160	160	40	160	
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	320	160	160	320	80	640	40	80	160	320	40	160	
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT3	640	320	320	160	80	160	1280	160	80	160	80	160	
A/Hong Kong/146/2013		3C.2	2013-01-11	E6	2560	640	80	640	40	640	320	640	320	640	80	320	
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	40	<	160	80	40	40	<	40	160	160	80	80	[N158n/k, T160t/r] (-cho), S312N
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK2	80	80	160	160	80	80	40	80	160	320	80	160	Parent
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	40	80	80	80	40	40	40	<	320	640	160	640	N96S, T160K (-CHO), L194P
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT5	80	80	160	160	80	80	<	80	320	320	160	640	R142K, N158H (-CHO), Q197R
TEST VIRUSES																	
A/Stockholm/41/2016		P9	2016-11-10	MDCK1/SIAT1	80	40	160	80	40	40	<	40	160	320	80	160	
A/Stockholm/46/2016		P9	2016-11-16	MDCK1/SIAT1	40	<	160	80	40	<	<	40	80	160	40	80	
A/Netherlands/3356/2016		P9	2016-11-24	SIAT2/MDCK1/SIAT1	40	<	160	80	80	<	<	80	160	160	80	80	
A/Kyrgyzstan/21/2016		3C.2a	2016-11-29	MDCKx/SIAT1	40	40	160	80	40	40	<	40	160	160	80	80	N121K, N122D (-CHO), S144K, T160t/k (-cho), S262N
A/St Petersburg/334/2016		3C.2a1	2016-12-02	MDCK1/SIAT2	<	<	<	40	<	40	<	<	80	160	40	80	D53N, N171K, S198P
A/Stockholm/52/2016		P9	2016-12-02	MDCK0/SIAT1	40	<	80	40	<	<	<	40	160	80	40	80	
A/Linkoping/5/2016		P9	2016-12-03	MDCK1/SIAT1	40	<	80	80	<	40	<	40	80	160	40	40	
A/Khabarovsk/1/2016		3C.2a	2016-12-06	MDCK2/SIAT2	40	<	80	40	40	<	<	40	80	160	40	80	N121K, S144K
A/Netherlands/3455/2016		P9	2016-12-10	SIAT2/MDCK1/SIAT1	80	40	320	160	80	40	40	160	160	320	160	160	
A/Hungary/192/2016		3C.2a1	2016-12-14	MDCK2/SIAT1	80	40	160	160	80	40	<	40	160	320	80	80	N121K, T135K (-CHO), N171K, E280G
A/Netherlands/3768/2016		P9	2016-12-20	SIAT2/MDCK1/SIAT1	80	40	160	80	160	40	<	80	160	160	80	80	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) [1] < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-02-10 + 2017-02-14

Viruses	Other information		Collection date	Passage history	Haemagglutination inhibition titre											
					Post-infection ferret antisera											
					A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/HK	A/HK	A/HK	A/Georgia	A/Oman	A/Nor
					50/12	73/13	6/14	6/14	9715293/13	9715293/13	5738/14	4801/14	4801/14	532/15	2585/16	4436/16
		Passage history			Egg	SIAT	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT	SIAT	SIAT
		Ferret number			F36/12*1	F35/15*1	F14/14*1	F20/14*1	F18/15[1]	F32/14*1	F30/14*1	F43/15*1	F12/15*1	F33/15*1	NIB F50/16*1	F03/17*1
		Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1
REFERENCE VIRUSES																
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	5120	640	160	640	80	1280	160	640	80	320	320	640
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT2	1280	640	640	640	160	640	640	1280	160	640	640	1280
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT3	80	80	320	320	160	160	160	320	80	160	320	320
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	640	160	160	320	80	640	160	640	40	80	320	320
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT2	80	40	320	160	160	160	160	160	80	160	160	320
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	320	160	160	320	80	640	160	320	80	160	320	320
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	80	40	160	160	80	80	160	320	160	160	320	320
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK2	160	80	320	160	160	80	160	640	160	160	320	320
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	40	80	80	80	80	40	320	640	160	640	640	320
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT5	160	80	320	160	80	80	320	640	160	1280	320	320
TEST VIRUSES																
A/Hong Kong/120/2017		P9	2017-01-04	MDCK1/SIAT1	40	<	160	80	<	<	80	160	40	40	160	160
A/Poland/106/2017		P9	2017-01-05	SIAT1	40	<	160	40	40	<	80	160	80	80	160	160
A/Niedersachsen/5/2017		P10	2017-01-05	C1/SIAT1	40	40	160	80	40	40	160	160	40	160	80	160
A/Israel/C-558/2017		P9	2017-01-09	C1/SIAT1	40	<	80	40	<	<	80	160	40	40	80	80
A/Israel/C-791/2017		P9	2017-01-13	C1/SIAT1	40	<	160	40	<	<	80	80	<	40	80	80
A/Sachsen-Anhalt/2/2017		P10	2017-01-16	C2/SIAT1	40	<	80	40	40	<	80	160	40	40	40	80
A/Kalamata GR/540/2017		P9	2017-01-19	SIAT2	40	40	320	160	160	40	160	320	80	80	320	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) [1] < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) with collection dates after 2016-08-31 - Summary

Viruses	Other information		Collection date	Passage history	A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Neth	A/HK	A/HK	A/HK	A/HK	A/Georgia	A/Oman	A/Nor
					Haemagglutination inhibition titre													
					Post-infection ferret antisera													
					50/12	73/13	6/14	6/14	9715293/13	9715293/13	525/14	146/13	5738/14	4801/14	4801/14	532/15	2585/16	4436/16
		Passage history			Egg	SIAT	SIAT	Egg	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT	SIAT	SIAT
		Ferret number			F36/12[*1]	F35/15[*1]	F14/14[*1]	F20/14[*1]	F18/15[1]	F32/14[*1]	F23/15[*1]	F10/15[*1]	F30/14[*1]	F43/15[*1]	F12/15[*1]	F33/15[*1]	NIB F50/16[*1]	F03/17[*1]
		Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1
REFERENCE VIRUSES																		
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	5120	640	160	640	80	640	320	640	160	640	80	320	320	640
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT2	1280	640	640	640	160	640	320	640	320	640	160	320	640	1280
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT3	80	40	320	160	160	160	40	80	160	160	80	160	320	320
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	640	80	160	320	80	640	80	80	160	640	40	80	320	320
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT3	40	40	320	160	160	160	40	40	160	160	40	160	160	320
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	320	160	160	320	80	640	40	80	160	320	40	160	320	320
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT3	640	320	320	160	80	160	1280	160	80	160	80	160	ND	ND
A/Hong Kong/146/2013		3C.2	2013-01-11	E6	2560	640	80	640	40	640	320	640	320	640	80	320	ND	ND
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	40	<	160	80	40	40	<	40	160	160	80	80	320	320
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK2	80	80	160	160	80	80	40	80	160	320	80	160	320	320
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	40	80	80	80	40	40	40	<	320	640	160	640	640	320
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT5	80	80	160	160	80	80	<	80	320	320	160	640	320	320
TEST VIRUSES																		
Number of viruses tested					56	56	56	56	56	56	49	49	56	56	56	56	7*	7*
No with titre reduction ≤2-fold							21	5	14				53	29	21			
%							37.5	8.9	25.0				94.6	51.8	37.5			
No with titre reduction =4-fold							26	30	27			1	2	25	27	6		
%							46.4	53.6	48.2			2.0	3.6	44.6	48.2	10.7		
No with titre reduction ≥8-fold					56	56	9	21	15	56	49	48	1	2	8	50		
%					100.0	100.0	16.1	37.5	26.8	100.0	100.0	98.0	1.8	3.6	14.3	89.3		
Number of clade 3C.2a viruses tested					18	18	18	18	18	18	18	18	18	18	18	18		
No with titre reduction ≤2-fold							5		1				17	10	3			
%							27.8		5.5				94.4	55.6	16.7			
No with titre reduction =4-fold							9	11	12				1	8	11	3		
%							50.0	61.1	66.7				5.6	44.4	61.1	16.7		
No with titre reduction ≥8-fold					18	18	4	7	5	18	18	18			4	15		
%					100.0	100.0	22.2	38.9	27.8	100.0	100.0	100.0			22.2	83.3		
Number of clade 3C.2a1 viruses tested					23	23	23	23	23	23	23	23	23	23	23	23		
No with titre reduction ≤2-fold							5	3	8				21	12	12			
%							21.8	13.0	34.8				91.4	52.2	52.2			
No with titre reduction =4-fold							13	11	10				1	10	8	1		
%							56.4	47.9	43.5				4.3	43.5	34.8	4.3		
No with titre reduction ≥8-fold					23	23	5	9	5	23	23	23	1	1	3	22		
%					100.0	100.0	21.8	39.1	21.7	100.0	100.0	100	4.3	4.3	13.0	95.7		

* Homologous HI titres not available

Vaccine

Reference virus results are taken from individual tables as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Table 9-1 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-01-18

Viruses	Passage history / Ferret number / Genetic group	Collection Date	Passage History	Neutralisation titre[1] Post-infection ferret antisera: A/HK 4801/14 Egg NIB F53/16 3C.2a		A/HK 7295/14 MDCK F02/15 3C.2a		A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a		A/Oman 2585/16 SIAT NIB F50/16 3C.2a1		A/Norway 3806/16 Egg NIB F45/16 3C.2a1		HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E7	640	780	640	601	320	272	320	465	320	431	N96S, T160K (-CHO), L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	61	640	570	160	144	320	311	80	71	none
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT2	80	118	1280	1768	2560	2384	1280	1880	160	168	N121K, S144K, S198P
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT2	160	203	320	386	640	508	2560	2115	80	119	N121K, N171K
A/Norway/3806/2016	3C.2a1	2016-06-13	E7	640	733	640	512	320	365	640	727	640	647	N121K, I140M, T160K (-CHO), N171K, L194P
TEST VIRUSES														
A/Saudi Arabia/191934/2016	3C.2a	2016-09-09	SIAT1	80	80	1280	1259	1280	1103	1280	1482	320	240	N121K, N122D (-CHO), S144K, S262N
A/Saudi Arabia/192245/2016	3C.2a	2016-09-10	SIAT1	320	256	640	845	1280	1114	1280	1324	160	207	N121K, S144K
A/Saudi Arabia/192516/2016	3C.2a	2016-09-12	SIAT1	160	189	640	793	640	774	1280	989	320	311	N121K, N122D (-CHO), S144K, S262N
A/Saudi Arabia/193221/2016	3C.2a	2016-09-18	SIAT1	320	256	1280	1408	1280	1240	1280	1629	320	256	R33Q, G49S, Y94F, N121R, S144K, A163S
A/Norway/4293/2016	3C.2a	2016-10-22	SIAT1	160	160	640	523	320	384	1280	1000	80	115	T131K, R142K, R261Q
A/Norway/4289/2016	3C.2a	2016-10-24	SIAT1/SIAT1	320	280	1280	1067	1280	1629	1280	1600	320	245	N121K, S144K
A/Paris/2907/2016	3C.2a	2016-10-29	MDCK1/SIAT1	20	20	320	247	640	737	640	838	80	78	N121K, T135A (-CHO), S144K, R261Q
A/Norway/4396/2016	3C.2a	2016-10-31	SIAT1	320	320	1280	1280	1280	1898	2560	1941	320	260	N121K, S144K
A/Norway/4437/2016	3C.2a	2016-11-02	SIAT1	640	640	1280	987	1280	1506	1280	1807	320	305	N121K, S144K, S219Y
A/England/64540250/2016	3C.2a	2016-11-07	SIAT1/SIAT1	320	246	1280	1255	2560	3107	1280	1849	320	320	N121K, S144K
A/Bretagne/2940/2016	3C.2a	2016-11-10	MDCK1/SIAT1	320	297	640	900	2560	2176	1280	1720	160	230	I58V, N121K, S144K, S219Y
A/England/64680165/2016	3C.2a	2016-11-14	MDCK1/SIAT1	320	444	640	669	1280	1896	1280	1239	160	160	N31D, N121K, S124R (-CHO), G129R, S144K, S219Y
A/Saudi Arabia/192078/2016	3C.2a1	2016-09-09	SIAT1	320	340	1280	1042	640	782	2560	1958	320	414	K92R, N121K, N171K, H311Q
A/Saudi Arabia/192254/2016	3C.2a1	2016-09-10	SIAT1	160	153	1280	1097	640	807	1280	1266	160	217	N31S, N171K
A/Saudi Arabia/192281/2016	3C.2a1	2016-09-11	SIAT1	160	204	640	880	640	773	1280	1898	320	263	K92R, N121K, N171K, H311Q
A/Norway/4255/2016	3C.2a1	2016-10-12	SIAT1/SIAT1	160	131	640	949	640	557	1280	1813	160	194	N121K, I140M, N171K
A/Norway/4294/2016	3C.2a1	2016-10-20	SIAT1/SIAT1	160	150	640	864	640	606	1280	1792	160	198	N121K, I140M, N171K
A/England/64460094/2016	3C.2a1	2016-10-28	SIAT1/SIAT1	80	99	640	924	640	517	1280	1440	160	136	N121K, I140M, N171K
A/Norway/4436/2016	3C.2a1	2016-11-03	SIAT1	160	120	1280	968	320	469	1280	1554	160	154	N121K, I140M, N171K
A/Bretagne/2934/2016	3C.2a1	2016-11-07	MDCK2/SIAT1	20	20	160	224	160	216	640	720	80	106	N121K, R142G, N171K
A/England/64560781/2016	3C.2a1	2016-11-08	MDCK1/SIAT1	80	70	1280	1101	640	550	2560	1956	160	203	N121K, I140M, N171K
A/England/64580216/2016	3C.2a1	2016-11-08	SIAT1/SIAT1	80	107	640	562	320	408	1280	1147	80	88	N121K, I140M, N171K, Q211K
A/Paris/2960/2016	3C.2a1	2016-11-14	MDCK2/SIAT1	20	20	80	74	80	74	160	236	40	40	R142G, N171K
A/England/64680170/2016	3C.2a1	2016-11-15	MDCK1/SIAT1	20	20	320	300	320	280	640	900	80	90	N121K, R142G, N171K
				Vaccine										

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

In phylogenetic trees

Table 9-2 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-01-23

Viruses	Passage history / Ferret number / Genetic group	Collection Date	Passage History	Neutralisation titre[1] Post-infection ferret antisera: A/HK 4801/14 Egg NIB F53/16 3C.2a		A/HK 7295/14 MDCK F02/15 3C.2a		A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a		A/Oman 2585/16 SIAT NIB F50/16 3C.2a1		A/Norway 3806/16 Egg NIB F45/16 3C.2a1		HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E7	640	780	640	601	320	272	320	465	320	431	N96S, T160K (-CHO), L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	61	640	570	160	144	320	311	80	71	none
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT2	80	118	1280	1768	2560	2384	1280	1880	160	168	N121K, S144K, S198P
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT2	160	203	320	386	640	508	2560	2115	80	119	N121K, N171K
A/Norway/3806/2016	3C.2a1	2016-06-13	E7	640	733	640	512	320	365	640	727	640	647	N121K, I140M, T160K (-CHO), N171K, L194P
TEST VIRUSES														
A/Jordan/30308/2016	3C.2a	2016-11-01	SIAT1	160	196	640	917	1280	1493	1280	1255	320	320	N121K, N122D (-CHO), S144K, P169L
A/Jordan/20655/2016	3C.2a	2016-11-14	SIAT1	160	156	1280	1067	1280	1431	1280	1237	320	290	N121K, S144K, V323(V/I)
A/Norway/4634/2016	3C.2a	2016-11-20	SIAT1/SIAT1	320	354	320	452	320	289	1280	1338	80	114	T131K, R142K, R261Q
A/Kyrgyzstan/68/2016	3C.2a	2016-11-21	SIAT1	160	143	1280	1075	1280	1090	1280	1387	320	336	N121K, N122D (-CHO), S144K, S262N
A/Jordan/4026/2016	3C.2a	2016-11-21	SIAT1	160	142	1280	1083	1280	1319	1280	1280	320	320	N121K, S144K, D291N
A/Norway/4676/2016	3C.2a	2016-11-23	SIAT1/SIAT1	160	156	640	600	320	393	1280	1344	80	77	T131K, R142K, R261Q
A/Norway/4693/2016	3C.2a	2016-11-24	SIAT1/SIAT1	160	136	640	611	320	472	1280	1280	80	72	T131K, R142K, R261Q
A/Kyrgyzstan/2/2016	3C.2a	2016-11-25	SIAT1	80	107	640	912	640	729	640	853	160	144	N121K, N122D (-CHO), S144K, S262N
A/Kyrgyzstan/108/2016	3C.2a	2016-11-28	SIAT1	320	320	1280	1486	1280	1047	1280	1792	160	152	N121K, N122D (-CHO), S144K, S262N
A/Kyrgyzstan/117/2016	3C.2a	2016-11-28	SIAT1	160	192	1280	1138	1280	1376	1280	1813	160	218	N121K, N122D (-CHO), S144K, S262N
A/Kyrgyzstan/130/2016	3C.2a	2016-11-28	SIAT1	160	179	1280	1280	1280	1417	1280	1111	160	206	N121K, N122D (-CHO), S144K, G240R, S262N, P293(P/L)
A/Kyrgyzstan/158/2016	3C.2a	2016-11-29	SIAT1	40	56	640	689	640	731	1280	1066	160	156	D53G, N121K, S144K, D291N
A/Norway/4766/2016	3C.2a	2016-11-29	SIAT1/SIAT1	160	128	640	567	1280	1197	1280	1013	160	126	N121K, S144K, S219Y
A/Norway/4784/2016	3C.2a	2016-11-30	SIAT1/SIAT1	160	139	320	286	320	262	640	567	80	91	G78D, T131K, R142K, R261Q
A/Norway/4790/2016	3C.2a	2016-11-30	SIAT1/SIAT1	160	124	640	564	160	236	640	808	80	67	T131K, R142K, V237I, R261Q
A/Kyrgyzstan/164/2016	3C.2a	2016-12-01	SIAT1	320	240	640	930	640	800	640	823	80	115	N121K, S144K
A/Norway/4858/2016	3C.2a	2016-12-02	SIAT1/SIAT1	160	160	320	427	320	394	640	929	160	174	T131K, R261Q
A/Norway/4795/2016	3C.2a1	2016-11-26	SIAT1/SIAT1	20	20	640	564	320	475	1280	1254	160	128	N121K, I140M, N171K
				Vaccine										

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-3 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-01-25

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre[1] Post-infection ferret antisera A/HK 4801/14 Egg NIB F53/16 3C.2a		A/HK 7295/14 MDCK F02/15 3C.2a		A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a		A/Oman 2585/16 SIAT NIB F50/16 3C.2a1		A/Norway 3806/16 Egg NIB F45/16 3C.2a1		HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E7	640	780	640	601	320	272	320	465	320	431	N96S, T160K (-CHO), L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	61	640	570	160	144	320	311	80	71	none
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT2	80	118	1280	1768	2560	2384	1280	1880	160	168	N121K, S144K, S198P
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT2	160	203	320	386	640	508	2560	2115	80	119	N121K, N171K
A/Norway/3806/2016	3C.2a1	2016-06-13	E7	640	733	640	512	320	365	640	727	640	647	N121K, I140M, T160K (-CHO), N171K, L194P
TEST VIRUSES														
A/Valladolid/287/2016	3C.2a	2016-11-18	SIAT1	160	220	1280	1240	1280	1750	2560	1978	320	320	N121K, N122D (-CHO), S144K, S262N
A/Albania/7011/2016	3C.2a	2016-12-02	SIAT1/SIAT1	320	462	2560	2071	1280	1670	2560	2074	640	563	N121K, N122D (-CHO), S144K, S262N
A/Norway/4963/2016	3C.2a	2016-12-10	SIAT1/SIAT1	1280	1006	1280	1280	1280	1845	2560	2106	640	775	N121K, S144K, S219Y
A/Albania/6998/2016	3C.2a1	2016-12-02	SIAT1/SIAT1	40	46	1280	1156	1280	1097	2560	3129	320	450	N121K, I140M, N171K
A/Norway/4883/2016	3C.2a1	2016-12-08	SIAT1/SIAT1	160	175	1280	1792	1280	1113	1280	1880	640	640	S47T, G78S, N171K
				Vaccine										

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

In phylogenetic trees

Table 9-4 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-01-30

Neutralisation titre[1] — Post-infection ferret antisera

Viruses	Passage history / Ferret number / Genetic group	Collection Date	Passage History	A/HK 4801/14 Egg NIB F53/16 3C.2a 2-fold	A/HK 4801/14 Read	A/HK 7295/14 MDCK F02/15 3C.2a 2-fold	A/HK 7295/14 Read	A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a 2-fold	A/Cote d'Ivoire 544/16 Read	A/Oman 2585/16 SIAT NIB F50/16 3C.2a1 2-fold	A/Oman 2585/16 Read	A/Norway 4436/16 SIAT F03/17 3C.2a1 2-fold	A/Norway 4436/16 Read	A/Norway 3806/16 Egg NIB F45/16 3C.2a1 2-fold	A/Norway 3806/16 Read	HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
REFERENCE VIRUSES																
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E7	640	780	640	601	320	272	320	465	320	320	320	431	N96S, T160K (-CHO), L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	61	640	570	160	144	320	311	320	251	80	71	none
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT2	80	118	1280	1768	2560	2384	1280	1880	1280	993	160	168	N121K, S144K, S198P
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT2	160	203	640	386	640	508	2560	2115	1280	993	80	119	N121K, N171K
A/Norway/4436/2016	3C.2a1	2016-11-03	SIAT1	160	122	320	775	640	481	1280	1406	1280	992	160	120	N121K, I140M, N171K
A/Norway/3806/2016	3C.2a1	2016-06-13	E7	640	733	640	512	320	365	640	727	640	506	640	647	N121K, I140M, T160K (-CHO), N171K, L194P
TEST VIRUSES																
A/Toulon/2291/2016	3C.2a	2016-11-16	MDCK2/SIAT1	40	59	640	578	640	731	640	807	640	742	80	115	N121K, S144K
A/Lyon/2313/2016	3C.2a	2016-11-22	MDCK2/SIAT1	160	122	320	404	320	256	640	830	1280	960	40	55	T131K, R142K, R261Q
A/Norway/4671/2016	3C.2a	2016-11-25	SIAT1	80	70	640	800	1280	1772	1280	1371	1280	1253	160	160	N121K, S144K
A/Estonia/103974/2016	3C.2a	2016-12-21	SIAT1/SIAT1	40	40	320	301	640	576	640	529	640	480	80	72	N121K, S144K, V309I
A/Albania/7010/2016	P9	2016-11-30	SIAT1	40	40	640	560	1280	985	640	775	640	786	160	144	
A/Norway/4659/2016	3C.2a1	2016-11-21	SIAT1	40	43	640	510	320	457	1280	1257	1280	1114	80	77	N121K, I140M, N171K
A/Lyon/2315/2016	3C.2a1	2016-11-21	MDCK2/SIAT1	40	52	640	495	320	257	1280	1019	640	891	80	100	N121K, I140M, N171K
A/Bourgoin-Jallieu/2294/2016	3C.2a1	2016-11-22	MDCK2/SIAT1	40	56	640	516	320	298	1280	1162	640	924	320	302	N121K, N171K
A/Lyon/CHU22.11.R96/2016	3C.2a1	2016-11-22	MDCK2/SIAT1	40	42	160	154	320	240	640	587	640	640	40	48	N171K
A/Norway/4669/2016	3C.2a1	2016-11-25	SIAT1	20	20	320	434	320	298	640	943	640	835	80	67	N121K, R142G, N171K
A/Spain/110153/2016	3C.2a1	2016-12-12	SIAT1	40	50	640	800	320	440	1280	1783	1280	1583	160	206	N121K, I140M, N171K, V323I
A/Estonia/103933/2016	3C.2a1	2016-12-21	SIAT1/SIAT1	40	40	320	352	160	172	640	918	640	620	80	60	R142G, N171K
A/Estonia/103960/2016	3C.2a1	2016-12-21	SIAT1/SIAT1	<	10	320	337	160	200	640	807	640	520	40	51	N121K, R142G, N171K
A/Estonia/103969/2016	3C.2a1	2016-12-22	SIAT1/SIAT1	40	44	320	275	160	181	640	738	640	493	80	66	N121K, I140M, N171K
A/Estonia/103980/2016	3C.2a1	2016-12-22	SIAT1/SIAT1	40	40	320	427	320	280	1280	1141	640	747	80	80	R142G, N171K
				Vaccine												

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

In phylogenetic trees

Table 9-5 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-02-01

Viruses	Passage history / Ferret number / Genetic group	Collection Date	Passage History	A/HK 4801/14 Egg NIB F53/16 3C.2a		A/HK 7295/14 MDCK F02/15 3C.2a		A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a		A/Oman 2585/16 SIAT NIB F50/16 3C.2a1		A/Norway 4436/16 SIAT F03/17 3C.2a1		HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				Neutralisation titre[1] – Post-infection ferret antisera										
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E7	640	780	640	601	320	272	320	465	320	320	N96S, T160K (-CHO), L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	61	640	570	160	144	320	311	320	251	none
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT2	80	118	1280	1768	2560	2384	1280	1880	1280	993	N121K, S144K, S198P
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT2	160	203	320	386	640	508	2560	2115	1280	993	N121K, N171K
A/Norway/4436/2016	3C.2a1	2016-11-03	SIAT1	160	122	640	775	640	481	1280	1406	1280	992	N121K, I140M, N171K
TEST VIRUSES														
A/Spain/108862/2016	3C.2a	2016-12-02	SIAT1	160	120	1280	1021	1280	1078	1280	1180	1280	1193	N121K, N122D (-CHO), S144K, S262N
A/Estonia/103732/2016	3C.2a	2016-12-12	SIAT1/SIAT1	80	78	1280	987	640	846	1280	1411	1280	1038	N121K, S144K
A/Iran/68081/2016	3C.2a	2016-12-14	MDCK1/SIAT1	80	89	1280	1088	1280	1213	1280	1164	1280	1157	A106T, N121K, S144K, I236V
A/Switzerland/19430223/2016	3C.2a	2016-12-16	SIAT1	160	151	320	398	320	307	640	815	640	630	T131K, R142K, R261Q
A/Switzerland/19430147/2016	3C.2a	2016-12-20	SIAT1	160	217	1280	1019	640	826	1280	971	1280	1280	N121K, N122D (-CHO), S144K, S262N
A/Iceland/15954/2016	3C.2a1	2016-09-14	SIAT2	80	74	640	724	320	459	1280	1192	1280	960	S47T, G78S, N171K, K310R
A/Iceland/16073/2016	3C.2a1	2016-09-15	SIAT2	160	144	640	807	320	400	1280	1463	1280	989	S47T, G78S, N171K, K310R
A/Iceland/16097/2016	3C.2a1	2016-09-15	SIAT2	160	229	640	950	640	576	1280	1657	1280	1242	S47T, G78S, N171K, K310R
A/Champagne Ardenne/3131/2016	3C.2a1	2016-12-05	MDCK1/SIAT1	40	40	640	526	320	320	640	809	640	929	N121E, I140M, N171K, V323I
A/Spain/109626/2016	3C.2a1	2016-12-05	SIAT1	40	47	640	747	640	576	1280	1219	1280	981	K27R, N171K
A/Nord Pas de Calais/3195/2016	3C.2a1	2016-12-06	MDCK1/SIAT1	80	61	640	574	320	378	1280	1032	640	844	D104N, N171K
A/Estonia/103677/2016	3C.2a1	2016-12-07	SIAT1/SIAT1	40	46	640	501	320	282	640	776	640	704	N121K, I140M, N171K
A/Paris/3199/2016	3C.2a1	2016-12-08	MDCK1/SIAT1	320	309	640	747	640	501	1280	1779	640	731	I34V, N171K
A/Estonia/103729/2016	3C.2a1	2016-12-12	SIAT1/SIAT1	80	74	1280	960	640	533	1280	1707	1280	1055	R142G, N171K
A/Estonia/103759/2016	3C.2a1	2016-12-12	SIAT1/SIAT1	80	72	640	640	320	369	1280	1254	640	795	R142G, N171K
A/Bretagne/3243/2016	3C.2a1	2016-12-12	MDCK1/SIAT1	80	63	640	569	160	213	1280	960	640	557	S47T, D53N, G78S, N171K
A/Paris/3248/2016	3C.2a1	2016-12-12	MDCK1/SIAT1	40	50	640	576	320	320	1280	1227	1280	960	N171K
A/Estonia/103796/2016	3C.2a1	2016-12-14	SIAT1/SIAT1	80	61	640	526	320	308	1280	1008	640	683	N121K, R142G, N171K
A/Iran/67534/2016	3C.2a1	2016-12-14	MDCK1/SIAT1	80	80	640	597	320	243	1280	1138	640	618	I3S, D53E, Q75H, K92R, N121K, N171K, S262N, H311Q
A/Iran/67441/2016	3C.2a1	2016-12-14	MDCK1/SIAT1	80	102	640	853	640	486	1280	1569	1280	1222	N121K, I140M, N171K
A/Iran/67702/2016	3C.2a1	2016-12-14	MDCK1/SIAT1	80	72	640	873	320	390	1280	1166	1280	1006	N171K
A/Iran/67829/2016	3C.2a1	2016-12-14	MDCK1/SIAT1	80	60	640	626	320	250	1280	1020	640	547	D53E, Q75H, K92R, A106T, N121K, N171K, H311Q
A/Iran/67362/2016	3C.2a1	2016-12-14	MDCK1/SIAT1	40	54	320	464	160	205	640	740	640	518	R33Q, N121K, N171K
A/Switzerland/19429699/2016	3C.2a1	2016-12-20	SIAT1	320	280	1280	1029	1280	1040	1280	1463	1280	1250	N171K
				Vaccine										

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

In phylogenetic trees

Table 9-6 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-02-06

Viruses	Passage history / Ferret number / Genetic group	Collection Date	Passage History	Neutralisation titre[1] Post-infection ferret antisera: A/HK 4801/14 Egg NIB F53/16 3C.2a		A/HK 7295/14 MDCK F02/15 3C.2a		A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a		A/Oman 2585/16 SIAT NIB F50/16 3C.2a1		A/Norway 4436/16 SIAT F03/17 3C.2a1		HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E7	640	780	640	601	320	272	320	465	320	320	N96S, T160K (-CHO), L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	61	640	570	160	144	320	311	320	251	none
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT2	80	118	1280	1768	2560	2384	1280	1880	1280	993	N121K, S144K, S198P
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT2	160	203	320	386	640	508	2560	2115	1280	993	N121K, N171K
A/Norway/4436/2016	3C.2a1	2016-11-03	SIAT1	160	122	640	775	640	481	1280	1406	1280	992	N121K, I140M, N171K
TEST VIRUSES														
A/La Rioja/2992/2016	3c.2a	2016-11-15	SIAT1/SIAT1	40	40	640	640	1280	996	1280	1030	640	891	D53N, N121K, S144K
A/Madrid/2920/2016	3c.2a	2016-11-16	SIAT1/SIAT1	80	92	320	395	160	172	640	800	640	931	G78D, T131K, R142K, R261Q
A/La Rioja/2994/2016	3c.2a	2016-11-17	SIAT1/SIAT1	40	20	640	630	1280	980	1280	1203	640	867	D53N, N121K, S144K
A/Asturias/3088/2016	3c.2a	2016-11-17	SIAT1/SIAT1	40	58	640	604	1280	1440	1280	1510	1280	1355	N121K, N122D (-CHO), S144K, S262N
A/Switzerland/19260442/2016	3c.2a	2016-12-03	SIAT1	80	95	640	581	2560	2211	1280	1280	1280	1216	I58V, N121K, S144K, S219Y
A/Switzerland/19347103/2016	3c.2a	2016-12-06	SIAT1	160	142	320	384	320	346	640	877	640	924	T131K, R142K, R261Q
A/Switzerland/18821514/2016	3c.2a1	2016-10-20	SIAT1	80	84	640	837	640	896	1280	1728	2560	1963	D77E, N121K, R142G, N171K
A/La/Rioja/2995/2016	3c.2a1	2016-11-14	SIAT1/SIAT1	40	58	640	507	640	544	1280	1152	640	832	K92R, N121K, N171K, H311Q
A/Madrid/2921/2016	3c.2a1	2016-11-15	SIAT1/SIAT1	80	76	640	640	640	775	2560	2194	1280	1177	R33Q, N171K
A/La Rioja/2998/2016	3c.2a1	2016-11-16	SIAT1/SIAT1	40	40	160	146	160	148	640	517	320	416	K92R, N121K, N171K, H311Q
A/La Rioja/2997/2016	3c.2a1	2016-11-18	SIAT1/SIAT1	40	53	640	860	640	800	1280	1440	1280	1093	K92R, N121K, N171K, H311Q
A/La Rioja/3000/2016	3c.2a1	2016-11-18	SIAT1/SIAT1	80	87	640	640	640	640	1280	1259	1280	1166	K92R, N121K, N171K, H311Q
A/Andalucia/3039/2016	3c.2a1	2016-11-18	SIAT1/SIAT1	80	80	640	940	640	597	2560	1978	1280	1829	N171K, R261Q
A/La Rioja/2999/2016	3c.2a1	2016-11-22	SIAT1/SIAT1	40	42	320	469	320	469	1280	987	640	907	K92R, N121K, N171K, H311Q
A/Madrid/3014/2016	3c.2a1	2016-11-28	SIAT1/SIAT1	640	590	640	782	640	594	1280	1600	1280	1248	K92R, N121K, N171K, H311Q

Vaccine

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

In phylogenetic trees

Table 9-7. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-02-08

Viruses	Passage history / Ferret number / Genetic group	Collection Date	Passage History	A/HK 4801/14 Egg NIB F53/16 3C.2a 2-fold	Read	A/HK 7295/14 MDCK F02/15 3C.2a 2-fold	Read	A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a 2-fold	Read	A/Oman 2585/16 SIAT NIB F50/16 3C.2a1 2-fold	Read	A/Norway 4436/16 SIAT F03/17 3C.2a1 2-fold	Read	HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				Neutralisation titre[1] – Post-infection ferret antisera										
REFERENCE VIRUSES														
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E7	640	780	640	601	320	272	320	465	320	320	N96S, T160K (-CHO), L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	61	640	570	160	144	320	311	320	251	none
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT2	80	118	1280	1768	2560	2384	1280	1880	1280	993	N121K, S144K, S198P
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT2	160	203	320	386	640	508	2560	2115	1280	993	N121K, N171K
A/Norway/4436/2016	3C.2a1	2016-04-06	SIAT1	160	122	640	775	640	481	1280	1406	1280	992	N121K, I140M, N171K
TEST VIRUSES														
A/Ghana/DILI-16-1024/2016	3C.2a	2016-10-12	SIAT1	160	183	1280	1513	2560	2103	2560	1991	1280	1760	N121K, S144K
A/Oman/7679/2016	3C.2a	2016-11-05	SIAT1/SIAT2	320	260	1280	1536	2560	2086	2560	2393	1280	1772	V112I, N121K, S144K
A/Tunis/14115/2016	3C.2a	2016-11-30	SIAT1	320	320	1280	1633	1280	1536	1280	1600	1280	1259	G78S, N121K, N122D (-CHO), S144K, S262N
A/Caen/3385/2016	3C.2a	2016-12-10	MDCK1/SIAT1	320	269	1280	1375	2560	2207	2560	2188	1280	1472	I58V, N121K, S144K, S219Y
A/Caen/3389/2016	3C.2a	2016-12-10	MDCK1/SIAT1	640	566	1280	960	640	668	1280	1428	1280	1072	T131K, R142K, R261Q
A/Paris/3513/2016	3C.2a	2016-12-21	MDCK1/SIAT1	320	340	2560	2103	2560	2393	2560	3614	5120	3891	Y94H, N121K, I140T, S144K
A/Basse Normandie/3553/2016	3C.2a	2016-12-26	MDCK1/SIAT1	80	65	1280	985	1280	1600	2560	2062	1280	1440	N121K, S144K
A/Bourgogne/3561/2016	3C.2a	2016-12-26	MDCK1/SIAT1	320	249	640	861	1280	1614	1280	1663	1280	1109	I58V, N121K, S144K, S219Y
A/Friuli-Venezia Giulia/31/2016	3C.2a	2016-12-29	SIAT1/SIAT1	320	312	1280	1254	2560	2363	2560	2240	2560	2906	H17h/r, S54G, N121K, S144K, S219Y
A/Milano/78/2016	3C.2a1	2016-11-22	SIAT2/SIAT1	640	751	1280	1213	1280	1024	2560	2004	1280	1894	N171K, R261Q
A/Lyon/2308/2016	3C.2a1	2016-11-23	MDCK2/SIAT1	160	206	1280	1036	640	701	2560	2038	1280	1646	N31S, N171K
A/Lyon/2352/2016	3C.2a1	2016-11-29	MDCK2/SIAT1	80	63	320	397	640	500	1280	987	640	943	N171K
A/Lyon/2402/2016	3C.2a1	2016-12-02	MDCK2/SIAT1	40	53	160	231	160	154	640	504	640	528	N171K
A/Lyon/2406/2016	3C.2a1	2016-12-05	MDCK2/SIAT1	40	20	80	80	80	69	160	174	160	192	N171K
A/Padova/29/2016	3C.2a1	2016-12-06	SIAT2/SIAT1	40	45	1280	1126	1280	981	1280	1477	1280	1280	S47T, G78S, N171K
A/Pays de Loire/3483/2016	3C.2a1	2016-12-14	MDCK1/SIAT1	320	344	1280	1229	640	815	1280	1920	1280	1341	N171K, S262N, R269K
A/Bulgaria/1229/2016	3C.2a1	2016-12-20	SIAT1	640	560	320	441	320	427	1280	1037	640	720	N121K, I140M, P162T, N171K
A/Bulgaria/1233/2016	3C.2a1	2016-12-21	SIAT1	160	120	320	405	320	328	1280	1067	1280	1147	S9N, K92R, N121K, N171K
A/Austria/960845/2016	3C.2a1	2016-12-22	C1/SIAT1	40	20	320	391	320	426	1280	1363	640	848	N121K, T135K (-CHO), N171K
A/Friuli-Venezia Giulia/29/2016	3C.2a1	2016-12-23	SIAT1/SIAT1	160	120	1280	1120	640	807	2560	2560	1280	1544	N121K, I140M, N171K
A/Pays de Loire/3517/2016	3C.2a1	2016-12-26	MDCK1/SIAT1	160	144	1280	1162	640	791	2560	2413	1280	1457	N121K, R142G, N171K
A/Nord Pas de Calais/3584/2016	3C.2a1	2016-12-27	MDCK1/SIAT1	160	150	640	853	640	702	1280	1547	1280	1076	K92R, N121K, N171K
A/Bulgaria/1263/2016	3C.2a1	2016-12-28	SIAT1	80	114	1280	1078	640	896	2560	2150	1280	1463	S9N, K92R, N121K, N171K
A/Bretagne/3548/2016	3C.2a1	2016-12-28	MDCK1/SIAT1	160	189	1280	1170	640	756	2560	2110	1280	1829	S47T, G78S, N171K, R261Q
				Vaccine										

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

In phylogenetic trees

Table 9-8 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-02-13

Viruses	Passage history / Ferret number / Genetic group	Collection Date	Passage History	A/HK 4801/14 Egg NIB F53/16 3C.2a 2-fold	A/HK 4801/14 Read	A/HK 7295/14 MDCK F02/15 3C.2a 2-fold	A/HK 7295/14 Read	A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a 2-fold	A/Cote d'Ivoire 544/16 Read	A/Oman 2585/16 SIAT NIB F50/16 3C.2a1 2-fold	A/Oman 2585/16 Read	A/Norway 4436/16 SIAT F03/17 3C.2a1 2-fold	A/Norway 4436/16 Read	HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
REFERENCE VIRUSES														
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E7	640	780	640	601	320	272	320	465	320	320	N96S, T160K (-CHO), L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	61	640	570	160	144	320	311	320	251	none
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT2	80	118	1280	1768	2560	2384	1280	1880	1280	993	N121K, S144K, S198P
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT2	160	203	320	386	640	508	2560	2115	1280	993	N121K, N171K
A/Norway/4436/2016	3C.2a1	2016-04-06	SIAT1	160	122	640	775	640	481	1280	1406	1280	992	N121K, I140M, N171K
TEST VIRUSES														
A/Montenegro/64/2016	3C.2a	2016-12-07	Mx/SIAT1	<	0	160	141	320	301	320	320	320	384	I58V, N121K, S144K, D188G, S219Y
A/Montenegro/112/2016	3C.2a	2016-12-15	Mx/SIAT1	640	496	40	40	160	127	160	138	160	143	I58V, N121K, S144K, S219Y
A/Nordrhein-Westfalen/77/2016	3C.2a	2016-12-16	C1/SIAT1	160	171	640	756	2560	2117	1280	1871	1280	1422	N121K, S144K, S219Y
A/Montenegro/142/2016	3C.2a	2016-12-19	Mx/SIAT1	320	365	80	105	160	169	320	267	320	292	I58V, N121K, S144K, D188G, S219Y
A/Latvia/01-000239/2017	3C.2a	2017-01-01	C1/SIAT1	<	10	320	335	640	555	640	539	640	619	N121K, S144K, T160t/k (-cho)
A/Slovenia/96/2017	3C.2a	2017-01-04	Mx/SIAT1	80	114	640	594	640	853	1280	1200	1280	1280	T131K, R142K, R261Q
A/Latvia/01-010093/2017	3C.2a	2017-01-04	C1/SIAT1	40	40	160	233	160	153	640	678	640	564	G78D, T131K, R142K, R261Q
A/Bosnia & Herzegovina/36-G/2017	3C.2a	2017-01-05	SIAT1	40	40	1280	1024	1280	1468	2560	1920	1280	1612	N121K, D122D (-CHO), S144K, S262N
A/Croatia/3586/2016	3C.2a1	2016-12-05	Mx/SIAT1	80	69	640	689	640	906	1280	1792	1280	1368	N171K
A/Croatia/3587/2016	3C.2a1	2016-12-05	Mx/SIAT1	80	62	320	377	640	725	1280	1138	640	800	N121K, R142G, M168I, N171K, Q197q/h
A/Kyiv/948/2016	3C.2a1	2016-12-13	SIAT1/SIAT1	<	0	640	533	320	295	1280	1135	640	910	R142G, N171K
A/Baden-Wurttemberg/125/2016	3C.2a1	2016-12-16	C1/SIAT1	40	20	640	582	640	524	1280	1321	640	863	N171K
A/Berlin/94/2016	3C.2a1	2016-12-16	C1/SIAT1	40	50	640	529	640	621	1280	1547	1280	1056	N121K, I140M, N171K, K189R, P289S
A/Kyiv/950/2016	3C.2a1	2016-12-16	SIAT2/SIAT1	<	0	320	377	640	582	1280	1222	640	788	N121K, T135K (-CHO), N171K, S279Y
A/Nordrhein-Westfalen/79/2016	3C.2a1	2016-12-19	C1/SIAT1	<	10	320	268	640	715	1280	1402	1280	1280	N121K, V130I, T135K (-CHO), N171K
A/Austria/960280/2016	3C.2a1	2016-12-20	C1/SIAT1	40	40	320	306	320	276	1280	1451	640	587	N121K, I140M, N171K, K189R
A/Khmelnitsky/2475/2016	3C.2a1	2016-12-21	M1/SIAT1	<	0	320	290	640	533	1280	1173	640	884	Q57K, R142G, N171K
A/Latvia/12-068351/2016	3C.2a1	2016-12-28	C2/SIAT1	40	20	80	103	80	102	320	440	640	526	N121K, I140M, (N158n/k, T160t/I (-cho)), N171K
A/Slovenia/91/2017	3C.2a1	2017-01-04	Mx/SIAT1	40	40	640	566	1280	960	1280	1629	1280	1109	N121K, T135K (-CHO), N171K
				Vaccine										

Neutralisation titre[1] — Post-infection ferret antisera

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

In phylogenetic trees

Table 9-9. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) - Summary

Viruses		Collection	Passage	Neutralisation titre[1]											
				Post-infection ferret antisera											
		Date	History	A/HK		A/HK		A/Cote d'Ivoire		A/Oman		A/Norway		A/Norway	
				4801/14		7295/14		544/16		2585/16		4436/16		3806/16	
	Passage history			Egg		MDCK		SIAT		SIAT		SIAT		Egg	
	Ferret number			NIB F53/16		F02/15		NIB F54/16		NIB F50/16		F03/17		NIB F45/16	
	Genetic group			3C.2a		3C.2a		3C.2a		3C.2a1		3C.2a1		3C.2a1	
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read
REFERENCE VIRUSES															
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E7	640	780	640	601	320	272	320	465	320	320	320	431
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	61	640	570	160	144	320	311	320	251	80	71
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT2	80	118	1280	1768	2560	2384	1280	1880	1280	993	160	168
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT2	160	203	640	386	640	508	2560	2115	1280	993	80	119
A/Norway/4436/2016	3C.2a1	2016-11-03	SIAT1	160	122	320	775	640	481	1280	1406	1280	992	160	120
A/Norway/3806/2016	3C.2a1	2016-06-13	E7	640	733	640	512	320	365	640	727	640	506	640	647
TEST VIRUS SUBCLADE	Number tested			64	79	64	79	64	79	64	79	32	64	36	25
				Number and % of viruses reacting ≤4-fold reduced compared to the homologous titre											
3C.2a	64			44	68.8%	62	96.9%	49	76.6%	61	95.3%	31	96.9%	25	69.4%
3C.2a1	79			21	26.6%	76	96.2%	40	50.6%	76	96.2%	63	98.4%	13	52.0%
	HA1 amino acid substitutions			Number ≤4-fold reduced compared to the homologous titre/number tested & % neutralised											
3C.2a	N121K, S144K			9/16	56.3%	16/16	100.0%	16/16	100.0%	16/16	100.0%	9/9	100.0%	7/10	70.0%
3C.2a	T131K, R142K, R261Q			10/13	76.9%	13/13	100.0%	2/13	15.4%	13/13	100.0%	7/7	100.0%	0/7	
3C.2a	N121K, S144K, S219Y			11/13	84.6%	11/13	84.6%	10/13	76.9%	10/13	76.9%	7/8	87.5%	5/5	100.0%
3C.2a	N121K, N122D (-CHO), S144K			13/16	81.3%	16/16	100.0%	16/16	100.0%	16/16	100.0%	5/5	100.0%	11/11	100.0%
3C.2a1	N171K			11/25	44.0%	24/25	96.0%	17/25	68.0%	24/25	96.0%	21/22	95.5%	3/4	75.0%
3C.2a1	N121K, N171K			5/14	35.7%	14/14	100.0%	8/14	57.1%	14/14	100.0%	12/12	100.0%	3/3	100.0%
3C.2a1	R142G, N171K			0/7		6/7	85.7%	2/7	28.6%	6/7	85.7%	6/6	100.0%	0/3	
3C.2a1	N121K, T135K (-CHO), N171K			0/4		4/4	100.0%	3/4	75.0%	4/4	100.0%	4/4	100.0%		
3C.2a1	N121K, R142G, N171K			1/8	12.5%	8/8	100.0%	3/8	37.5%	8/8	100.0%	6/6	100.0%	0/4	
3C.2a1	N121K, I140M, N171K			5/20	25.0%	19/20	95.0%	9/20	45.0%	19/20	95.0%	11/11	100.0%	9/13	69.2%

Vaccine

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Figure 7. Phylogenetic comparison of A(H3N2) HA genes

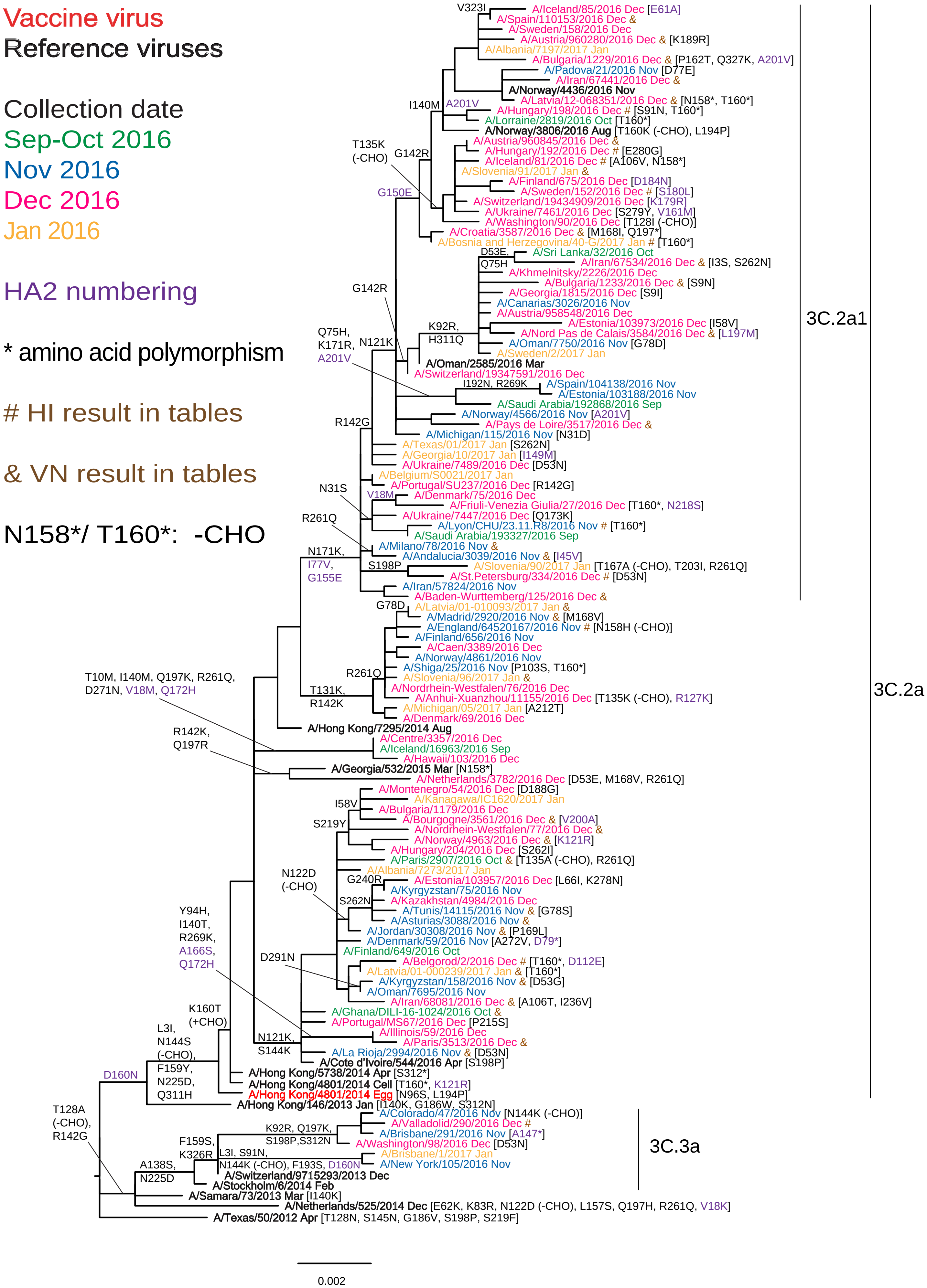


Figure 8. HA protein alignment for influenza A(H3N2) viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

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                                        10        20        30        40        50        60        70        80        90       100
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Texas/50/2012                 QKLPGNDNSTATLCLGHHAVPNGTIVKTITNDRIEVTNATELVQNSSIGEICDSPHQILDGENCTLIDALLGDPQCDGFQNKKWDLFVERSKAYSNCYPY 3C.1
A/Netherlands/525/2014          .............................................................K....................R................. 3C.3b
A/Samara/73/2013                .................................................................................................... 3C.3
A/Stockholm/6/2014              .................................................................................................... 3C.3a
A/Switzerland/9715293/2013      .................................................................................................... 3C.3a
A/Brisbane/1/2017               ..I.......................................................................................N......... 3C.3a
A/Valladolid/290/2016           ...........................................................................................R........ 3C.3a
A/Hong Kong/146/2013            .................................................................................................... 3C.2
A/Hong Kong/4801/2014 Egg       ..I............................................................................................S.... 3C.2a
A/Hong Kong/4801/2014 Cell      ..I................................................................................................. 3C.2a
A/Hong Kong/5738/2014           ..I................................................................................................. 3C.2a
A/Cote d'Ivoire/544/2016        ..I................................................................................................. 3C.2a
A/Paris/3513/2016               ..I..........................................................................................H...... 3C.2a
A/Kyrgyzstan/158/2016           ..I.................................................G............................................... 3C.2a
A/Jordan/30308/2016             ..I................................................................................................. 3C.2a
A/Estonia/103957/2016           ..I..............................................................I.................................. 3C.2a
A/Montenegro/54/2016            ..I......................................................V.......................................... 3C.2a
A/Georgia/532/2015              ..I................................................................................................. 3C.2a
A/Hong Kong/7295/2014           ..I................................................................................................. 3C.2a
A/Anhui-Xuanzhou/11155/2016     ..I................................................................................................. 3C.2a
A/Latvia/01-010093/2017         ..I..........................................................................D...................... 3C.2a
A/Slovenia/90/2017              ..I................................................................................................. 3C.2a
A/Andalucia/3039/2016           ..I................................................................................................. 3C.2a1
A/Friuli-Venezia Giulia/27/2016 ..I...........................S..................................................................... 3C.2a1
A/Ukraine/7489/2016             ..I.................................................N............................................... 3C.2a1
A/Estonia/103188/2016           ..I.......................................................................H......................... 3C.2a1
A/Oman/2585/2016                ..I................................................................................................. 3C.2a1
A/Estonia/103973/2016           ..I......................................................V.................................R........ 3C.2a1
A/Iran/67534/2016               ..S.................................................E.....................H................R........ 3C.2a1
A/Ukraine/7461/2016             ..I................................................................................................. 3C.2a1
A/Finland/675/2016              ..I................................................................................................. 3C.2a1
A/Norway/3806/2016              ..I................................................................................................. 3C.2a1
A/Norway/4436/2016              ..I................................................................................................. 3C.2a1
A/Austria/960280/2016           ..I................................................................................................. 3C.2a1

                                       110       120       130       140       150       160       170       180       190       200
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Texas/50/2012                 DVPDYASLRSLVASSGTLEFNNESFNWNGVTQNGTSSACIRRSNNSFFSRLNWLTHLNFKYPALNVTMPNNEQFDKLYIWGVHHPVTDKDQIFLYAQPSG
A/Netherlands/525/2014          .....................D.....A.............G..S...........S............................G..........HS..
A/Samara/73/2013                ...........................A...........K.G..S........................................G...........S..
A/Stockholm/6/2014              ...........................A.........S...G..S........................................G...........S..
A/Switzerland/9715293/2013      ...........................A.........S...G..S.............S..........................G...........S..
A/Brisbane/1/2017               ...........................A.........S...G.KS.............S..........................G......S....S..
A/Valladolid/290/2016           ...........................A.........S...G..S.............S..........................G..........K...
A/Hong Kong/146/2013            ...........................T...........K....S........................................W...........S..
A/Hong Kong/4801/2014 Egg       ...........................T...............SS.............Y..........................G.......P...S..
A/Hong Kong/4801/2014 Cell      ...........................T...............SS.............YX.........................G...........S..
A/Hong Kong/5738/2014           ...........................T...............SS.............YT.........................G...........S..
A/Cote d'Ivoire/544/2016        ....................K......T...............KS.............YT.........................G..............
A/Paris/3513/2016               ....................K......T...........T...KS.............YT.........................G...........S..
A/Kyrgyzstan/158/2016           ....................K......T...............KS.............YT.........................G...........S..
A/Jordan/30308/2016             ....................KD.....T...............KS.............YT........L................G...........S..
A/Estonia/103957/2016           ....................KD.....T...............KS.............YT.........................G...........S..
A/Montenegro/54/2016            ....................K......T...............KS.............YT.........................G.G.........S..
A/Georgia/532/2015              ...........................T.............K.SS............XYT.........................G..........RS..
A/Hong Kong/7295/2014           ...........................T...............SS.............YT.........................G...........S..
A/Anhui-Xuanzhou/11155/2016     ...........................T..K...K......K.SS.............YT.........................G...........S..
A/Latvia/01-010093/2017         ...........................T..K..........K.SS.............YT.........................G...........S..
A/Slovenia/90/2017              ...........................T...............SS.............YT......A...K..............G..............
A/Andalucia/3039/2016           ...........................T...............SS.............YT..........K..............G...........S..
A/Friuli-Venezia Giulia/27/2016 ...........................T...............SS.............YX..........K..............G...........S..
A/Ukraine/7489/2016             ...........................T.............G.SS.............YT..........K..............G...........S..
A/Estonia/103188/2016           ....................K......T.............G.SS.............YT..........R..............G.....N.....S..
A/Oman/2585/2016                ....................K......T...............SS.............YT..........K..............G...........S..
A/Estonia/103973/2016           ....................K......T...............SS.............YT..........K..............G...........S..
A/Iran/67534/2016               ....................K......T...............SS.............YT..........K..............G...........S..
A/Ukraine/7461/2016             ....................K......T......K........SS.............YT..........K..............G...........S..
A/Finland/675/2016              ....................K......T......K........SS.............YT..........K..............G...........S..
A/Norway/3806/2016              ....................K......T...........M...SS.............Y...........K..............G.......P...S..
A/Norway/4436/2016              ....................K......T...........M...SS.............YT..........K..............G...........S..
A/Austria/960280/2016           ....................K......T...........M...SS.............YT..........K..............G..R........S..

                                       210       220       230       240       250       260       270       280       290       300
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Texas/50/2012                 RITVSTKRSQQAVIPNIGFRPRIRNIPSRISIYWTIVKPGDILLINSTGNLIAPRGYFKIRSGKSSIMRSDAPIGKCKSECITPNGSIPNDKPFQNVNRI
A/Netherlands/525/2014          ..................S.........................................Q.......................................
A/Samara/73/2013                ..................S.................................................................................
A/Stockholm/6/2014              ..................S.....D...........................................................................
A/Switzerland/9715293/2013      ..................S.....D...........................................................................
A/Brisbane/1/2017               ..................S.....D...........................................................................
A/Valladolid/290/2016           ..................S.....D...........................................................................
A/Hong Kong/146/2013            ..................S.................................................................................
A/Hong Kong/4801/2014 Egg       ..................S.....D...........................................................................
A/Hong Kong/4801/2014 Cell      ..................S.....D...........................................................................
A/Hong Kong/5738/2014           ..................S.....D...........................................................................
A/Cote d'Ivoire/544/2016        ..................S.....D...........................................................................
A/Paris/3513/2016               ..................S.....D...........................................K...............................
A/Kyrgyzstan/158/2016           ..................S.....D.................................................................N.........
A/Jordan/30308/2016             ..................S.....D...........................................................................
A/Estonia/103957/2016           ..................S.....D..............R.....................N...............N......................
A/Montenegro/54/2016            ..................Y.....D...........................................................................
A/Georgia/532/2015              ..................S.....D...........................................................................
A/Hong Kong/7295/2014           ..................S.....D...........................................................................
A/Anhui-Xuanzhou/11155/2016     ..................S.....D...................................Q.......................................
A/Latvia/01-010093/2017         ..................S.....D...................................Q.......................................
A/Slovenia/90/2017              ..I...............S.....D...................................Q.......................................
A/Andalucia/3039/2016           ..................S.....D...................................Q.......................................
A/Friuli-Venezia Giulia/27/2016 ..................S.....D...........................................................................
A/Ukraine/7489/2016             ..................S.....D...........................................................................
A/Estonia/103188/2016           ..................S.....D...........................................K...............................
A/Oman/2585/2016                ..................S.....D...........................................................................
A/Estonia/103973/2016           ..................S.....D...........................................................................
A/Iran/67534/2016               ..................S.....D....................................N......................................
A/Ukraine/7461/2016             ..................S.....D.....................................................Y.....................
A/Finland/675/2016              ..................S.....D...........................................................................
A/Norway/3806/2016              ..................S.....D...........................................................................
A/Norway/4436/2016              ..................S.....D...........................................................................
A/Austria/960280/2016           ..................S.....D...........................................................................
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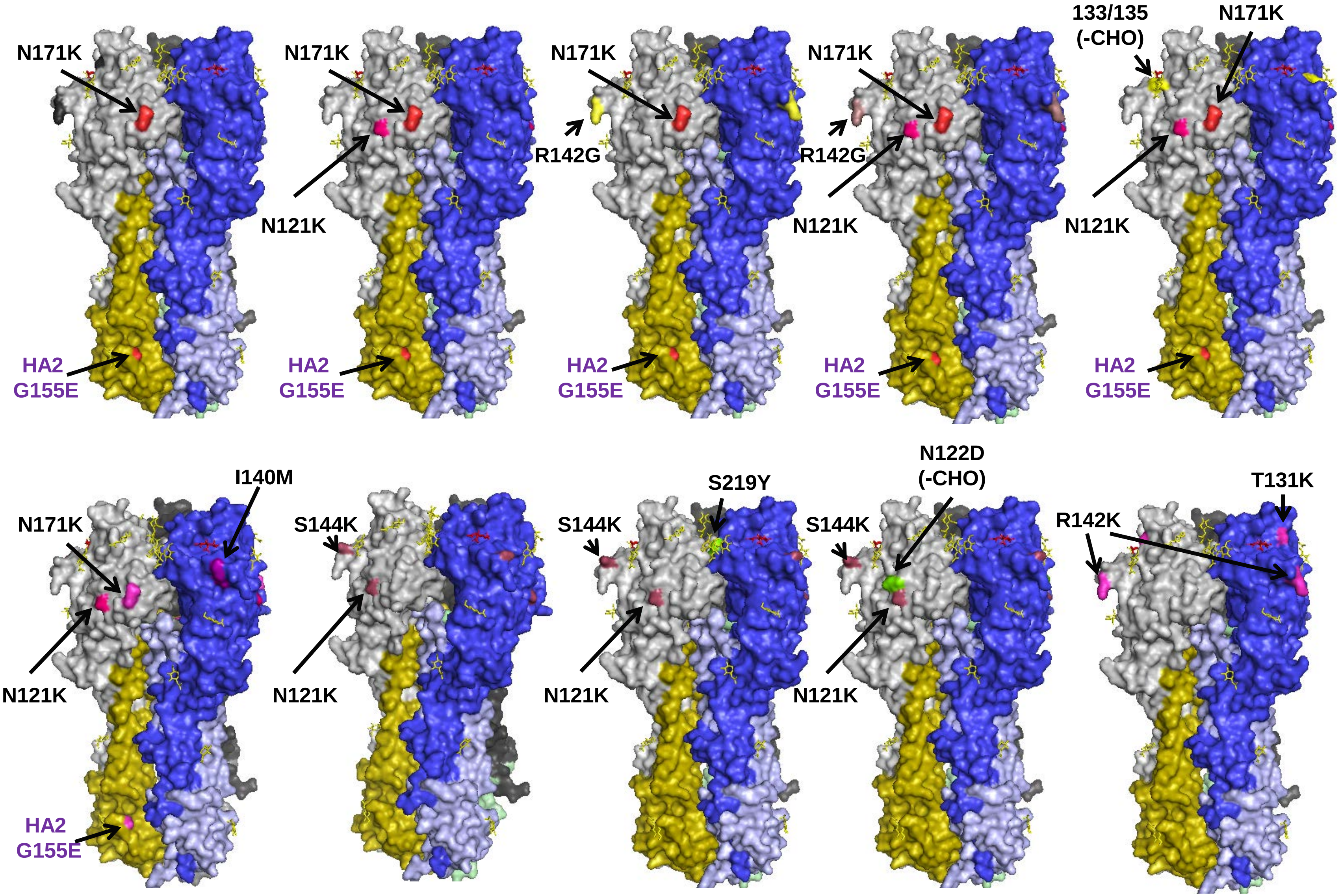
                                        310       320        1        11        21        31        41        51        61        71
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Texas/50/2012                 TYGACPRYVKQSTLKLATGMRNVPEKQTRGIFGAIAGFIENGWEGMVDGWYGFRHQNSEGRGQAADLKSTQAAIDQINGKLNRLIGKTNEKFHQIEKEFS
A/Netherlands/525/2014          ..............................................K.....................................................
A/Samara/73/2013                ....................................................................................................
A/Stockholm/6/2014              ....................................................................................................
A/Switzerland/9715293/2013      .........................R..........................................................................
A/Brisbane/1/2017               .........................R..........................................................................
A/Valladolid/290/2016           ...........N.............R..........................................................................
A/Hong Kong/146/2013            ...........N........................................................................................
A/Hong Kong/4801/2014 Egg       ..........H.........................................................................................
A/Hong Kong/4801/2014 Cell      ..........H.........................................................................................
A/Hong Kong/5738/2014           ..........HX........................................................................................
A/Cote d’Ivoire/544/2016        ..........H.........................................................................................
A/Paris/3513/2016               ..........H.........................................................................................
A/Kyrgyzstan/158/2016           ..........H.........................................................................................
A/Jordan/30308/2016             ..........H.........................................................................................
A/Estonia/103957/2016           ..........H.........................................................................................
A/Montenegro/54/2016            ..........H.........................................................................................
A/Georgia/532/2015              ..........H.........................................................................................
A/Hong Kong/7295/2014           ..........H.........................................................................................
A/Anhui-Xuanzhou/11155/2016     ..........H.........................................................................................
A/Latvia/01-010093/2017         ..........H.........................................................................................
A/Slovenia/90/2017              ..........H.........................................................................................
A/Andalucia/3039/2016           ..........H..............................................................V..........................
A/Friuli-Venezia Giulia/27/2016 ..........H...................................M.....................................................
A/Ukraine/7489/2016             ..........H.........................................................................................
A/Estonia/103188/2016           ..........H.........................................................................................
A/Oman/2585/2016                ..........H.........................................................................................
A/Estonia/103973/2016           ....................................................................................................
A/Iran/67534/2016               ....................................................................................................
A/Ukraine/7461/2016             ..........H.........................................................................................
A/Finland/675/2016              ..........H.........................................................................................
A/Norway/3806/2016              ..........H.........................................................................................
A/Norway/4436/2016              ..........H.........................................................................................
A/Austria/960280/2016           ..........H.........................................................................................

                                        81        91       101       111       121       131       141       151       161       171
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Texas/50/2012                 EVEGRIQDLEKYVEDTKIDLWSYNAELLVALENQHTIDLTDSEMNKLFEKTKKQLRENAEDMGNGCFKIYHKCDNACIGSIRNGTYDHDVYRDEALNNRF
A/Netherlands/525/2014          ....................................................................................................
A/Samara/73/2013                ....................................................................................................
A/Stockholm/6/2014              ....................................................................................................
A/Switzerland/9715293/2013      ....................................................................................................
A/Brisbane/1/2017               ........................................................................................N...........
A/Valladolid/290/2016           ....................................................................................................
A/Hong Kong/146/2013            ........................................................................................N...........
A/Hong Kong/4801/2014 Egg       ........................................................................................N...........
A/Hong Kong/4801/2014 Cell      .................................................R......................................N...........
A/Hong Kong/5738/2014           ........................................................................................N...........
A/Cote d’Ivoire/544/2016        ........................................................................................N...........
A/Paris/3513/2016               ........................................................................................N.....S.....
A/Kyrgyzstan/158/2016           ........................................................................................N...........
A/Jordan/30308/2016             ........................................................................................N...........
A/Estonia/103957/2016           ........................................................................................N...........
A/Montenegro/54/2016            ........................................................................................N...........
A/Georgia/532/2015              ........................................................................................N...........
A/Hong Kong/7295/2014           ........................................................................................N...........
A/Anhui-Xuanzhou/11155/2016     .......................................................K................................N...........
A/Latvia/01-010093/2017         ........................................................................................N...........
A/Slovenia/90/2017              .....V.............................................................................E....N...........
A/Andalucia/3039/2016           .....V.............................................................................E....N...........
A/Friuli-Venezia Giulia/27/2016 .....V.............................................................................E....N...........
A/Ukraine/7489/2016             .....V.............................................................................E....N...........
A/Estonia/103188/2016           .....V.............................................................................E....N...........
A/Oman/2585/2016                .....V.............................................................................E....N...........
A/Estonia/103973/2016           .....V.............................................................................E....N...........
A/Iran/67534/2016               .....V.............................................................................E....N...........
A/Ukraine/7461/2016             .....V........................................................................E....E....NM..........
A/Finland/675/2016              .....V........................................................................E....E....N...........
A/Norway/3806/2016              .....V........................................................................E....E....N...........
A/Norway/4436/2016              .....V........................................................................E....E....N...........
A/Austria/960280/2016           .....V........................................................................E....E....N...........

                                        181       191       201       211       221
                                ....|....|....|....|....|....|....|....|....|....|
A/Texas/50/2012                 QIKGVELKSGYKDWILWISFAISCFLLCVALLGFIMWACQKGNIRCNICI
A/Netherlands/525/2014          ..................................................
A/Samara/73/2013                ..................................................
A/Stockholm/6/2014              ..................................................
A/Switzerland/9715293/2013      ..................................................
A/Brisbane/1/2017               ..................................................
A/Valladolid/290/2016           ..................................................
A/Hong Kong/146/2013            ..................................................
A/Hong Kong/4801/2014 Egg       ..................................................
A/Hong Kong/4801/2014 Cell      ..................................................
A/Hong Kong/5738/2014           ..................................................
A/Cote d’Ivoire/544/2016        ..................................................
A/Paris/3513/2016               H.................................................
A/Kyrgyzstan/158/2016           ..................................................
A/Jordan/30308/2016             ..................................................
A/Estonia/103957/2016           ..................................................
A/Montenegro/54/2016            ..................................................
A/Georgia/532/2015              ..................................................
A/Hong Kong/7295/2014           ..................................................
A/Anhui-Xuanzhou/11155/2016     ..................................................
A/Latvia/01-010093/2017         ..................................................
A/Slovenia/90/2017              ..................................................
A/Andalucia/3039/2016           ..................................................
A/Friuli-Venezia Giulia/27/2016 ..............................................S...
A/Ukraine/7489/2016             ..................................................
A/Estonia/103188/2016           .............................V....................
A/Oman/2585/2016                ..................................................
A/Estonia/103973/2016           ..................................................
A/Iran/67534/2016               ..................................................
A/Ukraine/7461/2016             ..................................................
A/Finland/675/2016              ............N.....................................
A/Norway/3806/2016              ..................................................
A/Norway/4436/2016              ..................................................
A/Austria/960280/2016           ..................................................
```

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 9. Locations of the main groups of amino acid substitutions (Table 9-9) in 3C.2a viruses



3C.2a1 groups all carry HA1 N171K and HA2 G155E

Figure 10. Phylogenetic comparison of A(H3N2) NA genes

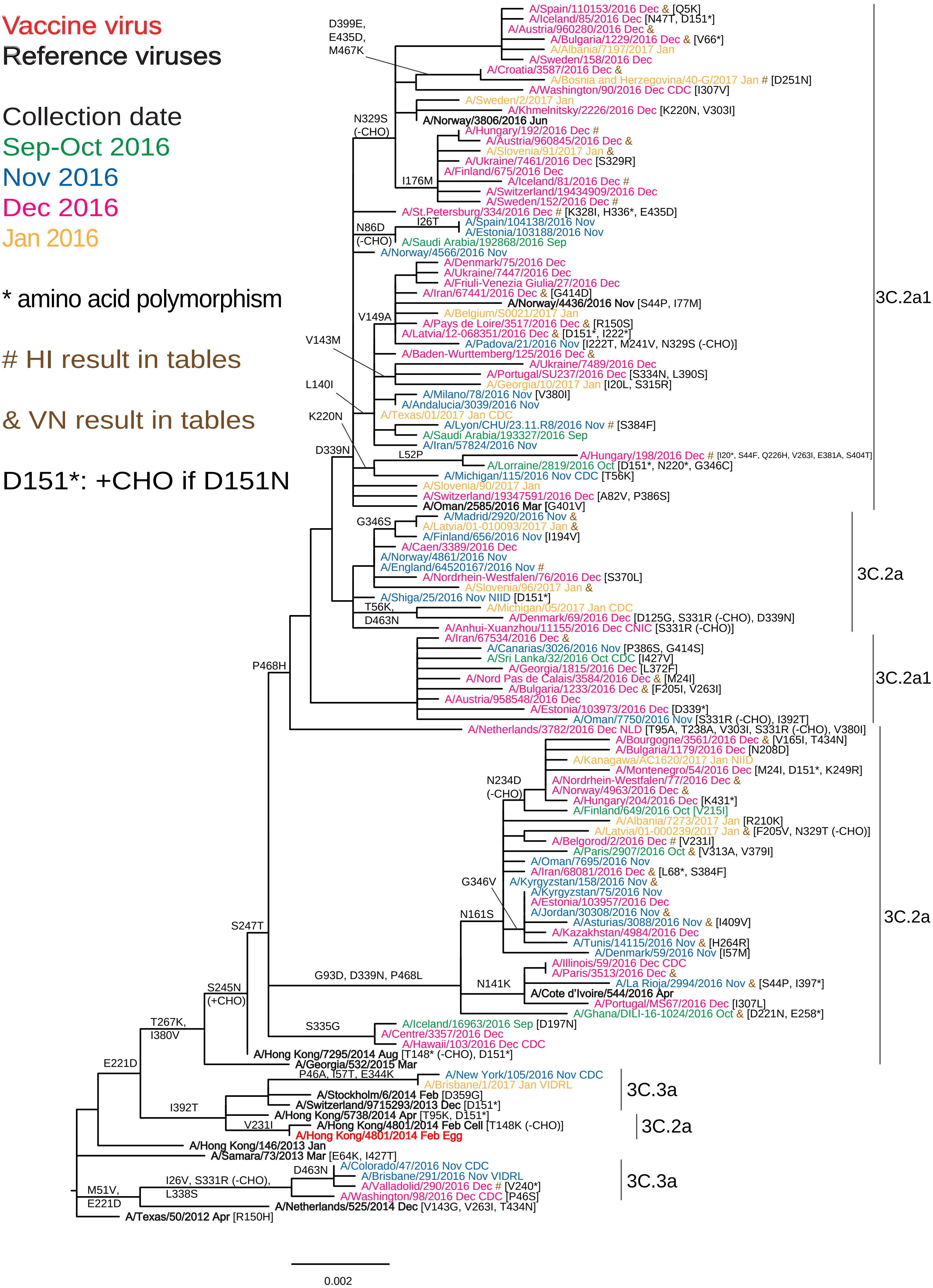


Figure 11. NA protein alignment for influenza A(H3N2) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

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                                         10        20        30        40        50        60        70        80        90       100
                                 ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Texas/50/2012                  MNPNQKIITIGSVSLTISTICFFMQIAILITTVTLHFKQYEFNSPPNNQVMLCEPTIIERNITEIVYLTNTTIEKEICPKPAEYRNWSKPQCGITGFAPF 3C.1
A/Netherlands/525/2014           ..................................................V................................................. 3C.3b
A/Samara/73/2013                 ...............................................................K.................................... 3C.3
A/Stockholm/6/2014               .................................................................................................... 3C.3a
A/Switzerland/9715293/2013       .................................................................................................... 3C.3a
A/Brisbane/1/2017                .............................................A..........T........................................... 3C.3a
A/Valladolid/290/2016            .........................V........................V................................................. 3C.3a
A/Hong Kong/146/2013             .................................................................................................... 3C.2
A/Hong Kong/4801/2014      Egg   .................................................................................................... 3C.2a
A/Hong Kong/4801/2014      Cell  .................................................................................................... 3C.2a
A/Hong Kong/5738/2014            ..............................................................................................K..... 3C.2a
A/Cote d'Ivoire/544/2016         ............................................................................................D....... 3C.2a
A/Paris/3513/2016                ............................................................................................D....... 3C.2a
A/Kyrgyzstan/158/2016            ............................................................................................D....... 3C.2a
A/Jordan/30308/2016              ............................................................................................D....... 3C.2a
A/Estonia/103957/2016            ............................................................................................D....... 3C.2a
A/Montenegro/54/2016             .......................I....................................................................D....... 3C.2a
A/Georgia/532/2015               .................................................................................................... 3C.2a
A/Hong Kong/7295/2014            .................................................................................................... 3C.2a
A/Anhui-Xuanzhou/11155/2016      .................................................................................................... 3C.2a
A/Latvia/01-010093/2017          .................................................................................................... 3C.2a
A/Slovenia/90/2017               .................................................................................................... 3C.2a
A/Andalucia/3039/2016            .................................................................................................... 3C.2a1
A/Friuli-Venezia Giulia/27/2016  .................................................................................................... 3C.2a1
A/Ukraine/7489/2016              .................................................................................................... 3C.2a1
A/Estonia/103188/2016            .........................T...........................................................D.............. 3C.2a1
A/Oman/2585/2016                 .................................................................................................... 3C.2a1
A/Estonia/103973/2016            .................................................................................................... 3C.2a1
A/Iran/67534/2016                .................................................................................................... 3C.2a1
A/Ukraine/7461/2016              .................................................................................................... 3C.2a1
A/Finland/675/2016               .................................................................................................... 3C.2a1
A/Norway/3806/2016               .................................................................................................... 3C.2a1
A/Norway/4436/2016               ...........................................P................................M....................... 3C.2a1
A/Austria/960280/2016            .................................................................................................... 3C.2a1

                                        110       120       130       140       150       160       170       180       190       200
                                 ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Texas/50/2012                  SKDNSIRLSAGGDIWVTREPYVSCDPDKCYQFALGQGTTLNNVHSNNTVHDRTPYRTLLMNELGVPFHLGTKQVCIAWSSSSCHDGKAWLHVCITGDDKN
A/Netherlands/525/2014           ..........................................G......R..................................................
A/Samara/73/2013                 .................................................R..................................................
A/Stockholm/6/2014               .................................................R..................................................
A/Switzerland/9715293/2013       .................................................RX.................................................
A/Brisbane/1/2017                .................................................R..................................................
A/Valladolid/290/2016            .................................................R..................................................
A/Hong Kong/146/2013             .................................................R..................................................
A/Hong Kong/4801/2014      Egg   .................................................R..................................................
A/Hong Kong/4801/2014      Cell  ...............................................K.R..................................................
A/Hong Kong/5738/2014            .................................................RX.................................................
A/Cote d'Ivoire/544/2016         ........................................K........R..................................................
A/Paris/3513/2016                ........................................K........R..................................................
A/Kyrgyzstan/158/2016            .................................................R..........S.......................................
A/Jordan/30308/2016              .................................................R..........S.......................................
A/Estonia/103957/2016            .................................................R..........S.......................................
A/Montenegro/54/2016             .................................................RX.........S.......................................
A/Georgia/532/2015               .................................................R..................................................
A/Hong Kong/7295/2014            ...............................................X.RX.................................................
A/Anhui-Xuanzhou/11155/2016      .................................................R..................................................
A/Latvia/01-010093/2017          .................................................R..................................................
A/Slovenia/90/2017               .................................................R..................................................
A/Andalucia/3039/2016            .......................................I.........R..................................................
A/Friuli-Venezia Giulia/27/2016  .......................................I........AR..................................................
A/Ukraine/7489/2016              .......................................I..M......R..................................................
A/Estonia/103188/2016            .................................................R..................................................
A/Oman/2585/2016                 .................................................R..................................................
A/Estonia/103973/2016            .................................................R..................................................
A/Iran/67534/2016                .................................................R..................................................
A/Ukraine/7461/2016              .................................................R.........................M........................
A/Finland/675/2016               .................................................R.........................M........................
A/Norway/3806/2016               .................................................R..................................................
A/Norway/4436/2016               .......................................I........AR..................................................
A/Austria/960280/2016            .................................................R..................................................

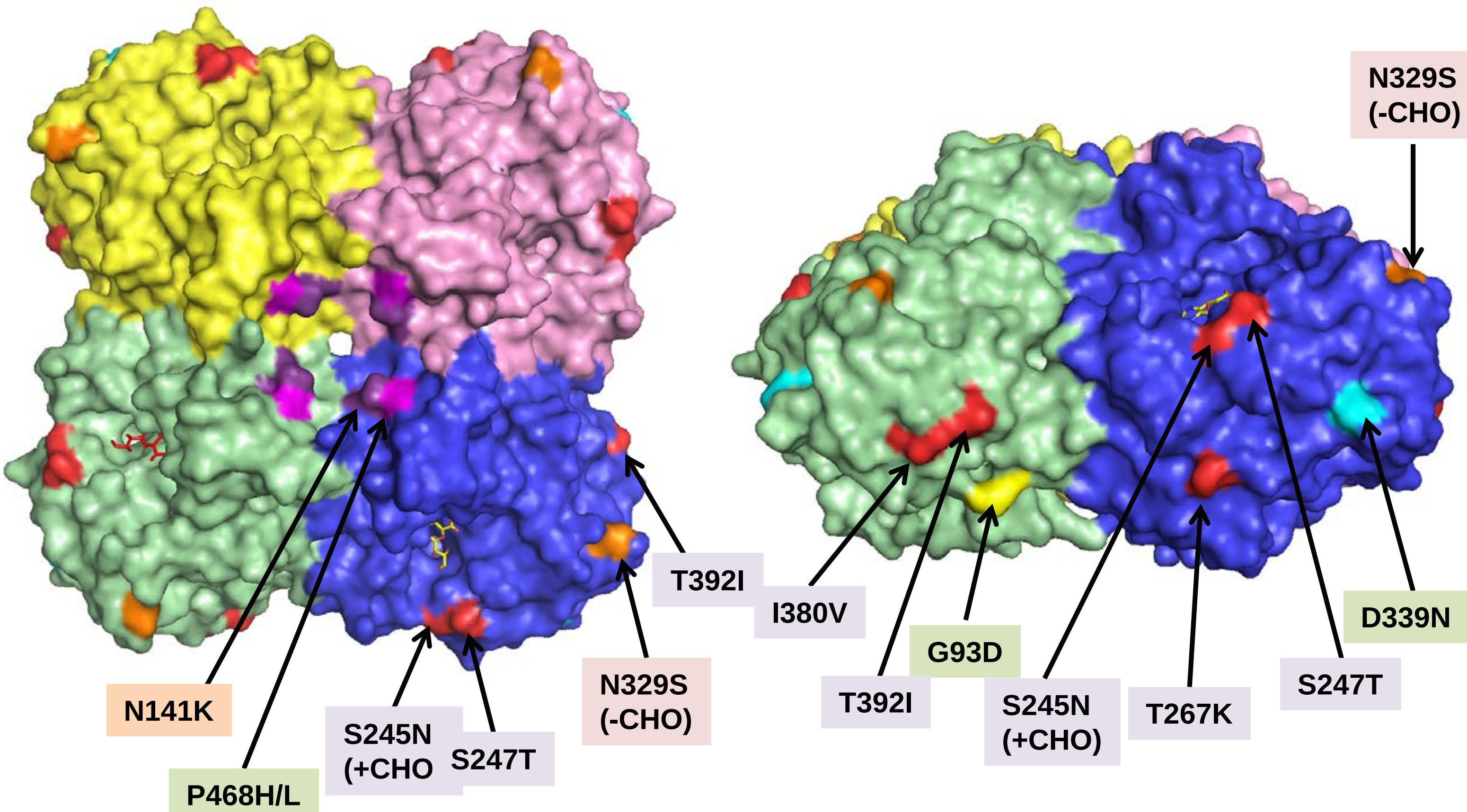
                                        210       220       230       240       250       260       270       280       290       300
                                 ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Texas/50/2012                  ATASFIYNGRLVDSVVSWSKEILRTQESECVCINGTCTVVMTDGSASGKADTKILFIEEGKIVHTSTLSGSAQHVEECSCYPRYPGVRCVCRDNWKGSNR
A/Netherlands/525/2014           ....................D.........................................I.....................................
A/Samara/73/2013                 ....................................................................................................
A/Stockholm/6/2014               ....................D...............................................................................
A/Switzerland/9715293/2013       ....................D...............................................................................
A/Brisbane/1/2017                ....................D...............................................................................
A/Valladolid/290/2016            ....................D..................X............................................................
A/Hong Kong/146/2013             ....................................................................................................
A/Hong Kong/4801/2014      Egg   ....................D.........I.....................................................................
A/Hong Kong/4801/2014      Cell  ....................D.........I.....................................................................
A/Hong Kong/5738/2014            ....................D...............................................................................
A/Cote d'Ivoire/544/2016         ....................D.......................N.T...................K.................................
A/Paris/3513/2016                ....................D.......................N.T...................K.................................
A/Kyrgyzstan/158/2016            ....................D.......................N.T...................K.................................
A/Jordan/30308/2016              ....................D.......................N.T...................K.................................
A/Estonia/103957/2016            ....................D.......................N.T...................K.................................
A/Montenegro/54/2016             ....................D............D..........N.T.R.................K.................................
A/Georgia/532/2015               ....................D.............................................K.................................
A/Hong Kong/7295/2014            ....................D.......................N.....................K.................................
A/Anhui-Xuanzhou/11155/2016      ....................D.......................N.T...................K.................................
A/Latvia/01-010093/2017          ....................D.......................N.T...................K.................................
A/Slovenia/90/2017               ....................D.......................N.T...................K.................................
A/Andalucia/3039/2016            ....................D.......................N.T...................K.................................
A/Friuli-Venezia Giulia/27/2016  ....................D.......................N.T...................K.................................
A/Ukraine/7489/2016              ....................D.......................N.T...................K.................................
A/Estonia/103188/2016            ....................D.......................N.T...................K.................................
A/Oman/2585/2016                 ....................D.......................N.T...................K.................................
A/Estonia/103973/2016            ....................D.......................N.T...................K.................................
A/Iran/67534/2016                ....................D.......................N.T...................K.................................
A/Ukraine/7461/2016              ....................D.......................N.T...................K.................................
A/Finland/675/2016               ....................D.......................N.T...................K.................................
A/Norway/3806/2016               ....................D.......................N.T...................K.................................
A/Norway/4436/2016               ....................D.......................N.T...................K.................................
A/Austria/960280/2016            ....................D.......................N.T...................K.................................
```

```
                                   310       320       330       340       350       360       370       380       390       400
                               ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Texas/50/2012                PIVDINIKDHSIVSSYVCSGLVGDTPRKNDSSSSSHCLDPNNEEGGHGVKGWAFDDGNDVWMGRTINETSRLGYETFKVIEGWSNPKSKLQINRQVIVDR
A/Netherlands/525/2014         ....................................................................................................
A/Samara/73/2013               ....................................................................................................
A/Stockholm/6/2014             ..........................................................G................................T........
A/Switzerland/9715293/2013     ...........................................................................................T........
A/Brisbane/1/2017              ...........................................K...............................................T........
A/Valladolid/290/2016          ..............................R......S..............................................................
A/Hong Kong/146/2013           ....................................................................................................
A/Hong Kong/4801/2014     Egg  ...........................................................................................T........
A/Hong Kong/4801/2014     Cell ...........................................................................................T........
A/Hong Kong/5738/2014          ...........................................................................................T........
A/Cote d'Ivoire/544/2016       ......................................N........................................V....................
A/Paris/3513/2016              ......................................N........................................V....................
A/Kyrgyzstan/158/2016          ......................................N........................................V....................
A/Jordan/30308/2016            ......................................N......V.................................V....................
A/Estonia/103957/2016          ......................................N......V.................................V....................
A/Montenegro/54/2016           ......................................N........................................V....................
A/Georgia/532/2015             ...............................................................................V....................
A/Hong Kong/7295/2014          ...............................................................................V....................
A/Anhui-Xuanzhou/11155/2016    ..............................R................................................V....................
A/Latvia/01-010093/2017        .............................................S.................................V....................
A/Slovenia/90/2017             ......................................N........................................V....................
A/Andalucia/3039/2016          ......................................N........................................V....................
A/Friuli-Venezia Giulia/27/2016 .....................................N........................................V....................
A/Ukraine/7489/2016            ......................................N........................................V....................
A/Estonia/103188/2016          ......................................N........................................V....................
A/Oman/2585/2016               ......................................N........................................V....................
A/Estonia/103973/2016          ......................................X........................................V....................
A/Iran/67534/2016              ...............................................................................V....................
A/Ukraine/7461/2016            ............................R.........N........................................V....................
A/Finland/675/2016             ............................S.........N........................................V....................
A/Norway/3806/2016             ............................S.........N........................................V....................
A/Norway/4436/2016             ......................................N........................................V....................
A/Austria/960280/2016          ............................S.........N........................................V....................

                                   410       420       430       440       450       460
                               ....|....|....|....|....|....|....|....|....|....|....|....|....|....
A/Texas/50/2012                GDRSGYSGIFSVEGKSCINRCFYVELIRGRKEETEVLWTSNSIVVFCGTSGTYGTGSWPDGADLNLMPI
A/Netherlands/525/2014         .................................N...................................
A/Samara/73/2013               ..........................T..........................................
A/Stockholm/6/2014             .....................................................................
A/Switzerland/9715293/2013     .....................................................................
A/Brisbane/1/2017              .....................................................................
A/Valladolid/290/2016          ..............................................................N......
A/Hong Kong/146/2013           .....................................................................
A/Hong Kong/4801/2014     Egg  .....................................................................
A/Hong Kong/4801/2014     Cell .....................................................................
A/Hong Kong/5738/2014          .....................................................................
A/Cote d'Ivoire/544/2016       ...................................................................L.
A/Paris/3513/2016              ...................................................................L.
A/Kyrgyzstan/158/2016          ...................................................................L.
A/Jordan/30308/2016            ...................................................................L.
A/Estonia/103957/2016          ...................................................................L.
A/Montenegro/54/2016           ...................................................................L.
A/Georgia/532/2015             .....................................................................
A/Hong Kong/7295/2014          .....................................................................
A/Anhui-Xuanzhou/11155/2016    ...................................................................H.
A/Latvia/01-010093/2017        ...................................................................H.
A/Slovenia/90/2017             ...................................................................H.
A/Andalucia/3039/2016          ...................................................................H.
A/Friuli-Venezia Giulia/27/2016 ..................................................................H.
A/Ukraine/7489/2016            ...................................................................H.
A/Estonia/103188/2016          ...................................................................H.
A/Oman/2585/2016               V..................................................................H.
A/Estonia/103973/2016          ...................................................................H.
A/Iran/67534/2016              ...................................................................H.
A/Ukraine/7461/2016            ...................................................................H.
A/Finland/675/2016             ...................................................................H.
A/Norway/3806/2016             ...................................................................H.
A/Norway/4436/2016             ...................................................................H.
A/Austria/960280/2016          ...................................................................H.
```

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Figure 12. Amino acid substitutions associated with NA subclade 3C.2a viruses relative to A/Hong Kong/4801/2014



All 3C.2a viruses have NA carrying the substitutions highlighted (I231V cannot be seen)

Other substitutions indicated individually (e.g. N141K or N329S [N161S cannot be seen]) or in combination (e.g. G93D + D339N + P468L) define genetic groupings as shown in Figure 10

Influenza B virus analyses.

Influenza B viruses represented 13.5% of the samples received with collection dates after 2016-08-31 **(Table 1)**. B/Victoria lineage viruses have predominated over those of the B/Yamagata lineage at a ratio of approximately 1.5:1.

B/Victoria lineage.

Influenza B viruses of the B/Victoria lineage were received from 16 countries or administrative regions in four WHO Regions **(Table 1)**: African (2: Ghana and Madagascar), European (10: Bosnia and Herzegovina, France, Georgia, Greece, Kazakhstan, Kyrgyzstan, Norway, Portugal, Russia, Switzerland), Eastern Mediterranean (3: Iran, Oman and Tunisia) and Western Pacific (1: Hong Kong SAR). The collection details and summary results for viruses that were recovered and assayed by HI are shown in **Table 10**.

HI results for the propagated viruses are shown in **Tables 11-1 to 11-4**: test viruses in each table are sorted by date of collection and those below the red horizontal lines in the tables are those with collection dates after 2016-08-31. The HA genetic group is indicated for those viruses that have been sequenced and those included in phylogenetic trees are highlighted. A summary of the recognition of the test viruses collected after 2016-08-31 is shown in **Table 11-5**. Only three (7.1%) of the 42 cell culture-propagated test viruses were recognised at titres within 2-fold of the homologous titre with the post-infection ferret antiserum raised against egg-propagated vaccine virus, B/Brisbane/60/2008, but a total of 26 (62%) test viruses were recognised at titres within 4-fold of the homologous titres. The cell culture-propagated test viruses were very poorly recognised by post-infection ferret antisera raised against other egg-propagated reference viruses: B/Malta/636714/2011, B/Johannesburg/3964/2012 and B/South Australia/81/2012, with less than 3% of the tested viruses being recognised at titres within 4-fold of the homologous titres of these antisera. In contrast, over 80% of the test viruses were recognised at titres within 2-fold of the titres with the corresponding homologous viruses for antisera raised against three viruses genetically closely related to B/Brisbane/60/2008 but propagated in tissue culture; such viruses are considered to be surrogate cell-propagated antigens representing the egg-propagated B/Brisbane/60/2008 prototype virus. These antisera were raised against B/Hong Kong/514/2009, B/Ireland/3154/2016 and B/Nordrhein-Westfalen/1/2016; the antisera recognised 81%, 90% and 83%, respectively, of test viruses at titres within 2-fold, and 87%, 93% and 95%, respectively, at titres within 4-fold of the titres of the antisera with their corresponding homologous viruses.

An antiserum raised against cell culture-propagated B/Formosa/V2367/2012 recognised the cell culture-propagated test viruses less well. Only 6 of 42 test viruses (14%) were recognised at titres within 2-fold of the titre of the antiserum for the homologous virus and 29 viruses (69%) were recognised at titres within 4-fold of the homologous titre, but 13 viruses (31% of the total) were recognised at titres ≥8-fold lower than the homologous titre of the antiserum. It is noteworthy that B/Formosa/V2367/2012 had become polymorphic for the egg-adaptive substitution resulting in loss of the HA1 197-199 glycosylation site, despite isolation and propagation exclusively in tissue culture.

Phylogenetic analysis of the HA genes of representative B/Victoria lineage viruses is shown in **Figure 13** and an alignment of the HA glycoproteins of a selection of these viruses is shown in **Figure 14**. All recently circulating viruses analysed carried HA genes that fell into genetic group 1A, the B/Brisbane/60/2008 genetic group, the vast majority falling into a genetic group defined by two HA1 amino acid substitutions compared with B/Brisbane/60/2008: I117V and N129D. A number of virus clusters have emerged within clade 1A: two carry **HA2 R151K** amino acid substitution, one with **HA1 P31S** substitution (viruses from Ghana and Norway) and the other with **HA1 I180V** substitution (viruses in the USA). Notably, a number of viruses in the latter cluster have a two amino acid deletion in **HA1 ∆162/163**.

Figure 15 shows a phylogenetic tree for the NA gene and **Figure 16** shows a corresponding amino acid sequence alignment for a selection of NA glycoproteins. The NA gene phylogeny is largely congruent with the HA phylogeny but a small group of reassortant viruses, from four WHO Regions, with HA genes from the B/Yamagata lineage but NA genes from the B/Victoria lineage is shown **(Figure 15)**.

B/Yamagata lineage.

Influenza B viruses of the B/Yamagata lineage were received from 15 countries or administrative regions in four WHO Regions **(Table 1)**: African (1: Ghana), Eastern Mediterranean (2: Iran and Saudi Arabia), European (11: Croatia, Denmark, Finland, Germany, Kyrgyzstan, Latvia, Luxembourg, Norway, Slovenia, Switzerland and United Kingdom) and Western Pacific (1: Hong Kong SAR). The collection details and summary results for viruses that were recovered and assayed by HI are shown in **Table 12**.

The results of HI analyses for propagated viruses of the B/Yamagata lineage are shown in **Tables 13-1 to 13-4**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines in the tables are viruses with collection dates after 2016-08-31. The clade into which the HA gene falls is shown where known, and viruses for which gene sequences are included in phylogenetic trees are highlighted. A summary of the results for all of the post-infection ferret antisera is shown in **Table 13-5**.

Post-infection ferret antiserum raised against the egg-propagated vaccine virus B/Phuket/3073/2013, recommended for use in quadrivalent vaccines for the Northern Hemisphere for 2016/2017 and Southern Hemisphere 2017, recognised 66% of the test viruses collected after 2016-08-31 at titres within 4-fold of the titre with the homologous virus, 34% within 2-fold.

Antisera raised against other viruses from the B/Phuket/3073/2013 clade (clade 3), egg-propagated B/Wisconsin/1/2010, B/Stockholm/12/2011 and B/Hong Kong/3417/2014 recognised 47%, 82%, 66% and 100%, respectively, of the test viruses collected after 2016-08-31 at titres within 4-fold of the titres of the antisera for their homologous viruses. The antisera raised against egg-propagated B/Wisconsin/1/2010 and B/Stockholm/12/2011, recognised, respectively 13%, and 42% of the test viruses collected after 2016-08-31 at titres within 2-fold of the homologous titres of the antisera while the antiserum raised against B/Hong Kong/3417/2014 recognised all the test viruses at titres within 2-fold of the titre of the antiserum for the homologous virus.

In contrast, a ferret antiserum raised against a cell culture-propagated cultivar of B/Phuket/3073/2013 recognised only 45% of the recently collected test viruses at titres within 4-fold, and only 13% within 2-fold, of its titre with the homologous virus.

An antiserum raised against egg-propagated B/Massachusetts/02/2012 recognised only 21% of recently collected test viruses at titres within 4-fold of the titre of the antiserum for the homologous virus, and only two viruses were recognised at titres within 2-fold of the homologous titre. Antiserum raised against the cell culture-propagated cultivar of B/Massachusetts/02/2012 recognised 29% of recently collected test viruses at a titre within 4-fold of its homologous titre, and 24% within 2-fold of its homologous titre. Test viruses were recognised somewhat better by an antiserum raised against cell culture-propagated B/Estonia/55669/2011, another virus from the B/Massachusetts/02/2012 clade (clade 2), with 27 of the 38 test viruses (71%) collected after 2016-08-31 being recognised at titres within 4-fold of the titre of the antiserum for the homologous virus (34% were within 2-fold of the homologous titre). However, the homologous titre of this antiserum was only 80 (low) in three of the four HI assays.

Figure 17 shows a phylogenetic analysis of the HA genes of representative B/Yamagata lineage viruses and an amino acid sequence alignment of a selection of the HA glycoproteins is shown in **Figure 18**. The vast majority of recently circulating viruses fall in clade 3, the B/Phuket/3073/2013 clade, with most falling into a genetic group defined by two HA1 amino acid substitutions compared with B/Phuket/3073/2013: **L172Q** and **M251V**. Reassortant viruses with HA genes of the B/Yamagata lineage but NA genes of the B/Victoria lineage, from four WHO Regions, are indicated.

Figure 19 shows a phylogenetic tree for NA genes of the same viruses represented in the HA phylogeny and **Figure 20** shows a corresponding amino acid sequence alignment for selected NA glycoproteins. As for the HA gene, the vast majority of NA genes of recently collected viruses fell into the B/Phuket/3073/2013 clade (clade 3) with a majority of viruses carrying **NA D342N** and **K373Q** amino acid substitutions compared to the B/Phuket/3073/2013 sequence.

Antiviral Analysis.

Susceptibility to the neuraminidase inhibitors zanamivir and oseltamivir was assessed phenotypically, backed up by full NA gene sequencing, for 71 influenza B viruses with collection dates after 2016-08-31 (**Table 14**). All viruses showed normal inhibition (no reduced susceptibility) by both antivirals.

Conclusion.

Viruses of the B/Victoria lineage and the B/Yamagata lineage have been received in similar proportions but in small numbers. Most viruses of the B/Victoria lineage were antigenically similar to viruses genetically closely related to B/Brisbane/60/2008. Viruses of the B/Yamagata lineage fell within genetic clade 3, the B/Phuket/3073/2013 clade, and the majority were recognised well by antisera raised against egg-propagated clade 3 vaccine/reference viruses.

Table 10. Summary of influenza B (Victoria-lineage) viruses analysed, collected after 2016-08-31

MONTH	B Victoria lineage				
Country	Number received[1]	Number propagated[2]	≤2 fold[3]	4 fold[3]	≥8 fold[3]
2016					
SEPTEMBER					
Madagascar	2	in process			
OCTOBER					
Ghana	12	in process	3		
Iran	1	1			1
Kyrgyzstan	1	0			
Madagascar	4	in process	1		
Oman	1	1		1	
NOVEMBER					
France	1	1		1	
Ghana	4	3	1	2	
Kyrgyzstan	14	in process	7		
Madagascar	1	1	1		
Norway	2	1		1	
Oman	1	1		1	
DECEMBER					
Bosnia and Herzegovina	1	1	1		
Georgia	1	1	1		
Ghana	1	1	1		
Hong Kong	1	1	1		
Iran	1	0			
Madagascar	3	in process	1		
Kazakhstan	2	2			2
Kyrgyzstan	1	1			1
Norway	2	1		1	
Russia	7	in process	5		
Switzerland	1	1		1	
Tunisia	2	1	1		
2017					
JANUARY					
Georgia	1	1	1		
Greece	7	in process	2	1	
Hong Kong	1	1	1		
Portugal	1	in process			
16 Countries	77	21	28	9	4
			68.3%	22.0%	9.7%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2. Propagated to sufficient titre to perform HI assay

3. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Hong Kong/514/2009, B/Ireland/3154/2016 & B/Nordrhein-Westfalen/1/2016)

Table 11-1. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-11-22 + 2016-12-20 + 2017-01-10

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre / Post-infection ferret antisera: B/Bris 60/08	B/Mal 2506/04	B/Bris 60/08	B/Malta 636714/11	B/Jhb 3964/12	B/For V2367/12	B/Sth Aus 81/12	B/HK 514/09	B/Ireland 3154/16	B/Nord-West 1/16
	Passage history			Egg	Egg	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	MDCK
	Ferret number			SH 539, 540, 543, 544, 570, 571 ,574*1,3	F41/14*2	F26/13*2	F29/13*2	F01/13*2	F04/13*2	F41/13*2	F09/13*2	F15/16*2	F16/16*2
	Genetic group			1A		1A	1A	1A	1A	1A	1B	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E6	2560	320	80	80	160	40	160	20	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	80	320	320	160	80	160	40	20	20
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	320	320	320	160	640	80	20	20
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	640	1280	1280	1280	640	1280	320	80	160
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	20	640	320	160	160	640	80	40	20
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2562	80	320	320	320	160	640	80	40	20
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	40	40	40	20	80	80	40	20
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	<	40	20	20	20	40	40	40	20
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK3	2560	<	40	20	20	20	40	40	40	40
TEST VIRUSES													
B/Kanagawa/IC45/2016	1A	2016-04-28	MDCK2/MDCK1	1280	<	80	40	20	40	10	40	40	40
B/Jordan/30155/2016	1A	2016-05-15	MDCK1	2560	<	80	40	20	40	10	20	40	40
B/Kyrgyzstan/36/2016	1A	2016-11-15	SIAT1	2560	<	80	40	20	20	10	20	40	40
B/Kyrgyzstan/54/2016	1A	2016-11-17	SIAT1	2560	<	80	40	40	40	20	40	80	40
B/Lyon/2383/2016	1A	2016-11-23	MDCK2/MDCK1	2560	20	80	20	20	20	10	20	40	20
B/Norway/4831/2016	1A	2016-11-30	MDCK1	2560	<	80	40	20	20	20	20	40	20

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):
[1] < = <40; [2] < = <10; [3] hyperimmune sheep serum

Vaccine
NH 2016-17
SH 2017

Sequences in phylogenetic trees

Table 11-2. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-01-17 + 2017-01-24

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre — Post-infection ferret antisera: B/Bris 60/08	B/Mal 2506/04	B/Bris 60/08	B/Malta 636714/11	B/Jhb 3964/12	B/For V2367/12	B/Sth Aus 81/12	B/HK 514/09	B/Ireland 3154/16	B/Nord-West 1/16
	Passage history			Egg	Egg	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	MDCK
	Ferret number			Sh 539, 540, 543, 544, 570, 571, 574*1,3	F41/14*2	F26/13*2	F29/13*2	F01/13*2	F04/13*2	F41/13*2	F09/13*2	F15/16*2	F16/16*2
	Genetic group			1A		1A	1A	1A	1A	1A	1B	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E6	2560	320	80	80	160	40	160	20	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	80	320	160	160	160	160	40	20	20
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	320	320	320	160	640	80	20	20
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	640	1280	1280	1280	640	1280	320	80	160
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	20	640	320	160	160	640	80	40	20
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2562	80	320	320	320	160	640	80	40	20
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	40	40	40	20	80	80	40	20
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	<	40	20	20	20	40	40	40	20
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK3	2560	<	40	20	20	20	40	40	40	40
TEST VIRUSES													
B/Oman/7281/2016	1A	2016-10-17	MDCK1/MDCK1	1280	10	40	<	<	20	<	20	20	40
B/Ghana/DILI-16-1091/2016	1A	2016-10-26	Cx/MDCK1	2560	<	80	<	<	20	10	40	40	40
B/Oman/7883/2016	1A	2016-11-13	MDCK1/MDCK1	1280	<	80	<	<	20	10	20	20	40
B/Norway/4857/2016	1A	2016-12-02	MDCK2	2560	<	160	<	20	10	10	20	40	40
B/Switzerland/19357129/2016	1A	2016-12-12	MDCK1	2561	<	80	<	20	10	10	20	40	10

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):
[1] < = <40; [2] < = <10; [3] hyperimmune sheep serum

Vaccine NH 2016-17 SH 2017

Sequences in phylogenetic trees

Table 11-3. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-02-07

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre: Post-infection ferret antisera: B/Bris 60/08	B/Mal 2506/04	B/Bris 60/08	B/Malta 636714/11	B/Jhb 3964/12	B/For V2367/12	B/Sth Aus 81/12	B/HK 514/09	B/Ireland 3154/16	B/Nord-West 1/16	
	Passage history			Egg	Egg	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	MDCK	
	Ferret number			Sh 539, 540, 543, 544, 570, 571, 574*[1,3]	F41/14*[2]	F26/13*[2]	F29/13*[1]	F01/13*[4]	F04/13*[2]	F41/13*[2]	F09/13*[2]	F15/16*[2]	F16/16*[2]	
	Genetic group			1A		1A	1A	1A	1A	1A	1B	1A	1A	
REFERENCE VIRUSES														
B/Malaysia/2506/2004		2004-12-06	E3/E7	1280	320	20	<	80	40	20	10	<	<	
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	80	640	160	320	160	160	40	<	20	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	40	640	320	320	160	160	40	<	20	
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	1280	1280	1280	1280	1280	320	160	320	
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	5120	20	640	160	160	160	640	80	40	40	
B/South Australia/81/2012	1A	2012-11-28	E4/E1	1280	40	320	160	160	80	640	40	40	<	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	80	<	<	20	10	40	40	40	
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	5110	<	80	<	<	40	20	40	80	80	
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK1	5120	<	80	<	40	40	20	40	80	80	
TEST VIRUSES														
B/Oman/6910/2016	1A		MDCK1/MDCK1	2560	<	80	<	<	40	10	40	40	40	
B/Iran/56446/2016	1A	2016-10-22	E1/E1	2560	40	640	320	320	320	160	80	<	20	
B/Ghana/FS-16-1620/2016	1A	2016-10-26	MDCKx/MDCK1	2560	<	160	<	20	40	20	40	40	40	
B/Ghana/FS-16-1642/2016	1A	2016-10-27	Cx/MDCK1	2560	<	160	<	20	20	20	40	40	40	
B/Ghana/FS-16-1636/2016	1A	2016-11-01	MDCK1	2560	<	160	<	20	40	20	40	40	20	
B/Ghana/FS-16-1769/2016	1A	2016-11-04	MDCKx/MDCK1	2560	<	160	<	20	40	20	40	80	40	
B/Kyrgyzstan/7/2016	1A	2016-11-26	MDCKx/MDCK1	2560	<	80	<	<	<	10	40	40	40	
B/Kyrgyzstan/19/2016	1A	2016-11-28	MDCKx/MDCK1	2560	<	160	<	20	40	20	40	80	40	
B/Kyrgyzstan/20/2016	1A	2016-11-28	MDCKx/MDCK1	2560	<	160	<	20	40	10	40	80	40	
B/Kyrgyzstan/13/2016	1A	2016-11-29	MDCKx/MDCK1	2560	<	80	<	40	40	10	40	80	40	
B/Kyrgyzstan/33/2016	1A	2016-11-29	MDCKx/MDCK1	5120	<	80	<	20	40	10	40	80	40	
B/Kyrgyzstan/15/2016	1A	2016-12-01	MDCKx/MDCK1	2560	<	80	<	20	40	10	40	80	40	
B/Astrakhan/63/2016	1A	2016-12-07	MDCK1/MDCK1	2560	<	160	<	40	40	20	40	40	40	
B/Kazakhstan/4964/2016	1A	2016-12-07	MDCK1	1280	<	<	<	<	<	<	<	<	<	K165N
B/Kazakhstan/4992/2016	1A	2016-12-13	MDCK1	1280	<	40	<	<	20	10	20	<	<	
B/Ghana/FS-16-1980/2016	1A	2016-12-14	MDCK1	2560	<	160	<	40	40	20	40	40	40	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):
[1] < = <40; [2] < = <10; [3] hyperimmune sheep serum; [4] < = <20

Vaccine
NH 2016-17
SH 2017

Sequences in phylogenetic trees

Table 11-4. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-02-14

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera								
				B/Bris 60/08	B/Mal 2506/04	B/Bris 60/08	B/Malta 636714/11	B/Jhb 3964/12	B/For V2367/12	B/Sth Aus 81/12	B/HK 514/09	B/Ireland 3154/16	B/Nord-West 1/16
	Passage history			Egg	Egg	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	MDCK
	Ferret number			Sh 539, 540, 543, 544, 570, 571, 574[*1,3]	F41/14[*2]	F26/13[*2]	F29/13[*1]	F01/13[*4]	F04/13[*2]	F41/13[*2]	F09/13[*2]	F15/16[*2]	F16/16[*2]
	Genetic group			1A		1A	1A	1A	1A	1A	1B	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	1280	320	20	<	80	40	20	10	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	80	640	320	320	160	160	80	40	20
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	40	640	320	320	160	160	40	<	20
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	1280	1280	1280	1280	1280	320	160	320
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	5120	20	640	160	160	160	640	80	40	40
B/South Australia/81/2012	1A	2012-11-28	E4/E1	1280	40	320	160	160	80	640	40	40	<
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	80	<	<	20	10	40	40	40
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	5110	<	80	<	<	40	20	40	80	80
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK1	5120	<	80	<	40	40	20	40	80	80
TEST VIRUSES													
B/Madagascar/6265/2016	P9	2016-10-24	MDCK1/MDCK1	2560	<	80	<	<	40	<	20	80	40
B/Madagascar/6556/2016	P9	2016-11-02	MDCK1/MDCK1	2560	<	80	<	<	40	10	40	40	40
B/Ghana/DILI-16-1161/2016	1A	2016-11-14	MDCK1	2560	<	160	<	20	40	10	20	40	20
B/Madagascar/7507/2016	P9	2016-12-01	MDCK1	5120	<	160	<	<	80	20	40	40	40
B/Moscow/88/2016	P9	2016-12-06	MDCK1/MDCK1	2560	<	80	<	<	80	10	20	40	40
B/Georgia/1675/2016	P9	2016-12-09	MDCK2	5120	<	160	<	20	40	10	40	40	40
B/Tunis/14747/2016	P9	2016-12-15	MDCK1	2560	<	80	<	20	40	10	40	40	40
B/Moscow/90/2016	P9	2016-12-21	MDCK1/MDCK1	5120	<	80	<	<	80	10	40	80	40
B/Bosnia & Herzegovina/13-G/2016	P9	2016-12-22	MDCK1	2560	<	160	<	40	80	20	40	80	40
B/Moscow/94/2016	P9	2016-12-22	MDCK1/MDCK1	5120	<	160	<	<	80	20	40	80	80
B/Hong Kong/8/2017	P9	2016-12-28	MDCK1/MDCK1	2560	<	160	<	<	40	20	40	40	40
B/Moscow/96/2016	P9	2016-12-28	MDCK1/MDCK1	2560	<	80	<	<	40	10	20	80	40
B/Athens GR/77/2017	P9	2017-01-04	MDCK1	2560	<	160	<	<	40	20	40	80	40
B/Hong Kong/15/2017	P9	2017-01-07	MDCK1/MDCK1	2560	<	80	<	<	40	10	40	40	40
B/Georgia/43/2017	P9	2017-01-10	MDCK1	2560	<	160	<	20	40	20	40	80	40
B/Athens GR/576/2017	P9	2017-01-19	MDCK1	2560	<	160	<	20	40	20	40	80	40
B/Athens GR/580/2017	P9	2017-01-19	MDCK1	1280	<	80	<	<	20	10	20	20	20

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):
[1] < = <40; [2] < = <10; [3] hyperimmune sheep serum; [4] < = <20

Vaccine
NH 2016-17
SH 2017

Sequences in phylogenetic trees

Table 11-5. Antigenic analyses of influenza B viruses (Victoria lineage) with collection dates after 2016-08-31 - Summary

Viruses	Other information	Passage history / Ferret number / Genetic group	Collection date	Passage history	B/Mal 2506/04 Egg F41/14[*2]	B/Bris 60/08 Egg F26/13[*2]	B/Malta 636714/11 Egg F29/13[*1]	B/Jhb 3964/12 Egg F01/13[*4]	B/For V2367/12 MDCK F04/13[*2]	B/Sth Aus 81/12 Egg F41/13[*2]	B/HK 514/09 MDCK F09/13[*2]	B/Ireland 3154/16 MDCK F15/16[*4]	B/Nord-West 1/16 MDCK F16/16[*4]
					Haemagglutination inhibition titre								
					Post-infection ferret antisera								
		Genetic group				1A	1A	1A	1A	1A	1B	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004			2004-12-06	E3/E7	320	20	<	80	40	20	10	<	<
B/Brisbane/60/2008		1A	2008-08-04	E4/E4	80	640	320	320	160	160	80	40	20
B/Malta/636714/2011		1A	2011-03-07	E4/E1	40	640	320	320	160	160	40	<	20
B/Johannesburg/3964/2012		1A	2012-08-03	E1/E2	320	1280	1280	1280	1280	1280	320	160	320
B/Formosa/V2367/2012		1A	2012-08-06	MDCK1/MDCK3	20	640	160	160	160	640	80	40	40
B/South Australia/81/2012		1A	2012-11-28	E4/E1	40	320	160	160	80	640	40	40	<
B/Hong Kong/514/2009		1B	2009-10-11	MDCK3	<	80	<	<	20	10	40	40	40
B/Ireland/3154/2016		1A	2016-01-14	MDCK1/MDCK4	<	80	<	<	40	20	40	80	80
B/Nordrhein-Westfalen/1/2016		1A	2016-01-04	C2/MDCK1	<	80	<	40	40	20	40	80	80
TEST VIRUSES													
TEST VIRUSES													
Number of viruses tested*					42	42	42	42	42	42	42	42	42
No with titre reduction ≤2-fold						3	1		6		34	38	35
%						7.1	2.4		14.2		81.0	90.4	83.3
No with titre reduction =4-fold						23		1	23	1	7	1	5
%						54.7		2.4	54.8	2.4	16.6	2.4	11.9
No with titre reduction ≥8-fold					42	16	41	41	13	41	1	3	2
%					100.0	38.2	97.6	97.6	31.0	97.6	2.4	7.2	4.8

* All viruses sequenced carried HA genes that fell in the B/Brisbane/60/2008 clade (clade 1A)

Vaccine NH 2016-17 SH 2017

Reference virus results are taken from individual tables as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Figure 13. Phylogenetic comparison of influenza B (Victoria-lineage) HA genes

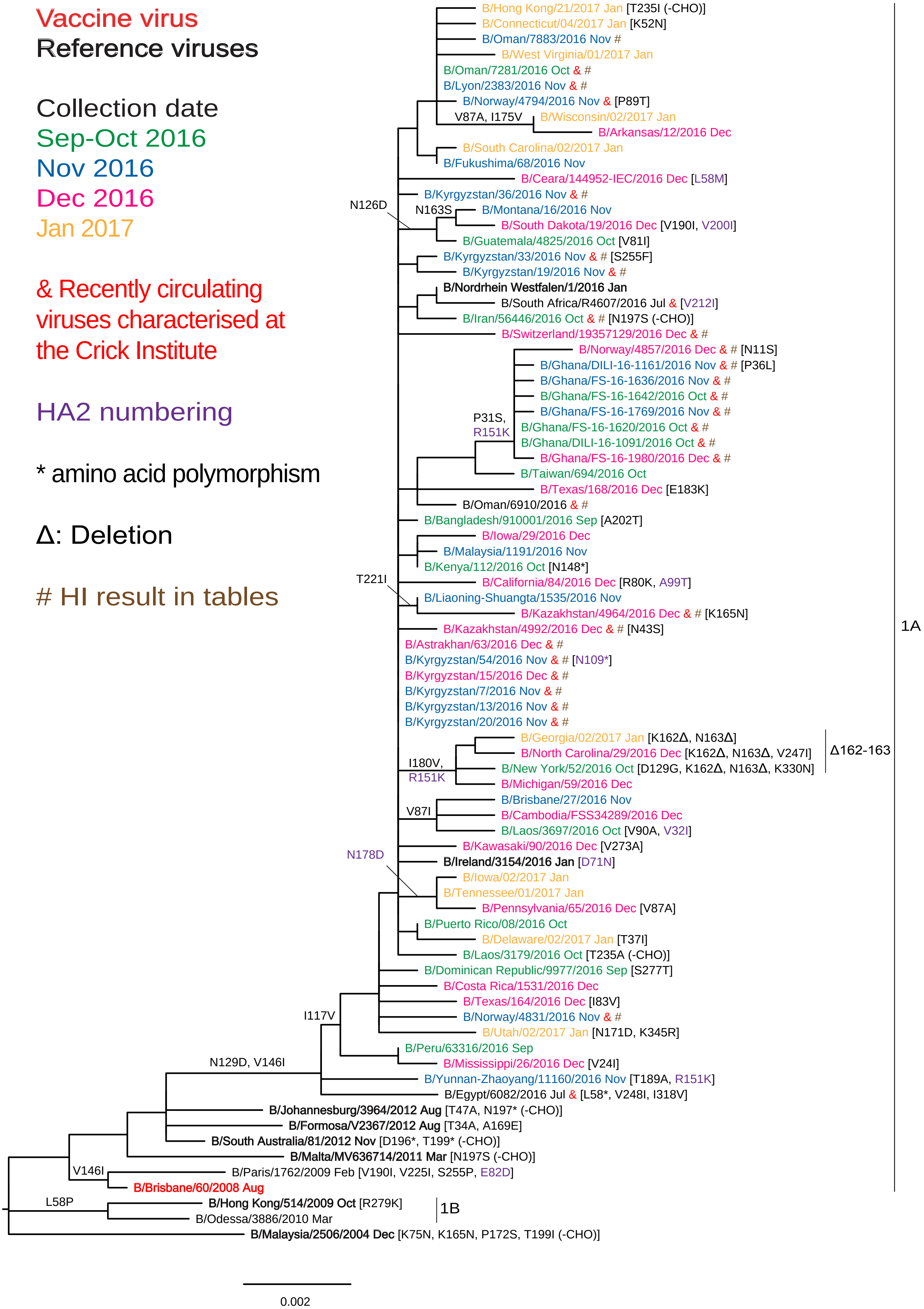


Figure 14. HA protein alignment for B (Victoria-lineage) viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

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                                    10        20        30        40        50        60        70        80        90       100
                               ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004           DRICTGITSSNSPHVVKTATQGEVNVTGVIPLTTTPTKSHFANLKGTETRGKLCPKCLNCTDLDVALGRPKCTGNIPSARVSILHEVRPVTSGCFPIMHD
B/Hong Kong/514/2009           .........................................................P................K......................... 1B
B/Brisbane/60/2008             ..........................................................................K......................... 1A
B/Malta/MV636714/2011          ..........................................................................K......................... 1A
B/South Australia/81/2012      ..........................................................................K......................... 1A
B/Formosa/V2367/2012           .................................A........................................K......................... 1A
B/Johannesburg/3964/2012       ..............................................A...........................K......................... 1A
B/Yunnan-Zhaoyang/11160/2016   ..........................................................................K......................... 1A
B/Norway/4831/2016             ..........................................................................K......................... 1A
B/Pennsylvania/65/2016         ..........................................................................K...........A............. 1A
B/Iowa/02/2017                 ..........................................................................K......................... 1A
B/Ireland/3154/2016            ..........................................................................K......................... 1A
B/Cambodia/FSS34289/2016       ..........................................................................K...........I............. 1A
B/Brisbane/27/2016             ..........................................................................K...........I............. 1A
B/Michigan/59/2016             ..........................................................................K......................... 1A
B/New York/52/2016             ..........................................................................K......................... 1A
B/Georgia/02/2017              ..........................................................................K......................... 1A
B/Kyrgyzstan/15/2016           ..........................................................................K......................... 1A
B/Astrakhan/63/2016            ..........................................................................K......................... 1A
B/Kazakhstan/4964/2016         ..........................................................................K......................... 1A
B/Kenya/112/2016               ..........................................................................K......................... 1A
B/Oman/6910/2016               ..........................................................................K......................... 1A
B/Ghana/FS-16-1980/2016        ..............................S...........................................K......................... 1A
B/Norway/4857/2016             ..........S...................S...........................................K......................... 1A
B/Switzerland/19357129/2016    ..........................................................................K......................... 1A
B/Nordrhein Westfalen/1/2016   ..........................................................................K......................... 1A
B/Guatemala/4825/2016          ..........................................................................K.....I................... 1A
B/South Dakota/19/2016         ..........................................................................K......................... 1A
B/Kyrgyzstan/36/2016           ..........................................................................K......................... 1A
B/Arkansas/12/2016             ..........................................................................K...........A............. 1A
B/Norway/4794/2016             ..........................................................................K.............T........... 1A
B/Oman/7883/2016               ..........................................................................K......................... 1A

                                   110       120       130       140       150       160       170       180       190       200
                               ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004           RTKIRQLPNLLRGYEHIRLSTHNVINAENAPGGPYKIGTSGSCPNVTNGNGFFATMAWAVPKNDNNKTATNSLTIEVPYICTEGEDQITVWGFHSDNEIQ
B/Hong Kong/514/2009           ................................................................K......P............................
B/Brisbane/60/2008             .............................................I..................K......P..........................T.
B/Malta/MV636714/2011          ................................................................K......P........................S.T.
B/South Australia/81/2012      ................................................................K......P.......................X..X.
B/Formosa/V2367/2012           ................................................................K...E..P..........................T.
B/Johannesburg/3964/2012       ................................................................K......P........................X.T.
B/Yunnan-Zhaoyang/11160/2016   ............................D................I..................K......P................A.........T.
B/Norway/4831/2016             ................V...........D................I..................K......P..........................T.
B/Pennsylvania/65/2016         ................V...........D................I..................K......P..........................T.
B/Iowa/02/2017                 ................V...........D................I..................K......P..........................T.
B/Ireland/3154/2016            ................V...........D................I..................K......P..........................T.
B/Cambodia/FSS34289/2016       ................V...........D................I..................K......P..........................T.
B/Brisbane/27/2016             ................V...........D................I..................K......P..........................T.
B/Michigan/59/2016             ................V...........D................I..................K......P.......V..................T.
B/New York/52/2016             ................V...........G................I...............--.K......P.......V..................T.
B/Georgia/02/2017              ................V...........D................I...............--.K......P.......V..................T.
B/Kyrgyzstan/15/2016           ................V...........D................I..................K......P..........................T.
B/Astrakhan/63/2016            ................V...........D................I..................K......P..........................T.
B/Kazakhstan/4964/2016         ................V...........D................I.........................P..........................T.
B/Kenya/112/2016               ................V...........D................I.X................K......P..........................T.
B/Oman/6910/2016               ................V...........D................I..................K......P..........................T.
B/Ghana/FS-16-1980/2016        ................V...........D................I..................K......P..........................T.
B/Norway/4857/2016             ................V...........D................I..................K......P..........................T.
B/Switzerland/19357129/2016    ................V...........D................I..................K......P..........................T.
B/Nordrhein Westfalen/1/2016   ................V...........D................I..................K......P..........................T.
B/Guatemala/4825/2016          ................V........D..D................I..................K......P..........................T.
B/South Dakota/19/2016         ................V........D..D................I................S.K......P.................I........T.
B/Kyrgyzstan/36/2016           ................V...........D................I..................K......P..........................T.
B/Arkansas/12/2016             ................V...........D................I..................K......P..V.......................T.
B/Norway/4794/2016             ................V...........D................I..................K......P..........................T.
B/Oman/7883/2016               ................V...........D................I..................K......P..........................T.

                                   210       220       230       240       250       260       270       280       290       300
                               ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004           MAKLYGDSKPQKFTSSANGVTTHYVSQIGGFPNQTEDGGLPQSGRIVVDYMVQKSGKTGTITYQRGILLPQKVWCASGRSKVIKGSLPLIGEADCLHEKY
B/Hong Kong/514/2009           ..............................................................................K.....................
B/Brisbane/60/2008             ....................................................................................................
B/Malta/MV636714/2011          ....................................................................................................
B/South Australia/81/2012      ....................................................................................................
B/Formosa/V2367/2012           ....................................................................................................
B/Johannesburg/3964/2012       ....................................................................................................
B/Yunnan-Zhaoyang/11160/2016   ....................................................................................................
B/Norway/4831/2016             ....................................................................................................
B/Pennsylvania/65/2016         ....................................................................................................
B/Iowa/02/2017                 ....................................................................................................
B/Ireland/3154/2016            ....................................................................................................
B/Cambodia/FSS34289/2016       ....................................................................................................
B/Brisbane/27/2016             ....................................................................................................
B/Michigan/59/2016             ....................................................................................................
B/New York/52/2016             ....................................................................................................
B/Georgia/02/2017              ....................................................................................................
B/Kyrgyzstan/15/2016           ....................................................................................................
B/Astrakhan/63/2016            ....................................................................................................
B/Kazakhstan/4964/2016         ....................I...............................................................................
B/Kenya/112/2016               ....................................................................................................
B/Oman/6910/2016               ....................................................................................................
B/Ghana/FS-16-1980/2016        ....................................................................................................
B/Norway/4857/2016             ....................................................................................................
B/Switzerland/19357129/2016    ....................................................................................................
B/Nordrhein Westfalen/1/2016   ....................................................................................................
B/Guatemala/4825/2016          ....................................................................................................
B/South Dakota/19/2016         ....................................................................................................
B/Kyrgyzstan/36/2016           ....................................................................................................
B/Arkansas/12/2016             ....................................................................................................
B/Norway/4794/2016             ....................................................................................................
B/Oman/7883/2016               ....................................................................................................
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                                     310       320       330       340         3        13        23        33        43        53
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          GGLNKSKPYYTGEHAKAIGNCPIWVKTPLKLANGTKYRPPAKLLKERGFFGAIAGFLEGGWEGMIAGWHGYTSHGAHGVAVAADLKSTQEAINKITKNLN
B/Hong Kong/514/2009          ....................................................................................................
B/Brisbane/60/2008            ....................................................................................................
B/Malta/MV636714/2011         ....................................................................................................
B/South Australia/81/2012     ....................................................................................................
B/Formosa/V2367/2012          ....................................................................................................
B/Johannesburg/3964/2012      ....................................................................................................
B/Yunnan-Zhaoyang/11160/2016  ....................................................................................................
B/Norway/4831/2016            ....................................................................................................
B/Pennsylvania/65/2016        ....................................................................................................
B/Iowa/02/2017                ....................................................................................................
B/Ireland/3154/2016           ....................................................................................................
B/Cambodia/FSS34289/2016      ....................................................................................................
B/Brisbane/27/2016            ....................................................................................................
B/Michigan/59/2016            ....................................................................................................
B/New York/52/2016            ..............................N.....................................................................
B/Georgia/02/2017             ....................................................................................................
B/Kyrgyzstan/15/2016          ....................................................................................................
B/Astrakhan/63/2016           ....................................................................................................
B/Kazakhstan/4964/2016        ....................................................................................................
B/Kenya/112/2016              ....................................................................................................
B/Oman/6910/2016              ....................................................................................................
B/Ghana/FS-16-1980/2016       ....................................................................................................
B/Norway/4857/2016            ....................................................................................................
B/Switzerland/19357129/2016   ....................................................................................................
B/Nordrhein Westfalen/1/2016  ....................................................................................................
B/Guatemala/4825/2016         ....................................................................................................
B/South Dakota/19/2016        ....................................................................................................
B/Kyrgyzstan/36/2016          ....................................................................................................
B/Arkansas/12/2016            ....................................................................................................
B/Norway/4794/2016            ....................................................................................................
B/Oman/7883/2016              ....................................................................................................
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                                      63        73        83        93       103       113       123       133       143       153
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          SLSELEVKNLQRLSGAMDELHNEILELDEKVDDLRADTISSQIELAVLLSNEGIINSEDEHLLALERKLKKMLGPSAVEIGNGCFETKHKCNQTCLDRIA
B/Hong Kong/514/2009          ....................................................................................................
B/Brisbane/60/2008            ....................................................................................................
B/Malta/MV636714/2011         ....................................................................................................
B/South Australia/81/2012     ....................................................................................................
B/Formosa/V2367/2012          ....................................................................................................
B/Johannesburg/3964/2012      ....................................................................................................
B/Yunnan-Zhaoyang/11160/2016  .................................................................................................K..
B/Norway/4831/2016            ....................................................................................................
B/Pennsylvania/65/2016        ....................................................................................................
B/Iowa/02/2017                ....................................................................................................
B/Ireland/3154/2016           .................N..................................................................................
B/Cambodia/FSS34289/2016      ....................................................................................................
B/Brisbane/27/2016            ....................................................................................................
B/Michigan/59/2016            .................................................................................................K..
B/New York/52/2016            .................................................................................................K..
B/Georgia/02/2017             .................................................................................................K..
B/Kyrgyzstan/15/2016          ....................................................................................................
B/Astrakhan/63/2016           ....................................................................................................
B/Kazakhstan/4964/2016        ....................................................................................................
B/Kenya/112/2016              ....................................................................................................
B/Oman/6910/2016              ....................................................................................................
B/Ghana/FS-16-1980/2016       .................................................................................................K..
B/Norway/4857/2016            .................................................................................................K..
B/Switzerland/19357129/2016   ....................................................................................................
B/Nordrhein Westfalen/1/2016  ....................................................................................................
B/Guatemala/4825/2016         ....................................................................................................
B/South Dakota/19/2016        ....................................................................................................
B/Kyrgyzstan/36/2016          ....................................................................................................
B/Arkansas/12/2016            ....................................................................................................
B/Norway/4794/2016            ....................................................................................................
B/Oman/7883/2016              ....................................................................................................
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                                     163       173       183       193       203       213       223
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          AGTFDAGEFSLPTFDSLNITAASLNDDGLDNHTILLYYSTAASSLAVTLMIAIFVVYMVSRDNVSCSICL
B/Hong Kong/514/2009          ......................................................................
B/Brisbane/60/2008            ......................................................................
B/Malta/MV636714/2011         ......................................................................
B/South Australia/81/2012     ......................................................................
B/Formosa/V2367/2012          ......................................................................
B/Johannesburg/3964/2012      ......................................................................
B/Yunnan-Zhaoyang/11160/2016  ......................................................................
B/Norway/4831/2016            ......................................................................
B/Pennsylvania/65/2016        ........................D.............................................
B/Iowa/02/2017                ........................D.............................................
B/Ireland/3154/2016           ......................................................................
B/Cambodia/FSS34289/2016      ......................................................................
B/Brisbane/27/2016            ......................................................................
B/Michigan/59/2016            ......................................................................
B/New York/52/2016            ......................................................................
B/Georgia/02/2017             ......................................................................
B/Kyrgyzstan/15/2016          ......................................................................
B/Astrakhan/63/2016           ......................................................................
B/Kazakhstan/4964/2016        ......................................................................
B/Kenya/112/2016              ......................................................................
B/Oman/6910/2016              ......................................................................
B/Ghana/FS-16-1980/2016       ......................................................................
B/Norway/4857/2016            ......................................................................
B/Switzerland/19357129/2016   ......................................................................
B/Nordrhein Westfalen/1/2016  ......................................................................
B/Guatemala/4825/2016         ......................................................................
B/South Dakota/19/2016        ..............................................I.......................
B/Kyrgyzstan/36/2016          ......................................................................
B/Arkansas/12/2016            ......................................................................
B/Norway/4794/2016            ......................................................................
B/Oman/7883/2016              ......................................................................
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Vaccine virus: **Reference virus**: **Genetic group**: **HA1/HA2 boundary**: **Potential N-linked glycosylation site**

Figure 15 . Phylogenetic comparison of influenza B (Victoria-lineage) NA genes

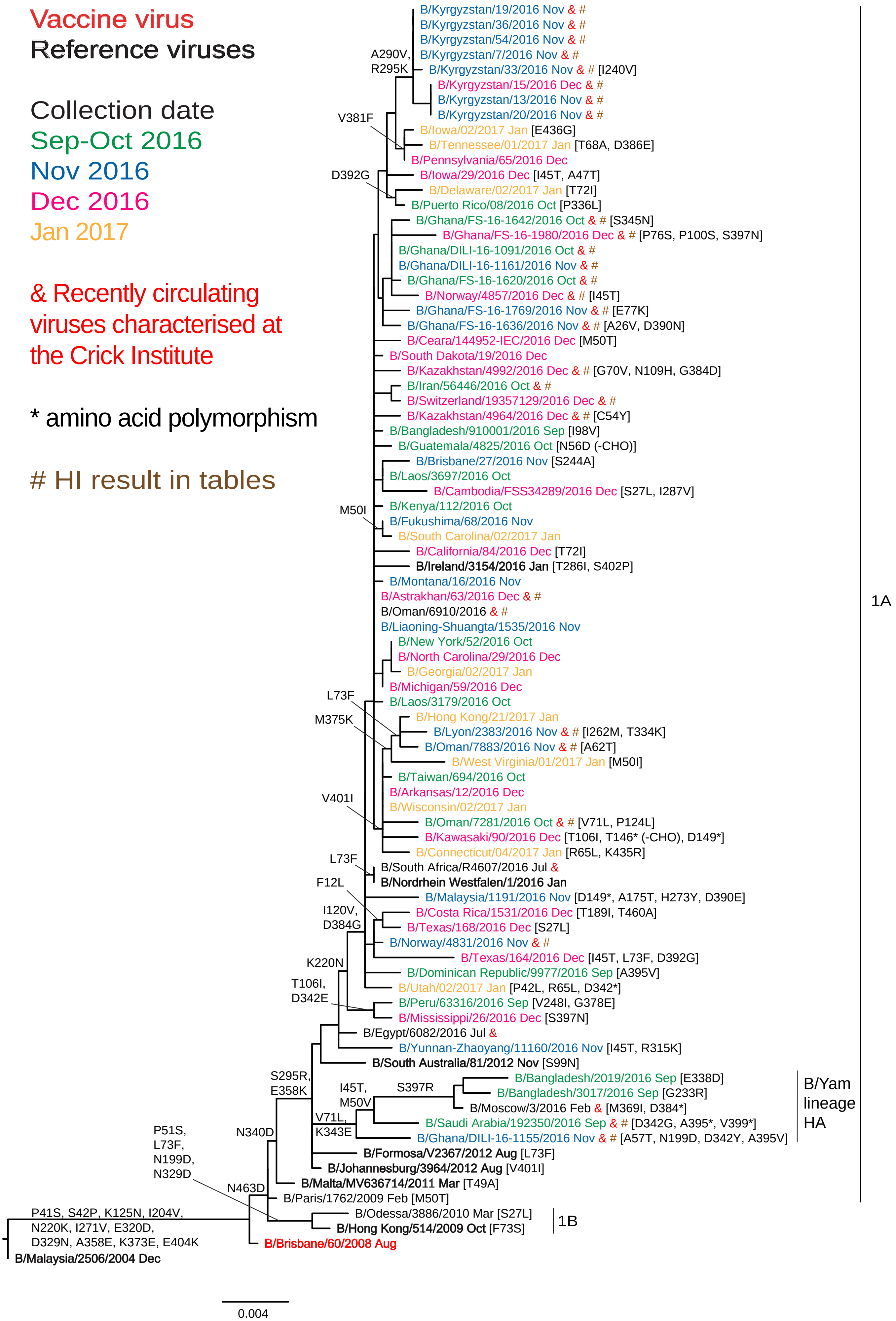

Figure 16. NA protein alignment for B (Victoria-lineage) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

```
                                       10        20        30        40        50        60        70        80        90       100
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004            MLPSTIQTLTLFLTSGGVLLSLYVSASLSYLLYSDILLKFPSTEITAPTMPLDCANASNVQAVNRSATKGVTLLLPEPEWTYPRLSCPGSTFQKALLISP
B/Hong Kong/514/2009            ........................................SP........S.....................S........................... 1B
B/Brisbane/60/2008              ........................................SP.......................................................... 1A
B/Malta/MV636714/2011           ........................................SP......A................................................... 1A
B/South Australia/81/2012       ........................................SP........................................................N. 1A
B/Formosa/V2367/2012            ........................................SP..............................F........................... 1A
B/Johannesburg/3964/2012        ........................................SP.......................................................... 1A
B/Yunnan-Zhaoyang/11160/2016    ........................................SP..T....................................................... 1A
B/Norway/4831/2016              ........................................SP.......................................................... 1A
B/Pennsylvania/65/2016          ....................................X...SP.......................................................... 1A
B/Iowa/02/2017                  ........................................SP.......................................................... 1A
B/Ireland/3154/2016             ........................................SP.......................................................... 1A
B/Cambodia/FSS34289/2016        ..........................L.............SP.......................................................... 1A
B/Brisbane/27/2016              ........................................SP.......................................................... 1A
B/Michigan/59/2016              ........................................SP.......................................................... 1A
B/New York/52/2016              ........................................SP.......................................................... 1A
B/Georgia/02/2017               ........................................SP.......................................................... 1A
B/Kyrgyzstan/15/2016            ........................................SP.......................................................... 1A
B/Astrakhan/63/2016             ........................................SP.......................................................... 1A
B/Kazakhstan/4964/2016          ........................................SP...........Y.............................................. 1A
B/Kenya/112/2016                ........................................SP.......................................................... 1A
B/Oman/6910/2016                ........................................SP.......................................................... 1A
B/Ghana/FS-16-1980/2016         ........................................SP.................................S.......................S 1A
B/Norway/4857/2016              ........................................SP..T....................................................... 1A
B/Switzerland/19357129/2016     ........................................SP.......................................................... 1A
B/Nordrhein Westfalen/1/2016    ........................................SP..............................F........................... 1A
B/Guatemala/4825/2016           ........................................SP.............D............................................ 1A
B/South Dakota/19/2016          ........................................SP.......................................................... 1A
B/Kyrgyzstan/36/2016            ........................................SP.......................................................... 1A
B/Arkansas/12/2016              ........................................SP.......................................................... 1A
B/Lyon/2383/2016                ........................................SP..............................F........................... 1A
B/Oman/7883/2016                ........................................SP...................T..........F........................... 1A

                                      110       120       130       140       150       160       170       180       190       200
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004            HRFGETKGNSAPLIIREPFIACGPKECKHFALTHYAAQPGGYYNGTRGDRNKLRHLISVKLGKIPTVENSIFHMAAWSGSACHDGKEWTYIGVDGPDNNA
B/Hong Kong/514/2009            ........................N.........................................................................D.
B/Brisbane/60/2008              ........................N...........................................................................
B/Malta/MV636714/2011           ........................N...........................................................................
B/South Australia/81/2012       ........................N...........................................................................
B/Formosa/V2367/2012            ........................N...........................................................................
B/Johannesburg/3964/2012        ........................N...........................................................................
B/Yunnan-Zhaoyang/11160/2016    ........................N...........................................................................
B/Norway/4831/2016              ...................V....N...........................................................................
B/Pennsylvania/65/2016          ...................V....N...........................................................................
B/Iowa/02/2017                  ...................V....N...........................................................................
B/Ireland/3154/2016             ...................V....N...........................................................................
B/Cambodia/FSS34289/2016        ...................V....N...........................................................................
B/Brisbane/27/2016              ...................V....N...........................................................................
B/Michigan/59/2016              ...................V....N...........................................................................
B/New York/52/2016              ...................V....N...........................................................................
B/Georgia/02/2017               ...................V....N...........................................................................
B/Kyrgyzstan/15/2016            ...................V....N...........................................................................
B/Astrakhan/63/2016             ...................V....N...........................................................................
B/Kazakhstan/4964/2016          ...................V....N...........................................................................
B/Kenya/112/2016                ...................V....N...........................................................................
B/Oman/6910/2016                ...................V....N...........................................................................
B/Ghana/FS-16-1980/2016         ...................V....N...........................................................................
B/Norway/4857/2016              ...................V....N...........................................................................
B/Switzerland/19357129/2016     ...................V....N...........................................................................
B/Nordrhein Westfalen/1/2016    ...................V....N...........................................................................
B/Guatemala/4825/2016           ...................V....N...........................................................................
B/South Dakota/19/2016          ...................V....N...........................................................................
B/Kyrgyzstan/36/2016            ...................V....N...........................................................................
B/Arkansas/12/2016              ...................V....N...........................................................................
B/Lyon/2383/2016                ...................V....N...........................................................................
B/Oman/7883/2016                ...................V....N...........................................................................

                                      210       220       230       240       250       260       270       280       290       300
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004            LLKIKYGEAYTDTYHSYANNILRTQESACNCIGGNCYLMITDGSASGVSECRFLKIREGRIIKEIFPTGRIKHTEECTCGFASNKTIECACRDNSYTAKR
B/Hong Kong/514/2009            ...V...............K..................................................V.............................
B/Brisbane/60/2008              ...V...............K..................................................V.............................
B/Malta/MV636714/2011           ...V...............K..................................................V.............................
B/South Australia/81/2012       ...V...............K..................................................V.......................R.....
B/Formosa/V2367/2012            ...V...............K..................................................V.......................R.....
B/Johannesburg/3964/2012        ...V...............K..................................................V.......................R.....
B/Yunnan-Zhaoyang/11160/2016    ...V...............K..................................................V.......................R.....
B/Norway/4831/2016              ...V..................................................................V.......................R.....
B/Pennsylvania/65/2016          ...V..................................................................V.......................R.....
B/Iowa/02/2017                  ...V..................................................................V.......................R.....
B/Ireland/3154/2016             ...V..................................................................V..............I........R.....
B/Cambodia/FSS34289/2016        ...V..................................................................V...............V.......R.....
B/Brisbane/27/2016              ...V.......................................A..........................V.......................R.....
B/Michigan/59/2016              ...V..................................................................V.......................R.....
B/New York/52/2016              ...V..................................................................V.......................R.....
B/Georgia/02/2017               ...V..................................................................V.......................R.....
B/Kyrgyzstan/15/2016            ...V..................................................................V..................V....K.....
B/Astrakhan/63/2016             ...V..................................................................V.......................R.....
B/Kazakhstan/4964/2016          ...V..................................................................V.......................R.....
B/Kenya/112/2016                ...V..................................................................V.......................R.....
B/Oman/6910/2016                ...V..................................................................V.......................R.....
B/Ghana/FS-16-1980/2016         ...V..................................................................V.......................R.....
B/Norway/4857/2016              ...V..................................................................V.......................R.....
B/Switzerland/19357129/2016     ...V..................................................................V.......................R.....
B/Nordrhein Westfalen/1/2016    ...V..................................................................V.......................R.....
B/Guatemala/4825/2016           ...V..................................................................V.......................R.....
B/South Dakota/19/2016          ...V..................................................................V.......................R.....
B/Kyrgyzstan/36/2016            ...V..................................................................V..................V....K.....
B/Arkansas/12/2016              ...V..................................................................V.......................R.....
B/Lyon/2383/2016                ...V.........................................................M........V.......................R.....
B/Oman/7883/2016                ...V..................................................................V.......................R.....
```

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                                        310       320       330       340       350       360       370       380       390       400
                                   ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004               PFVKLNVETDTAEIRLMCTETYLDTPRPDDGSITGPCESNGDKGSGGIKGGFVHQRMASKIGRWYSRTMSKTKRMGMGLYVKYDGDPWADSDALAFSGVM
B/Hong Kong/514/2009               ...................D.....................................E..............E...........................
B/Brisbane/60/2008                 ...................D........N............................E..............E...........................
B/Malta/MV636714/2011              ...................D........N............................E..............E...........................
B/South Australia/81/2012          ...................D........N..........D.................K..............E...........................
B/Formosa/V2367/2012               ...................D........N..........D.................K..............E...........................
B/Johannesburg/3964/2012           ...................D........N..........D.................K..............E...........................
B/Yunnan-Zhaoyang/11160/2016       ..............K....D........N..........D.................K..............E...........................
B/Norway/4831/2016                 ...................D........N..........D.................K..............E..........G................
B/Pennsylvania/65/2016             ...................D........N..........D.................K..............E.......F..G................
B/Iowa/02/2017                     ...................D........N..........D.................K..............E.......F..G................
B/Ireland/3154/2016                ...................D........N..........D.................K..............E..........G................
B/Cambodia/FSS34289/2016           ...................D........N..........D.................K..............E..........G................
B/Brisbane/27/2016                 ...................D........N..........D.................K..............E..........G................
B/Michigan/59/2016                 ...................D........N..........D.................K..............E..........G................
B/New York/52/2016                 ...................D........N..........D.................K..............E..........G................
B/Georgia/02/2017                  ...................D........N..........D.................K..............E..........G................
B/Kyrgyzstan/15/2016               ...................D........N..........D.................K..............E..........G................
B/Astrakhan/63/2016                ...................D........N..........D.................K..............E..........G................
B/Kazakhstan/4964/2016             ...................D........N..........D.................K..............E..........G................
B/Kenya/112/2016                   ...................D........N..........D.................K..............E..........G................
B/Oman/6910/2016                   ...................D........N..........D.................K..............E..........G................
B/Ghana/FS-16-1980/2016            ...................D........N..........D.................K..............E..........G............N...
B/Norway/4857/2016                 ...................D........N..........D.................K..............E..........G................
B/Switzerland/19357129/2016        ...................D........N..........D.................K..............E..........G................
B/Nordrhein Westfalen/1/2016       ...................D........N..........D.................K..............E..........G................
B/Guatemala/4825/2016              ...................D........N..........D.................K..............E..........G................
B/South Dakota/19/2016             ...................D........N..........D.................K..............E..........G................
B/Kyrgyzstan/36/2016               ...................D........N..........D.................K..............E..........G................
B/Arkansas/12/2016                 ...................D........N..........D.................K..............E..........G................
B/Lyon/2383/2016                   ...................D........N....K.....D.................K..............E.K........G................
B/Oman/7883/2016                   ...................D........N..........D.................K..............E.K........G................

                                        410       420       430       440       450       460
                                   ....|....|....|....|....|....|....|....|....|....|....|....|....|.
B/Malaysia/2506/2004               VSMEEPGWYSFGFEIKDKKCDVPCIGIEMVHDGGKETWHSAATAIYCLMGSGQLLWDTVTGVNMAL
B/Hong Kong/514/2009               ...K..........................................................D...
B/Brisbane/60/2008                 ...K..............................................................
B/Malta/MV636714/2011              ...K..........................................................D...
B/South Australia/81/2012          ...K..........................................................D...
B/Formosa/V2367/2012               ...K..........................................................D...
B/Johannesburg/3964/2012           I..K..........................................................D...
B/Yunnan-Zhaoyang/11160/2016       ...K..........................................................D...
B/Norway/4831/2016                 ...K..........................................................D...
B/Pennsylvania/65/2016             ...K..........................................................D...
B/Iowa/02/2017                     ...K...............................G..........................D...
B/Ireland/3154/2016                .P.K..........................................................D...
B/Cambodia/FSS34289/2016           ...K..........................................................D...
B/Brisbane/27/2016                 ...K..........................................................D...
B/Michigan/59/2016                 ...K..........................................................D...
B/New York/52/2016                 ...K..........................................................D...
B/Georgia/02/2017                  ...K..........................................................D...
B/Kyrgyzstan/15/2016               ...K..........................................................D...
B/Astrakhan/63/2016                ...K..........................................................D...
B/Kazakhstan/4964/2016             ...K..........................................................D...
B/Kenya/112/2016                   ...K..........................................................D...
B/Oman/6910/2016                   ...K..........................................................D...
B/Ghana/FS-16-1980/2016            ...K..........................................................D...
B/Norway/4857/2016                 ...K..........................................................D...
B/Switzerland/19357129/2016        ...K..........................................................D...
B/Nordrhein Westfalen/1/2016       ...K..........................................................D...
B/Guatemala/4825/2016              ...K..........................................................D...
B/South Dakota/19/2016             ...K..........................................................D...
B/Kyrgyzstan/36/2016               ...K..........................................................D...
B/Arkansas/12/2016                 I..K..........................................................D...
B/Lyon/2383/2016                   I..K..........................................................D...
B/Oman/7883/2016                   I..K..........................................................D...
```

Vaccine virus: **Reference virus**: **Genetic group**: **Potential N-linked glycosylation site**

Table 12. Summary of influenza B (Yamagata-lineage) viruses analysed, collected after 2016-08-31

MONTH / Country	Number received[1]	Number propagated[2]	≤2 fold[3]	4 fold[3]	≥8 fold[3]
	B Yamagata lineage				
2016					
SEPTEMBER					
Saudi Arabia	1	1	1		
OCTOBER					
Denmark	1	1			1
Ghana	2	in process			
Kyrgyzstan	1	1		1	
United Kingdom	1	1	1		
NOVEMBER					
Finland	1	1			1
Ghana	6	4	4		
Kyrgyzstan	1	1	1		
Norway	4	3	1	1	1
DECEMBER					
Croatia	2	in process			
Iran	1	1			1
Kyrgyzstan	1	1		1	
Latvia	3	3	3		
Luxembourg	1	1		1	
Norway	3	2	1	1	
Slovenia	4	4		1	3
Switzerland	1	1			1
2017					
JANUARY					
Germany	4	4	1	2	1
Hong Kong	2	2		2	
Latvia	1	1		1	
Slovenia	5	5		1	4
15 Countries	46	38	13	12	13
			34.2%	31.6%	34.2%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared
2. Propagated to sufficient titre to perform HI assay
3. Reduction in HI titre with ferret antiserum raised against B/Phuket/3073/2013 (egg-grown vaccine virus)

Table 13-1. Antigenic analyses of influenza B viruses (Yamagata lineage) 2016-11-22 + 2016-12-20 + 2017-01-10 + 2017-01-17

Viruses	Other information		Collection date	Passage history	Haemagglutination inhibition titre										
					Post-infection ferret antisera										
					B/Phuket 3073/13	B/Fl 4/06	B/Bris 3/07	B/Estonia 55669/11	B/Mass 02/12	B/Mass 02/12	B/Wis 1/10	B/Stock 12/11	B/Phuket 3073/13	B/Phuket 3073/13	B/HK 3417/14
		Passage history			Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg
		Ferret number			SH 614*1,3	F17/13*1	F38/14*2	F32/12*2	F05/15*2	F42/14*2	F10/13*2	F06/15*2	F35/14*2	F36/14*2	St Judes F715/14*2,4
		Genetic Group			3	1	2	2	2	2	3	3	3	3	3
REFERENCE VIRUSES															
B/Florida/4/2006		1	2006-12-15	E7/E1	2560	1280	640	80	80	640	160	160	20	160	160
B/Brisbane/3/2007		2	2007-09-03	E2/E3	1280	640	320	40	80	640	80	80	10	80	80
B/Estonia/55669/2011		2	2011-03-14	MDCK2/MDCK3	1280	80	80	80	160	80	40	20	40	40	80
B/Massachusetts/02/2012		2	2012-03-13	MDCK1/C2/MDCK3	2560	640	640	320	320	640	320	160	80	160	320
B/Massachusetts/02/2012		2	2012-03-13	E3/E3	1280	1280	320	40	80	640	160	160	20	160	160
B/Wisconsin/1/2010		3	2010-02-20	E3/E2	2560	320	320	20	20	320	320	160	40	160	160
B/Stockholm/12/2011		3	2011-03-28	E4/E1	1280	160	160	10	<	160	80	160	20	80	160
B/Phuket/3073/2013		3	2013-11-21	MDCK2/MDCK2	5120	320	160	320	640	320	320	160	640	640	320
B/Phuket/3073/2013		3	2013-11-21	E4/E2	2560	160	320	20	20	320	320	160	40	160	160
B/Hong Kong/3417/2014		3	2014-06-04	E4/E3	1280	80	40	10	<	40	80	20	10	40	160
TEST VIRUSES															
B/Arizona/10/2015		3	2015-11-02	C1/MDCK1	1280	80	20	20	<	20	40	20	80	40	80
B/Fukushima/23/2016		3	2016-03-25	MDCK2/MDCK2/MDCK1	5120	320	80	40	<	160	320	80	80	160	320
B/Jordan/20519/2016		3	2016-05-29	MDCK1	1280	160	40	20	<	40	80	20	40	40	80
B/Saudi Arabia/192350/2016		3	2016-09-12	MDCK1	2560	320	80	80	<	160	160	80	80	160	80
B/England/964/2016		3	2016-10-11	SIAT1/MDCK1	2560	160	80	40	<	80	160	40	80	80	80
B/Kyrgyzstan/1/2016		3	2016-10-27	MDCK1	1280	160	40	20	<	40	160	20	40	40	80
B/Kyrgyzstan/52/2016		3	2016-11-17	SIAT1	2560	160	40	40	<	80	160	40	80	80	160
B/Norway/4872/2016		3	2016-11-24	MDCK2	1280	40	20	10	<	10	40	20	40	10	80
B/Norway/4783/2016		3	2016-11-28	MDCK1	2560	320	80	80	80	160	80	40	160	160	160
B/Norway/4854/2016		3	2016-11-29	MDCK1	2560	160	40	20	10	40	80	20	40	40	160
B/Norway/4811/2016		3	2016-12-02	MDCK1	2560	320	40	40	20	80	80	40	80	80	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):
1 < = <40; 2 < = <10; 3 hyperimmune sheep serum; 4 RDE serum pre-adsorbed with TRBC

Vaccine#
NH 2016-17
SH 2017

B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 13-2. Antigenic analyses of influenza B viruses (Yamagata lineage) 2017-01-24

					Haemagglutination inhibition titre										
					Post-infection ferret antisera										
Viruses	Other information		Collection date	Passage history	B/Phuket 3073/13	B/Fl 4/06	B/Bris 3/07	B/Estonia 55669/11	B/Mass 02/12	B/Mass 02/12	B/Wis 1/10	B/Stock 12/11	B/Phuket 3073/13	B/Phuket 3073/13	B/HK 3417/14
		Passage history			Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg
		Ferret number			SH614*1,3	F17/13*1	F38/14*2	F32/12*2	F15/13*2	F42/14*2	F10/13*2	F06/15*2	F35/14*2	F36/14*2	St Judes F715/14*2,4
		Genetic Group			3	1	2	2	2	2	3	3	3	3	3
REFERENCE VIRUSES															
B/Florida/4/2006		1	2006-12-15	E7/E1	1280	640	320	40	80	640	80	160	40	80	160
B/Brisbane/3/2007		2	2007-09-03	E2/E2	1280	1280	640	80	160	1280	160	160	20	160	320
B/Estonia/55669/2011		2	2011-03-14	MDCK2/MDCK3	640	80	40	80	320	40	20	10	10	20	160
B/Massachusetts/02/2012		2	2012-03-13	MDCK1/C2/MDCK3	1280	320	160	160	320	320	80	40	20	80	320
B/Massachusetts/02/2012		2	2012-03-13	E3/E3	1280	640	320	40	80	640	80	80	10	80	160
B/Wisconsin/1/2010		3	2010-02-20	E3/E2	5120	640	320	20	40	640	640	160	40	320	320
B/Stockholm/12/2011		3	2011-03-28	E4/E1	1280	160	80	10	20	80	40	80	10	40	80
B/Phuket/3073/2013		3	2013-11-21	MDCK2/MDCK2	5120	160	160	80	80	320	320	80	320	320	160
B/Phuket/3073/2013		3	2013-11-21	E4/E3	2560	320	160	10	40	160	160	80	40	160	160
B/Hong Kong/3417/2014		3	2014-06-04	E4/E1	1280	80	40	<	20	40	40	20	<	20	160
TEST VIRUSES															
B/Ghana/DILI-16-1149/2016		3	2016-11-09	Cx/MDCK1	5120	160	160	160	320	320	160	80	640	320	320
B/Ghana/FS-16-1747/2016		3	2016-11-10	Cx/MDCK1	2560	80	40	40	40	40	80	40	80	80	160
B/Ghana/DILI-16-1155/2016		3	2016-11-10	Cx/MDCK1	5120	160	160	160	320	160	160	80	640	320	160
B/Finland/657/2016		3	2016-11-15	MDCK1/MDCK1	1280	40	20	10	20	10	40	20	20	20	80
B/Ghana/FS-16-1880/2016		3	2016-11-25	Cx/MDCK1	5120	160	80	160	80	160	160	80	160	160	160
B/Switzerland/19309131/2016		3	2016-12-06	MDCK1	1280	40	20	10	20	10	20	20	20	20	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):
1 < = <40; 2 < = <10; 3 hyperimmune sheep serum; 4 RDE serum pre-adsorbed with TRBC

Vaccine#
NH 2016-17
SH 2017

\# B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 13-3. Antigenic analyses of influenza B viruses (Yamagata lineage) 2017-02-07

					Haemagglutination inhibition titre										
						Post-infection ferret antisera									
Viruses	Other information		Collection date	Passage history	B/Phuket 3073/13	B/Fl 4/06	B/Bris 3/07	B/Estonia 55669/11	B/Mass 02/12	B/Mass 02/12	B/Wis 1/10	B/Stock 12/11	B/Phuket 3073/13	B/Phuket 3073/13	B/HK 3417/14
Passage history		Passage history			Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg
Ferret number		Ferret number			SH614[*1,3]	F17/13[*1]	F38/14[*2]	F32/12[*2]	F05/15[*2]	F42/14[*2]	F10/13[*2]	F06/15[*2]	F35/14[*2]	F37/14[*2]	St Judes F715/14[*1,4]
Genetic Group		Genetic Group			3	1	2	2	2	2	3	3	3	3	3
REFERENCE VIRUSES															
B/Florida/4/2006		1	2006-12-15	E7/E1	1280	640	320	40	80	640	80	160	40	80	160
B/Brisbane/3/2007		2	2007-09-03	E2/E2	1280	1280	640	80	160	1280	160	160	20	160	320
B/Estonia/55669/2011		2	2011-03-14	MDCK2/MDCK3	640	80	40	80	320	40	20	10	10	20	160
B/Massachusetts/02/2012		2	2012-03-13	MDCK1/C2/MDCK3	1280	320	160	160	320	320	80	40	20	80	320
B/Massachusetts/02/2012		2	2012-03-13	E3/E3	1280	640	640	80	1280	640	160	160	10	160	320
B/Wisconsin/1/2010		3	2010-02-20	E3/E2	5120	640	320	20	40	640	640	160	40	320	320
B/Stockholm/12/2011		3	2011-03-28	E4/E1	1280	160	80	10	20	80	40	80	10	40	80
B/Phuket/3073/2013		3	2013-11-21	MDCK2/MDCK2	5120	160	160	80	80	320	320	80	320	320	160
B/Phuket/3073/2013		3	2013-11-21	E4/E3	2560	320	160	10	20	160	160	160	40	320	160
B/Hong Kong/3417/2014		3	2014-06-04	E4/E1	1280	80	40	<	20	40	40	20	<	20	160
TEST VIRUSES															
B/Denmark/36/2016		3	2016-10-07	MDCK3/MDCK1	2560	80	40	20	40	40	160	40	80	40	160
B/Kyrgyzstan/16/2016		3	2016-12-01	MDCKx/MDCK1	2560	80	40	20	20	80	160	40	80	80	160
A/Norway/4958/2016		3	2016-12-10	SIAT1	1280	20	20	20	20	40	80	40	80	80	160
B/Slovenia/2727/2016		3	2016-12-22	MDCK1/MDCK1	1280	80	40	20	40	40	80	20	40	40	160
B/Latvia/12-059779/2016		3	2016-12-23	C2/MDCK1	5120	160	320	320	1280	320	320	160	640	320	320
B/Latvia/12-067257/2016		3	2016-12-27	C2/MDCK1	5120	160	80	160	1280	160	160	160	320	160	320
B/Luxembourg/65508/2016		3	2016-12-27	MDCK1/MDCK1	2560	80	40	20	10	80	80	40	40	80	160
B/Latvia/12-064992/2016		3	2016-12-28	C2/MDCK1	5120	160	80	80	1280	160	160	80	160	160	160
B/Latvia/01-004346/2017		3	2017-01-02	C1/MDCK1	1280	80	40	20	160	40	80	40	80	80	160
B/Slovenia/107/2017		3	2017-01-03	MDCKx/MDCK1	1280	80	20	20	40	20	80	20	40	40	160
B/Slovenia/26/2017		3	2017-01-03	MDCKx/MDCK1	1280	40	20	40	320	20	80	20	80	80	160
B/Slovenia/99/2017		3	2017-01-04	MDCKx/MDCK1	1280	80	40	20	40	40	80	20	40	40	160
B/Slovenia/92/2017		3	2017-01-04	MDCKx/MDCK1	1280	80	20	10	40	40	80	20	40	40	160
B/Slovenia/210/2017		3	2017-01-09	MDCKx/MDCK1	2560	80	40	20	40	40	160	40	80	40	160
B/Hong Kong/19/2017		3	2017-01-10	MDCK1/MDCK1	2560	80	40	20	20	80	80	40	80	80	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):
1 < = <40; 2 < = <10; 3 hyperimmune sheep serum; 4 RDE serum pre-adsorbed with TRBC

Vaccine[#]
NH 2016-17
SH 2017

[#] B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 13-4. Antigenic analyses of influenza B viruses (Yamagata lineage) 2017-02-14

				Haemagglutination inhibition titre										
					Post-infection ferret antisera									
Viruses	Other information	Collection date	Passage history	B/Phuket 3073/13	B/Fl 4/06	B/Bris 3/07	B/Estonia 55669/11	B/Mass 02/12	B/Mass 02/12	B/Wis 1/10	B/Stock 12/11	B/Phuket 3073/13	B/Phuket 3073/13	B/HK 3417/14
Passage history	Passage history			Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg
Ferret number	Ferret number			SH614*1,3	F17/13*1	F38/14*2	F27/13*1	F05/15*1	F42/14*2	F10/13*2	F06/15*2	F35/14*2	F37/14*2	St Judes F715/14*1,4
Genetic Group	Genetic Group			3	1	2	2	2	2	3	3	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	640	640	320	160	640	160	160	20	160	160
B/Brisbane/3/2007	2	2007-09-03	E2/E2	2560	1280	1280	320	160	1280	320	320	20	320	320
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	80	80	640	160	40	40	20	40	40	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	320	320	640	320	320	160	80	40	160	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	1280	640	640	320	80	640	80	160	10	160	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	5120	640	320	320	40	640	640	320	80	640	640
B/Stockholm/12/2011	3	2011-03-28	Ex/E2	1280	160	80	40	<	80	80	160	20	80	80
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	320	320	320	640	320	320	160	640	320	320
B/Phuket/3073/2013	3	2013-11-21	E4/E3	2560	320	320	80	20	320	320	160	40	320	160
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	1280	80	40	40	10	80	80	40	10	80	160
TEST VIRUSES														
B/Slovenia/2604/2016	P9	2016-12-14	MDCKx/MDCK1	2560	80	40	80	40	40	80	20	40	40	160
B/Slovenia/2649/2016	P9	2016-12-19	MDCKx/MDCK1	2560	80	40	80	40	40	80	40	40	80	320
B/Slovenia/2666/2016	P9	2016-12-21	MDCKx/MDCK1	2560	80	40	80	<	40	80	20	40	40	160
B/Iran/70853/2016		2016-12-25	E1/E1	1280	10	40	80	<	40	40	20	<	40	160
B/Hong Kong/14/2017	P9	2017-01-06	MDCK1/MDCK1	2560	80	80	160	80	80	160	40	160	80	320
B/Berlin/1/2017	P9	2017-01-11	C1/MDCK1	2560	80	40	80	40	40	80	40	40	80	160
B/Nordrhein-Westfalen/1/2017	P9	2017-01-23	C1/MDCK1	2560	160	80	160	160	80	160	80	160	160	320
B/Niedersachsen/1/2017	P9	2017-01-23	C1/MDCK1	2560	80	40	80	40	80	80	40	80	80	160
B/Nordrhein-Westfalen/2/2017	P9	2017-01-26	C1/MDCK1	2560	80	40	80	40	40	80	20	40	40	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):
1 < = <40; 2 < = <10; 3 hyperimmune sheep serum; 4 RDE serum pre-adsorbed with TRBC

Vaccine#
NH 2016-17
SH 2017

B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 13-5. Antigenic analyses of influenza B viruses (Yamagata lineage) with collection dates after 2016-08-31- Summary

					Haemagglutination inhibition titre									
					Post-infection ferret antisera									
Viruses	Other information		Collection date	Passage history	B/Fl 4/06	B/Bris 3/07	B/Estonia 55669/11	B/Mass 02/12	B/Mass 02/12	B/Wis 1/10	B/Stock 12/11	B/Phuket 3073/13	B/Phuket 3073/13	B/HK 3417/14
Passage history		Passage history			Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg
Ferret number		Ferret number			F17/13*1	F38/14*2	F27/13*1	F05/15*1	F42/14*2	F10/13*2	F06/15*2	F35/14*2	F37/14*2	St Judes F715/14*1,4
Genetic Group		Genetic Group			1	2	2	2	2	3	3	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006		1	2006-12-15	E7/E1	640	640	320	160	640	160	160	20	160	160
B/Brisbane/3/2007		2	2007-09-03	E2/E2	1280	1280	320	160	1280	320	320	20	320	320
B/Estonia/55669/2011		2	2011-03-14	MDCK2/MDCK3	80	80	640	160	40	40	20	40	40	160
B/Massachusetts/02/2012		2	2012-03-13	MDCK1/C2/MDCK3	320	320	640	320	320	160	80	40	160	320
B/Massachusetts/02/2012		2	2012-03-13	E3/E3	640	640	320	80	640	80	160	10	160	160
B/Wisconsin/1/2010		3	2010-02-20	E3/E2	640	320	320	40	640	640	320	80	640	640
B/Stockholm/12/2011		3	2011-03-28	Ex/E2	160	80	40	<	80	80	160	20	80	80
B/Phuket/3073/2013		3	2013-11-21	MDCK2/MDCK2	320	320	320	640	320	320	160	640	320	320
B/Phuket/3073/2013		3	2013-11-21	E4/E3	320	320	80	20	320	320	160	40	320	160
B/Hong Kong/3417/2014		3	2014-06-04	E4/E1	80	40	40	10	80	80	40	10	80	160
TEST VIRUSES														
Number of viruses tested*					38	38	38	38	38	38	38	38	38	38
No with titre reduction ≤2-fold						1	13	9	2	5	16	6	13	38
%						2.6	34.2	23.7	5.2	13.2	42.1	15.8	34.2	100.0
No with titre reduction =4-fold					10	5	14	2	6	13	15	11	12	
%					26.3	13.2	36.8	5.2	15.9	34.2	39.5	28.9	31.6	
No with titre reduction ≥8-fold					28	32	11	27	30	20	7	21	13	
%					73.7	84.2	29.0	71.1	78.9	52.6	18.4	55.3	34.2	

Vaccine[#]
NH 2016-17
SH2017

* All viruses sequenced in time for the VCM fell in clade 3

[#] B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Reference virus results are taken from individual tables as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Figure 17. Phylogenetic comparison of influenza B (Yamagata-lineage) HA genes

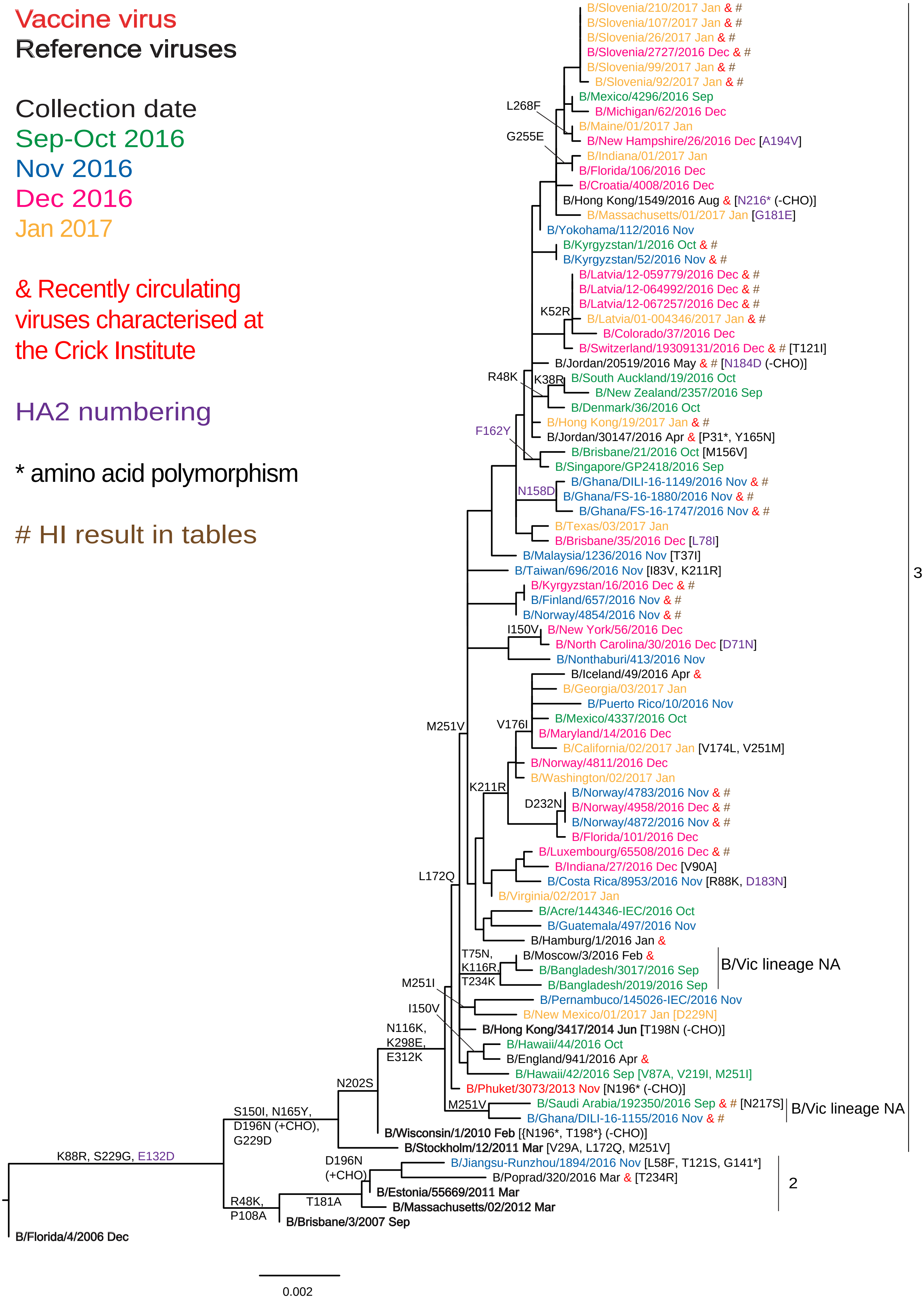

Figure 18. HA protein alignment for B (Yamagata-lineage) viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

```
                                    10        20        30        40        50        60        70        80        90       100
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006            DRICTGITSSNSPHVVKTATQGEVNVTGVIPLTTTPTKSYFANLKGTRTRGKLCPDCLNCTDLDVALGRPMCVGTTPSAKASILHEVKPVTSGCFPIMHD
B/Brisbane/3/2007           ...............................................K.......................................R............ 2
B/Massachusetts/02/2012     ...............................................K.......................................R............ 2
B/Estonia/55669/2011        ...............................................K.......................................R............ 2
B/Jiangsu-Runzhou/1894/2016 ...............................................K.........F.............................R............ 2
B/Stockholm/12/2011         ............................A..........................................................R............ 3
B/Wisconsin/1/2010          .......................................................................................R............ 3
B/Ghana/DILI-16-1155/2016   .......................................................................................R............ 3
B/Phuket/3073/2013          .......................................................................................R............ 3
B/England/941/2016          .......................................................................................R............ 3
B/Hong Kong/3417/2014       .......................................................................................R............ 3
B/Bangladesh/2019/2016      ..........................................................................N............R............ 3
B/Luxembourg/65508/2016     .......................................................................................R............ 3
B/Norway/4958/2016          .......................................................................................R............ 3
B/Norway/4811/2016          .......................................................................................R............ 3
B/California/02/2017        .......................................................................................R............ 3
B/North Carolina/30/2016    .......................................................................................R............ 3
B/Kyrgyzstan/16/2016        .......................................................................................R............ 3
B/Taiwan/696/2016           ..................................................................................V....R............ 3
B/Brisbane/35/2016          .......................................................................................R............ 3
B/Ghana/DILI-16-1149/2016   .......................................................................................R............ 3
B/Brisbane/21/2016          .......................................................................................R............ 3
B/Jordan/30147/2016         ..............................X........................................................R............ 3
B/Denmark/36/2016           ...............................................K.......................................R............ 3
B/South Auckland/19/2016    .....................................R.........K.......................................R............ 3
B/Switzerland/19309131/2016 .......................................................................................R............ 3
B/Latvia/01-004346/2017     ...................................................R...................................R............ 3
B/Kyrgyzstan/1/2016         .......................................................................................R............ 3
B/Massachusetts/01/2017     .......................................................................................R............ 3
B/Indiana/01/2017           .......................................................................................R............ 3
B/Maine/01/2017             .......................................................................................R............ 3
B/Slovenia/210/2017         .......................................................................................R............ 3

                                   110       120       130       140       150       160       170       180       190       200
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006            RTKIRQLPNLLRGYENIRLSTQNVIDAEKAPGGPYRLGTSGSCPNATSKSGFFATMAWAVPKDNNKNATNPLTVEVPYICTEGEDQITVWGFHSDDKTQM
B/Brisbane/3/2007           .......A............................................................................................
B/Massachusetts/02/2012     .......A........................................................................A...................
B/Estonia/55669/2011        .......A........................................................................A..............N....
B/Jiangsu-Runzhou/1894/2016 .......A............S...................X.......................................A..............N....
B/Stockholm/12/2011         .................................................I..............Y......Q.......................N....
B/Wisconsin/1/2010          .................................................I..............Y..............................X.X..
B/Ghana/DILI-16-1155/2016   ...............K.................................I..............Y..............................N....
B/Phuket/3073/2013          ...............K.................................I..............Y..............................X....
B/England/941/2016          ...............K.................................V..............Y......Q.......................N....
B/Hong Kong/3417/2014       ...............K.................................I..............Y......Q.......................N.N..
B/Bangladesh/2019/2016      ...............R.................................I..............Y......Q.......................N....
B/Luxembourg/65508/2016     ...............K.................................I..............Y......Q.......................N....
B/Norway/4958/2016          ...............K.................................I..............Y......Q.......................N....
B/Norway/4811/2016          ...............K.................................I..............Y......Q.......................N....
B/California/02/2017        ...............K.................................I..............Y......Q.L.I...................N....
B/North Carolina/30/2016    ...............K.................................V..............Y......Q.......................N....
B/Kyrgyzstan/16/2016        ...............K.................................I..............Y......Q.......................N....
B/Taiwan/696/2016           ...............K.................................I..............Y......Q.......................N....
B/Brisbane/35/2016          ...............K.................................I..............Y......Q.......................N....
B/Ghana/DILI-16-1149/2016   ...............K.................................I..............Y......Q.......................N....
B/Brisbane/21/2016          ...............K.................................I.....V........Y......Q.......................N....
B/Jordan/30147/2016         ...............K.................................I.....................Q.......................N....
B/Denmark/36/2016           ...............K.................................I..............Y......Q.......................N....
B/South Auckland/19/2016    ...............K.................................I..............Y......Q.......................N....
B/Switzerland/19309131/2016 ...............K....I............................I..............Y......Q.......................N....
B/Latvia/01-004346/2017     ...............K.................................I..............Y......Q.......................N....
B/Kyrgyzstan/1/2016         ...............K.................................I..............Y......Q.......................N....
B/Massachusetts/01/2017     ...............K.................................I..............Y......Q.......................N....
B/Indiana/01/2017           ...............K.................................I..............Y......Q.......................N....
B/Maine/01/2017             ...............K.................................I..............Y......Q.......................N....
B/Slovenia/210/2017         ...............K.................................I..............Y......Q.......................N....

                                   210       220       230       240       250       260       270       280       290       300
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006            KNLYGDSNPQKFTSSANGVTTHYVSQIGSFPDQTEDGGLPQSGRIVVDYMMQKPGKTGTIVYQRGVLLPQKVWCASGRSKVIKGSLPLIGEADCLHEKYG
B/Brisbane/3/2007           ............................G.......................................................................
B/Massachusetts/02/2012     ............................G.......................................................................
B/Estonia/55669/2011        ............................G.......................................................................
B/Jiangsu-Runzhou/1894/2016 ............................G.......................................................................
B/Stockholm/12/2011         ............................D.....................V.................................................
B/Wisconsin/1/2010          .S..........................D.......................................................................
B/Ghana/DILI-16-1155/2016   .S..........................D.....................V..............................................E..
B/Phuket/3073/2013          .S..........................D....................................................................E..
B/England/941/2016          .S..........................D....................................................................E..
B/Hong Kong/3417/2014       .S..........................D....................................................................E..
B/Bangladesh/2019/2016      .S..........................D....K...............................................................E..
B/Luxembourg/65508/2016     .S..........................D.....................V..............................................E..
B/Norway/4958/2016          .S........R.................D..N..................V..............................................E..
B/Norway/4811/2016          .S........R.................D.....................V..............................................E..
B/California/02/2017        .S........R.................D....................................................................E..
B/North Carolina/30/2016    .S..........................D.....................V..............................................E..
B/Kyrgyzstan/16/2016        .S..........................D.....................V..............................................E..
B/Taiwan/696/2016           .S........R.................D.....................V..............................................E..
B/Brisbane/35/2016          .S..........................D.....................V..............................................E..
B/Ghana/DILI-16-1149/2016   .S..........................D.....................V..............................................E..
B/Brisbane/21/2016          .S..........................D.....................V..............................................E..
B/Jordan/30147/2016         .S..........................D.....................V..............................................E..
B/Denmark/36/2016           .S..........................D.....................V..............................................E..
B/South Auckland/19/2016    .S..........................D.....................V..............................................E..
B/Switzerland/19309131/2016 .S..........................D.....................V..............................................E..
B/Latvia/01-004346/2017     .S..........................D.....................V..............................................E..
B/Kyrgyzstan/1/2016         .S..........................D.....................V..............................................E..
B/Massachusetts/01/2017     .S..........................D.....................V..............................................E..
B/Indiana/01/2017           .S..........................D.....................V...E..........................................E..
B/Maine/01/2017             .S..........................D.....................V................F.............................E..
B/Slovenia/210/2017         .S..........................D.....................V..............................................E..
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                                     310       320       330       340         4        14        24        34        44        54
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006              GLNKSKPYYTGEHAKAIGNCPIWVKTPLKLANGTKYRPPAKLLKERGFFGAIAGFLEGGWEGMIAGWHGYTSHGAHGVAVAADLKSTQEAINKITKNLNS
B/Brisbane/3/2007             ....................................................................................................
B/Massachusetts/02/2012       ....................................................................................................
B/Estonia/55669/2011          ....................................................................................................
B/Jiangsu-Runzhou/1894/2016   ....................................................................................................
B/Stockholm/12/2011           ....................................................................................................
B/Wisconsin/1/2010            ....................................................................................................
B/Ghana/DILI-16-1155/2016     ...........K........................................................................................
B/Phuket/3073/2013            ...........K........................................................................................
B/England/941/2016            ...........K........................................................................................
B/Hong Kong/3417/2014         ...........K........................................................................................
B/Bangladesh/2019/2016        ...........K........................................................................................
B/Luxembourg/65508/2016       ...........K........................................................................................
B/Norway/4958/2016            ...........K........................................................................................
B/Norway/4811/2016            ...........K........................................................................................
B/California/02/2017          ...........K........................................................................................
B/North Carolina/30/2016      ...........K........................................................................................
B/Kyrgyzstan/16/2016          ...........K........................................................................................
B/Taiwan/696/2016             ...........K........................................................................................
B/Brisbane/35/2016            ...........K........................................................................................
B/Ghana/DILI-16-1149/2016     ...........K........................................................................................
B/Brisbane/21/2016            ...........K........................................................................................
B/Jordan/30147/2016           ...........K........................................................................................
B/Denmark/36/2016             ...........K........................................................................................
B/South Auckland/19/2016      ...........K........................................................................................
B/Switzerland/19309131/2016   ...........K........................................................................................
B/Latvia/01-004346/2017       ...........K........................................................................................
B/Kyrgyzstan/1/2016           ...........K........................................................................................
B/Massachusetts/01/2017       ...........K........................................................................................
B/Indiana/01/2017             ...........K........................................................................................
B/Maine/01/2017               ...........K........................................................................................
B/Slovenia/210/2017           ...........K........................................................................................

                                      64        74        84        94       104       114       124       134       144       154
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006              LSELEVKNLQRLSGAMDELHNEILELDEKVDDLRADTISSQIELAVLLSNEGIINSEDEHLLALERKLKKMLGPSAVEIGNGCFETKHKCNQTCLDRIAA
B/Brisbane/3/2007             .............................................................................D......................
B/Massachusetts/02/2012       .............................................................................D......................
B/Estonia/55669/2011          .............................................................................D......................
B/Jiangsu-Runzhou/1894/2016   .............................................................................D......................
B/Stockholm/12/2011           .............................................................................D......................
B/Wisconsin/1/2010            .............................................................................D......................
B/Ghana/DILI-16-1155/2016     .............................................................................D......................
B/Phuket/3073/2013            .............................................................................D......................
B/England/941/2016            .............................................................................D......................
B/Hong Kong/3417/2014         .............................................................................D......................
B/Bangladesh/2019/2016        .............................................................................D......................
B/Luxembourg/65508/2016       .............................................................................D......................
B/Norway/4958/2016            .............................................................................D......................
B/Norway/4811/2016            .............................................................................D......................
B/California/02/2017          .............................................................................D......................
B/North Carolina/30/2016      ................N............................................................D......................
B/Kyrgyzstan/16/2016          .............................................................................D......................
B/Taiwan/696/2016             .............................................................................D......................
B/Brisbane/35/2016            .......................I.....................................................D......................
B/Ghana/DILI-16-1149/2016     .............................................................................D......................
B/Brisbane/21/2016            .............................................................................D......................
B/Jordan/30147/2016           .............................................................................D......................
B/Denmark/36/2016             .............................................................................D......................
B/South Auckland/19/2016      .............................................................................D......................
B/Switzerland/19309131/2016   .............................................................................D......................
B/Latvia/01-004346/2017       .............................................................................D......................
B/Kyrgyzstan/1/2016           .............................................................................D......................
B/Massachusetts/01/2017       .............................................................................D......................
B/Indiana/01/2017             .............................................................................D......................
B/Maine/01/2017               .............................................................................D......................
B/Slovenia/210/2017           .............................................................................D......................

                                     164       174       184       194       204       214
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....
B/Florida/4/2006              GTFNAGEFSLPTFDSLNITAASLNDDGLDNHTILLYYSTAASSLAVTLMLAIFIVYMVSRDNVSCSICL
B/Brisbane/3/2007             .....................................................................
B/Massachusetts/02/2012       .....................................................................
B/Estonia/55669/2011          .....................................................................
B/Jiangsu-Runzhou/1894/2016   .....................................................................
B/Stockholm/12/2011           .....................................................................
B/Wisconsin/1/2010            .....................................................................
B/Ghana/DILI-16-1155/2016     .....................................................................
B/Phuket/3073/2013            .....................................................................
B/England/941/2016            .....................................................................
B/Hong Kong/3417/2014         .....................................................................
B/Bangladesh/2019/2016        .....................................................................
B/Luxembourg/65508/2016       .....................................................................
B/Norway/4958/2016            .....................................................................
B/Norway/4811/2016            .....................................................................
B/California/02/2017          .....................................................................
B/North Carolina/30/2016      .....................................................................
B/Kyrgyzstan/16/2016          .....................................................................
B/Taiwan/696/2016             .....................................................................
B/Brisbane/35/2016            .....................................................................
B/Ghana/DILI-16-1149/2016     ...D.................................................................
B/Brisbane/21/2016            .......Y.............................................................
B/Jordan/30147/2016           .....................................................................
B/Denmark/36/2016             .....................................................................
B/South Auckland/19/2016      .....................................................................
B/Switzerland/19309131/2016   .....................................................................
B/Latvia/01-004346/2017       .....................................................................
B/Kyrgyzstan/1/2016           .....................................................................
B/Massachusetts/01/2017       ..........................E..........................................
B/Indiana/01/2017             .....................................................................
B/Maine/01/2017               .....................................................................
B/Slovenia/210/2017           .....................................................................
```

Vaccine virus: **Reference virus**: **Genetic group**: **HA1/HA2 boundary**: **Potential N-linked glycosylation site**

Figure 19. Phylogenetic comparison of influenza B (Yamagata-lineage) NA genes

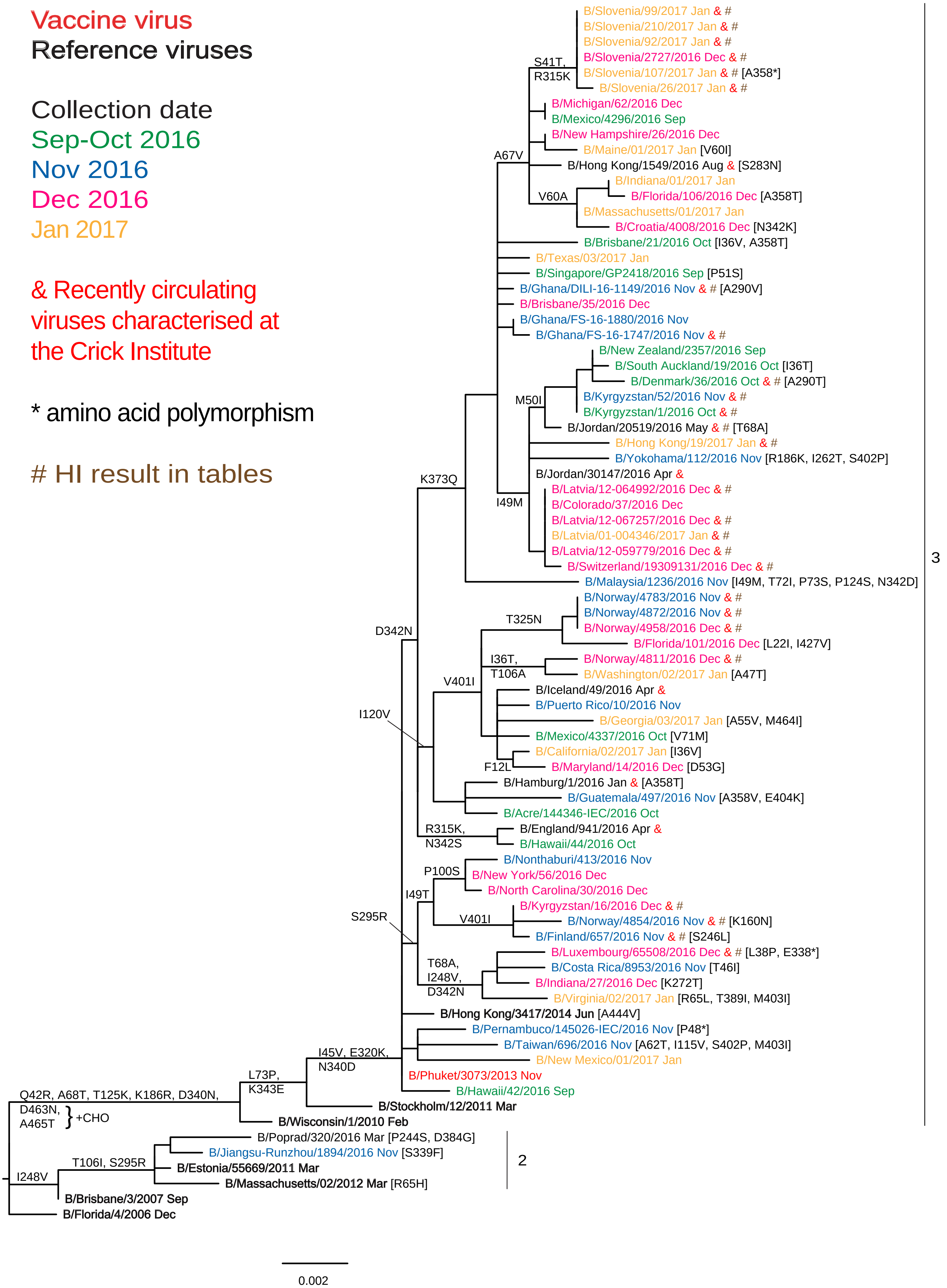

Figure 20. NA protein alignment for B (Yamagata-lineage) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

```
                                      10        20        30        40        50        60        70        80        90       100
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006             MLPSTIQTLTLFLTSGGVLLSLYVSASLSYLLYSDILLKFSQTEITAPIMPLDCANASNVQAVNRSAAKGVTLLLPEPEWTYPRLSCPGSTFQKALLISP
B/Brisbane/3/2007            .................................................................................................... 2
B/Massachusetts/02/2012      ................................................................H................................... 2
B/Estonia/55669/2011         .................................................................................................... 2
B/Jiangsu-Runzhou/1894/2016  .................................................................................................... 2
B/Stockholm/12/2011          .........................................R.........................T....P........................... 3
B/Wisconsin/1/2010           .........................................R.........................T................................ 3
B/Ghana/FS-16-1880/2016      .........................................R..V......................T....P........................... 3
B/Phuket/3073/2013           .........................................R..V......................T....P........................... 3
B/England/941/2016           .........................................R..V......................T....P........................... 3
B/Hong Kong/3417/2014        .........................................R..V......................T....P........................... 3
B/New Mexico/01/2017         .........................................R..V......................T....P........................... 3
B/Luxembourg/65508/2016      .....................................P...R..V...........................P........................... 3
B/Norway/4958/2016           .........................................R..V......................T....P........................... 3
B/Norway/4811/2016           ...................................T.....R..V......................T....P........................... 3
B/California/02/2017         ...........L.......................V.....R..V......................T....P........................... 3
B/North Carolina/30/2016     .........................................R..V...T..................T....P...........................S 3
B/Kyrgyzstan/16/2016         .........................................R..V...T..................T....P........................... 3
B/Taiwan/696/2016            .........................................R..V................T.....T....P........................... 3
B/Brisbane/35/2016           .........................................R..V......................T....P........................... 3
B/Ghana/DILI-16-1149/2016    .........................................R..V......................T....P........................... 3
B/Brisbane/21/2016           ...................................V.....R..V......................T....P........................... 3
B/Jordan/30147/2016          .........................................R..V...M..................T....P........................... 3
B/Denmark/36/2016            .........................................R..V...MI.................T....P........................... 3
B/South Auckland/19/2016     ...................................T.....R..V...MI.................T....P........................... 3
B/Switzerland/19309131/2016  .........................................R..V...M..................T....P........................... 3
B/Latvia/01-004346/2017      .........................................R..V...M..................T....P........................... 3
B/Kyrgyzstan/1/2016          .........................................R..V...MI.................T....P........................... 3
B/Massachusetts/01/2017      .........................................R..V..............A......VT....P........................... 3
B/Indiana/01/2017            .........................................R..V..............A......VT....P........................... 3
B/Maine/01/2017              .........................................R..V..............I......VT....P........................... 3
B/Slovenia/210/2017          ........................................TR..V.....................VT....P........................... 3

                                     110       120       130       140       150       160       170       180       190       200
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006             HRFGETKGNSAPLIIREPFIACGPTECKHFALTHYAAQPGGYYNGTREDRNKLRHLISVKLGKIPTVENSIFHMAAWSGSACHDGKEWTYIGVDGPDSNA
B/Brisbane/3/2007            ....................................................................................................
B/Massachusetts/02/2012      .....I..............................................................................................
B/Estonia/55669/2011         .....I..............................................................................................
B/Jiangsu-Runzhou/1894/2016  .....I..............................................................................................
B/Stockholm/12/2011          ........................K............................................................R..............
B/Wisconsin/1/2010           ........................K............................................................R..............
B/Ghana/FS-16-1880/2016      ........................K............................................................R..............
B/Phuket/3073/2013           ........................K............................................................R..............
B/England/941/2016           ........................K............................................................R..............
B/Hong Kong/3417/2014        ........................K............................................................R..............
B/New Mexico/01/2017         ........................K............................................................R..............
B/Luxembourg/65508/2016      ........................K............................................................R..............
B/Norway/4958/2016           ...................V....K............................................................R..............
B/Norway/4811/2016           .....A.............V....K............................................................R..............
B/California/02/2017         ...................V....K............................................................R..............
B/North Carolina/30/2016     ........................K............................................................R..............
B/Kyrgyzstan/16/2016         ........................K............................................................R..............
B/Taiwan/696/2016            ..............V.........K............................................................R..............
B/Brisbane/35/2016           ........................K............................................................R..............
B/Ghana/DILI-16-1149/2016    ........................K............................................................R..............
B/Brisbane/21/2016           ........................K............................................................R..............
B/Jordan/30147/2016          ........................K............................................................R..............
B/Denmark/36/2016            ........................K............................................................R..............
B/South Auckland/19/2016     ........................K............................................................R..............
B/Switzerland/19309131/2016  ........................K............................................................R..............
B/Latvia/01-004346/2017      ........................K............................................................R..............
B/Kyrgyzstan/1/2016          ........................K............................................................R..............
B/Massachusetts/01/2017      ........................K............................................................R..............
B/Indiana/01/2017            ........................K............................................................R..............
B/Maine/01/2017              ........................K............................................................R..............
B/Slovenia/210/2017          ........................K............................................................R..............

                                     210       220       230       240       250       260       270       280       290       300
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006             LLKIKYGEAYTDTYHSYAKNILRTQESACNCIGGDCYLMITDGPASGISECRFLKIREGRIIKEIFPTGRVKHTEECTCGFASNKTIECACRDNSYTAKR
B/Brisbane/3/2007            ...............................................V....................................................
B/Massachusetts/02/2012      ...............................................V..............................................R.....
B/Estonia/55669/2011         ...............................................V..............................................R.....
B/Jiangsu-Runzhou/1894/2016  ...............................................V..............................................R.....
B/Stockholm/12/2011          ....................................................................................................
B/Wisconsin/1/2010           ....................................................................................................
B/Ghana/FS-16-1880/2016      ....................................................................................................
B/Phuket/3073/2013           ....................................................................................................
B/England/941/2016           ....................................................................................................
B/Hong Kong/3417/2014        ....................................................................................................
B/New Mexico/01/2017         ....................................................................................................
B/Luxembourg/65508/2016      ...............................................V..............................................R.....
B/Norway/4958/2016           ....................................................................................................
B/Norway/4811/2016           ....................................................................................................
B/California/02/2017         ....................................................................................................
B/North Carolina/30/2016     ..............................................................................................R.....
B/Kyrgyzstan/16/2016         ..............................................................................................R.....
B/Taiwan/696/2016            ....................................................................................................
B/Brisbane/35/2016           ....................................................................................................
B/Ghana/DILI-16-1149/2016    .........................................................................................V..........
B/Brisbane/21/2016           ....................................................................................................
B/Jordan/30147/2016          ....................................................................................................
B/Denmark/36/2016            .........................................................................................T..........
B/South Auckland/19/2016     ....................................................................................................
B/Switzerland/19309131/2016  ....................................................................................................
B/Latvia/01-004346/2017      ....................................................................................................
B/Kyrgyzstan/1/2016          ....................................................................................................
B/Massachusetts/01/2017      ....................................................................................................
B/Indiana/01/2017            ....................................................................................................
B/Maine/01/2017              ....................................................................................................
B/Slovenia/210/2017          ....................................................................................................
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                                     310       320       330       340       350       360       370       380       390       400
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006              PFVKLNVETDTAEIRLMCTETYLDTPRPNDGSITGPCESDGDKGSGGIKGGFVHQRMASKIGRWYSRTMSKTKRMGMGLYVKYDGDPWTDSEALALSGVM
B/Brisbane/3/2007             ....................................................................................................
B/Massachusetts/02/2012       ....................................................................................................
B/Estonia/55669/2011          ....................................................................................................
B/Jiangsu-Runzhou/1894/2016   ......................................F.............................................................
B/Stockholm/12/2011           .......................................N..E.........................................................
B/Wisconsin/1/2010            .......................................N............................................................
B/Ghana/FS-16-1880/2016       ...................K.....................NE.............................Q...........................
B/Phuket/3073/2013            ...................K......................E.........................................................
B/England/941/2016            ..............K....K.....................SE.........................................................
B/Hong Kong/3417/2014         ...................K......................E.........................................................
B/New Mexico/01/2017          ...................K......................E.........................................................
B/Luxembourg/65508/2016       ...................K.................X...NE.........................................................
B/Norway/4958/2016            ...................K....N................NE.........................................................
B/Norway/4811/2016            ...................K.....................NE.........................................................
B/California/02/2017          ...................K.....................NE.........................................................
B/North Carolina/30/2016      ...................K......................E.........................................................
B/Kyrgyzstan/16/2016          ...................K......................E.........................................................
B/Taiwan/696/2016             ...................K......................E.........................................................
B/Brisbane/35/2016            ...................K.....................NE.............................Q...........................
B/Ghana/DILI-16-1149/2016     ...................K.....................NE.............................Q...........................
B/Brisbane/21/2016            ...................K.....................NE..............T..............Q...........................
B/Jordan/30147/2016           ...................K.....................NE.............................Q...........................
B/Denmark/36/2016             ...................K.....................NE.............................Q...........................
B/South Auckland/19/2016      ...................K.....................NE.............................Q...........................
B/Switzerland/19309131/2016   ...................K.....................NE.............................Q...........................
B/Latvia/01-004346/2017       ...................K.....................NE.............................Q...........................
B/Kyrgyzstan/1/2016           ...................K.....................NE.............................Q...........................
B/Massachusetts/01/2017       ...................K.....................NE.............................Q...........................
B/Indiana/01/2017             ...................K.....................NE.............................Q...........................
B/Maine/01/2017               ...................K.....................NE.............................Q...........................
B/Slovenia/210/2017           ..............K....K.....................NE.............................Q...........................

                                     410       420       430       440       450       460
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|.
B/Florida/4/2006              VSMEEPGWYSFGFEIKDKKCDVPCIGIEMVHDGGKTTWHSAATAIYCLMGSGQLLWDTVTGVDMAL
B/Brisbane/3/2007             ..................................................................
B/Massachusetts/02/2012       ..................................................................
B/Estonia/55669/2011          ..................................................................
B/Jiangsu-Runzhou/1894/2016   ..................................................................
B/Stockholm/12/2011           ..............................................................N.T.
B/Wisconsin/1/2010            ..............................................................N.T.
B/Ghana/FS-16-1880/2016       ..............................................................N.T.
B/Phuket/3073/2013            ..............................................................N.T.
B/England/941/2016            ..............................................................N.T.
B/Hong Kong/3417/2014         ...........................................V..................N.T.
B/New Mexico/01/2017          ..............................................................N.T.
B/Luxembourg/65508/2016       ..............................................................N.T.
B/Norway/4958/2016            I.............................................................N.T.
B/Norway/4811/2016            I.............................................................N.T.
B/California/02/2017          I.............................................................N.T.
B/North Carolina/30/2016      ..............................................................N.T.
B/Kyrgyzstan/16/2016          I.............................................................N.T.
B/Taiwan/696/2016             .PI...........................................................N.T.
B/Brisbane/35/2016            ..............................................................N.T.
B/Ghana/DILI-16-1149/2016     ..............................................................N.T.
B/Brisbane/21/2016            ..............................................................N.T.
B/Jordan/30147/2016           ..............................................................N.T.
B/Denmark/36/2016             ..............................................................N.T.
B/South Auckland/19/2016      ..............................................................N.T.
B/Switzerland/19309131/2016   ..............................................................N.T.
B/Latvia/01-004346/2017       ..............................................................N.T.
B/Kyrgyzstan/1/2016           ..............................................................N.T.
B/Massachusetts/01/2017       ..............................................................N.T.
B/Indiana/01/2017             ..............................................................N.T.
B/Maine/01/2017               ..............................................................N.T.
B/Slovenia/210/2017           ..............................................................N.T.
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Table 14. Antiviral (NAI) testing of isolates with collection dates after 2016-08-31*

Month of Collection	Number of viruses tested for phenotype[1]						
	A(H1N1)pdm09		A(H3N2)[2]		Influenza B		
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	Totals
2016							
September	2		55		1		58
October	1		29		7		37
November	3		124		20		147
December	6		204		26		236
2017							
January	6		38		17		61
Totals	18		450		71		539

[1] Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

[2] Amino acid substitutions (polymorphism) at residues NA 148 and 151: 405 sequences available.

NA sequence	No Sequences	Mean IC_{50}	
		Oseltamivir	Zanamivir
No polymorphism	360	0.39	0.37
148/151 polymorphism	45	0.46	0.45

* As of 2017-02-17